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Urinary Incontinence

Number: 0223

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Scope of Policy

This Clinical Policy Bulletin addresses urinary incontinence.

I. **Medical Necessity**

Aetna considers multi-channel urodynamic studies medically necessary when the member has both symptoms and physical findings of urinary incontinence/voiding dysfunctions (such as stress incontinence, overactive bladder, lower urinary tract symptoms) and there is consideration by the provider to perform invasive, potentially morbid or irreversible treatments after conservative management has been tried and failed.

Aetna considers the following urinary incontinence interventions medically necessary when criteria are met:

A. **Artificial Urinary Sphincter**

Aetna considers the implantation of an artificial urinary sphincter (AUS) medically necessary for the treatment of urinary incontinence (UI) due to intrinsic urethral sphincter deficiency (IUSD) for members with *any* of the following indications:

1. Children with intractable UI due to IUSD who are refractory to behavioral or pharmacological therapies and are unsuitable candidates for other types of surgical procedures for correction of UI; or
2. Members who are 6 or more months post-prostatectomy who have had no improvement in the severity of UI despite trials of behavioral and/or pharmacological therapies; or
3. Members with epispadias-exstrophy in whom bladder neck reconstruction has failed; or
4. Women with intractable UI who have failed behavioral or pharmacological, and other surgical treatments.

Aetna considers the artificial urinary sphincter experimental, investigational, or unproven for all other indications because its effectiveness for indications other than the ones listed above has not

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been established.

B. **Peri-Urethral Injections of Bulking Agents**

Aetna considers peri-urethral injections of bulking agents that are cleared by the Food and Drug Administration (FDA) for UI (e.g., Bulkamid (polyacrylamide hydrogel), Coaptite [calcium hydroxylapatite], Contigen [glutaraldehyde crossed-linked collagen], Durasphere [carbon-coated spheres/beads], Macroplastique [polydimethylsiloxane], Uryx [ethylene vinyl alcohol copolymer]) medically necessary for the management of members with UI resulting from intrinsic sphincter deficiency that is refractory to conservative management (e.g., Kegel exercises, biofeedback, electrical stimulation, and/or pharmacotherapies).

Members whose incontinence does not improve after 3 treatments with bulking agents are considered treatment failures and are not likely to respond to this therapy. In such cases, further treatment with bulking agents is not considered medically necessary.

Aetna considers injection of peri-urethral bulking agents for UI experimental, investigational, or unproven for neurogenic bladder and all other indications.

C. **Implantable Sacral Nerve Stimulators (e.g., Axonics and InterStim)**

Aetna considers permanent implantation (Stage 2) of FDA-approved implantable sacral nerve stimulators (e.g., Axonics and InterStim) medically necessary for the treatment of urge UI or symptoms of urge-frequency when *all* of the following criteria are met:

1. The member has experienced urge UI or symptoms of urge-frequency for at least 6 months and the condition has resulted in significant disability (the frequency and/or severity of symptoms are limiting the member's ability to participate in daily activities); *and*
2. Pharmacotherapies (i.e., at least 2 different anti-cholinergic drugs or an anti-cholinergic and a beta-3 adrenergic receptor agonist (mirabegron)) as well as behavioral treatments (e.g., pelvic floor exercise, biofeedback, timed voids, and fluid management) have failed; *and*
3. Test stimulation (Stage 1) provides at least 50 % decrease in symptoms.

A test stimulation (Stage 1) of the device is considered medically necessary for members who meet selection criteria a and b above.

Aetna also considers permanent implantation (Stage 2) of a sacral nerve stimulator medically necessary for the treatment of non-obstructive urinary retention when *all* of the following criteria are met:

1. The member has experienced urinary retention for at least 6 months and the condition has resulted in significant disability (the frequency and/or severity of symptoms are limiting the member's ability to participate in daily activities); *and*
2. Pharmacotherapies (e.g., alpha blockers and antibiotics for urinary tract infections) as well as intermittent catheterization have failed or are not well-tolerated; *and*
3. A test stimulation (Stage 1) of the device has provided at least 50 % decrease in residual urine volume.

Aetna considers test stimulation (Stage 1) of the right and left sides (where leads are placed bilaterally; and each side is tested sequentially during a single visit) medically necessary for members who meet selection criteria a and b above for treatment of urgent incontinence and non-obstructive urinary retention. No more than 6 total test stimulations (Stage 1) are considered medically necessary. Aetna considers bilateral implantation of a permanent sacral nerve stimulator experimental, investigational, or unproven for the treatment of UI and non-

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obstructive urinary retention because the effectiveness of bilateral implantation of a permanent sacral nerve stimulator has not been established.

Aetna considers removal of an implantable sacral nerve stimulator medically necessary even where the initial implantation of the implantable sacral nerve stimulator was not indicated.

Aetna considers the use of implantable sacral nerve stimulator experimental, investigational, or unproven for all other indications because its effectiveness for indications other than the ones listed above has not been established.

According to the product labeling, implantable sacral nerve stimulator is contraindicated and has no proven value for individuals who have not demonstrated an appropriate response to test stimulation (Stage 1) or are unable to operate the neurostimulator.

Exclusions

Implantable sacral nerve stimulator has no proven value for individuals with mechanical obstruction such as benign prostatic hypertrophy, or urethral stricture; persons with stress incontinence; and individuals with neurologic disease origins, such as multiple sclerosis or diabetes with peripheral nerve involvement. Implantable sacral nerve stimulator has not been shown to be effective for urinary retention, urinary frequency-urgency syndrome or urge urinary incontinence due to these causes.

D. **Vaginal Cones**

Aetna considers weighted vaginal cones (vaginal weights) medically necessary DME when they are used in combination with a structured pelvic floor muscle exercise (Kegel's exercise) program for the treatment of simple (pure) stress UI.

Aetna considers vaginal cones experimental, investigational, or unproven for other indications because their effectiveness for indications other than the ones listed above has not been established.

E. **Pessary (Bladder Neck Support Prosthesis)**

Aetna considers a pessary, a plastic device that fits into the vagina to help support the uterus and bladder, medically necessary DME for the treatment of women with stress or mixed UI, and for the treatment of pelvic organ (uterine) prolapse.

Aetna considers a pessary experimental, investigational, or unproven for other indications because its effectiveness for indications other than the ones listed above has not been established.

F. **Tension-Free Vaginal Tape Procedure**

Aetna considers the tension-free vaginal tape (TVT) procedure medically necessary for the treatment of stress UI when women with intractable UI have failed behavioral and/or pharmacological treatments.

Aetna considers the TVT procedure experimental, investigational, or unproven for other indications (except for the treatment of pelvic organ prolapse complicated by stress UI - see CPB 0858 - Organ Prolapse: Selected Procedures) because its effectiveness for indications other than the one listed above has not been established.

G. **Transobturator Tape Procedure**

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Aetna considers the transobturator tape (TOT) procedure medically necessary for the treatment of stress UI when women with intractable stress UI have failed behavioral and/or pharmacological treatments.

Aetna considers the TOT procedure experimental, investigational, or unproven for urge urinary incontinence and other indications because its effectiveness for indications other than the one listed above has not been established.

H. **Colposuspension and Sling Procedures**

Aetna considers colposuspension and conventional sub-urethral sling procedures (e.g., the Solyx single-incision sling) medically necessary for persons with stress UI that is refractory to conservative management (e.g., pelvic floor muscle training, electrical stimulation, and biofeedback).

Aetna considers the colposuspension and sub-urethral sling procedures experimental, investigational, or unproven for other indications because their effectiveness for indications other than the one listed above has not been established.

I. **Biofeedback**

For biofeedback for UI, see CPB 0132 - Biofeedback.

J. **Percutaneous Tibial Nerve Stimulation**

Aetna considers percutaneous tibial nerve stimulation (PTNS) (e.g., the NURO Percutaneous Tibial Neuromodulation System (Medtronic, Minneapolis MN), and the Urgent PC Neuromodulation System (Uroplasty, Inc., Minneapolis, MN)) medically necessary for the treatment of members with idiopathic overactive bladder (urge UI or urge-frequency) when they meet the first 2 criteria listed for Implantable Sacral Nerve Stimulators (e.g., Axonics and InterStim) (policy section I.B.3a and I.B.3b for the treatment of urge urinary incontinence or symptoms of urge-frequency). In general, 12 treatments (once-weekly) with PTNS are needed for symptom relief. If the member fails to improve after 12 PTNS treatments, continued treatment is considered not medically necessary. If the member improves after 12 PTNS treatments, continued monthly treatments are considered medically necessary as long as the member's symptoms remain improved.

Aetna considers percutaneous tibial nerve stimulation experimental, investigational, or unproven when criteria are not met.

K. **Transurethral Radiofrequency Therapy (Renessa Procedure)**

Aetna considers transurethral radiofrequency therapy (Renessa procedure) medically necessary for the treatment of stress UI in non-pregnant women who are either not able or not willing to undergo surgery for their condition.

L. **Urethral Inserts**

Aetna considers urethral inserts medically necessary for the treatment of female stress UI.

Aetna considers urethral inserts experimental, investigational, or unproven for other indications because their effectiveness for indications other than the one listed above has not been established.

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M. **Cunningham Clamp**

Aetna considers the Cunningham clamp medically necessary for the treatment of post-prostatectomy urinary incontinence in men with stress incontinence and good bladder storage function.

N. **Intravaginal Electrical Stimulation**

Aetna considers intravaginal electrical stimulation of the pelvic floor medically necessary for women with stress, urgency or mixed urinary incontinence.

II. Experimental, Investigational, or Unproven

A. Aetna considers the following UI interventions/managements experimental, investigational, or unproven because the effectiveness of the treatment has not been established:

- Adjustable retropubic sub-urethral sling in the treatment of stress urinary incontinence
- Adjustable Trans-obturator Male System for the treatment of stress urinary incontinence (SUI)
- Autologous myoblast transplantation
- Autologous muscle-derived cell therapy
- Bariatric surgery as a treatment of urinary incontinence in persons who would otherwise not meet medical necessity criteria for obesity surgery in CPB 0157 - Obesity Surgery
- Collagen porcine dermis mesh
- Dynamometry for quantification of pelvic floor muscle strength in female urinary incontinence
- Electro-acupuncture for the treatment of neurogenic bladder
- Flyte System (mechanotherapy) for the treatment of SUI
- Genetic testing for stress urinary incontinence
- High-intensity focused electromagnetic therapy for the treatment of stress urinary incontinence
- Implantable sacral nerve stimulator for the treatment of neurogenic bladder
- Laser Therapy: The Genityte Procedure (laser therapy) and FemiLift (CO2 laser)
- Magnetically controlled endo-urethral artificial urinary sphincter
- Magnetic stimulation for the treatment of women with SUI
- Micro-ablative radiofrequency for the treatment of over-active bladder
- Moxibustion for the treatment of post-stroke UI and SUI
- Neocontrol System, which uses extracorporeal magnetic innervation (ExMI)
- Platelet-rich plasma
- Protect PNS System (wireless microtechnology neuromodulation of the posterior tibial nerve) for the treatment of overactive bladder and the associated symptoms of urinary urgency, urinary frequency, and urge incontinence
- Pudendal nerve stimulation
- Radiofrequency micro-remodeling with the SURx System (paraurethral or transvaginal)
- Subcutaneous and subfascial tibial nerve stimulation (INTIBIA (Coloplast, Humlebaek, Denmark))
- Subcutaneous tibial nerve stimulation (e.g., the eCoin Peripheral Neurostimulator System (Valencia Technologies, Valencia, CA); Altaviva Implantable Tibial Neuromodulation Device (ITNM) (Medtronic, Minneapolis, MN))
- Subfascial tibial nerve stimulation (e.g., BlueWind Revi) for the treatment of overactive bladder syndrome, and urge urinary incontinence
- Stem cell therapy (including mesenchymal stem/stromal cells)
- Transcutaneous electrical nerve stimulation (TENS) in the treatment of overactive bladder
- Transcutaneous tibial nerve stimulation (e.g., the Vivally System, and ZIDA Wearable Neuromodulation System) for the treatment of overactive bladder (OAB) and the

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associated symptoms of urinary urgency, urinary frequency, and urge incontinence
- Transperineal implantation of permanent adjustable balloon continence device (e.g., ACT, ProACT Therapy System, Uromedica, Inc.)
- Vibratory perineal stimulation
- Peri-urethral injections of bulking agents for *any* of the following circumstances:

    - Members undergoing or planning to undergo desensitization injections to meat products; *or*
    - Members with an acute condition involving cystitis, urethritis, or infection; *or*
    - Members with severe allergies manifested by a history of anaphylaxis, or history or presence of multiple severe allergies; *or*
    - Previous pelvic radiation therapy; *or*
    - Unstable or noncompliant bladder

III. Policy Limitations and Exclusions

Pelvic Muscle Trainers

Note: Aetna does not cover the Athena pelvic muscle trainer, Gyneflex, Kegelmaster, or similar devices for the treatment of UI because these devices are considered exercise machines, and they do not meet Aetna's definition of covered durable medical equipment (DME). Most Aetna plans exclude coverage of exercise devices. Please check benefit plan descriptions for details. In addition, such exercise devices do not meet Aetna's definition of covered DME because they are not primarily medical in nature and/or are normally of use to persons who do not have an illness or injury.

IV. Related Policies

- CPB 0132 - Biofeedback
- CPB 0157 - Obesity Surgery
- CPB 0858 - Organ Prolapse: Selected Procedures

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
***Multi-channel Urodynamic Studies*:**	
CPT codes covered if selection criteria are met:	
51726	Complex cystometrogram (ie, calibrated electronic equipment)
51727	with urethral pressure profile studies (ie, urethral closure pressure profile), any technique
51728	with voiding pressure studies (ie, bladder voiding pressure), any technique
51729	with voiding pressure studies (ie, bladder voiding pressure) and urethral pressure profile studies (ie, urethral closure pressure profile), any technique
51741	Complex uroflowmetry (eg, calibrated electronic equipment)

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Code	Code Description
51784	Electromyography studies (EMG) of anal or urethral sphincter, other than needle, any technique
51785	Needle electromyography studies (EMG) of anal or urethral sphincter, any technique
51792	Stimulus evoked response (eg, measurement of bulbocavernosus reflex latency time)
51797	Voiding pressure studies, intra-abdominal (ie, rectal, gastric, intraperitoneal) (List separately in addition to code for primary procedure)
51798	Measurement of post-voiding residual urine and/or bladder capacity by ultrasound, non-imaging

ICD-10 codes covered if selection criteria are met:

N32.0 - N32.9	Other disorders of bladder
N39.3 - N39.9	Urinary incontinence
R32	Unspecified urinary incontinence
R39.81 - R39.89	Other symptoms and signs involving the genitourinary system

Genetic Testing:

CPT codes not covered for indications listed in this CPB:

Genetic testing for stress urinary incontinence - no specific code:

ICD-10 codes not covered for indications listed in the CPB:

N39.3	Stress incontinence (female) (male)
N39.46	Mixed incontinence

Artificial Urinary Sphincter [Not covered for magnetically controlled endourethral artificial urinary sphincter]:

CPT codes covered if selection criteria are met:

53444	Insertion of tandem cuff (dual cuff)
53445	Insertion of inflatable urethral/bladder neck sphincter, including placement of pump, reservoir, and cuff
53446	Removal of inflatable urethral/bladder neck sphincter, including pump, reservoir, and cuff
53447	Removal and replacement of inflatable urethral/bladder neck sphincter including, pump, reservoir, and cuff at the same operative session
53449	Repair of inflatable urethral/bladder neck sphincter, including pump, reservoir, and cuff

HCPCS codes covered if selection criteria are met:

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Code	Code Description
C1815	Prosthesis, urinary sphincter (implantable)

ICD-10 codes covered if selection criteria are met:

N36.42	Intrinsic sphincter deficiency (ISD)
N39.3 - N39.9, R32	Urinary incontinence
Q64.0	Epispadias
Q64.10 - Q64.19	Exstrophy of urinary bladder
Q62.5, Q64.5 - Q64.9	Other specified anomalies of bladder and urethra
Z85.46	Personal history of malignant neoplasm of prostate

Periurethral Injections of Bulking Agents:

CPT codes covered if selection criteria are met:

11950	Subcutaneous injection of filling material (e.g., collagen); 1 cc or less
11951	1.1 to 5.0 cc
11952	5.1 to 10.0 cc
11954	over 10.0 cc
51715	Endoscopic injection of implant material into the submucosal tissues of the urethra and/or bladder neck

HCPCS codes covered if selection criteria are met:

L8603	Injectable bulking agent, collagen implant, urinary tract, 2.5 ml syringe, includes shipping and necessary supplies
L8604	Injectable bulking agent, dextranomer/hyaluronic acid copolymer implant, urinary tract, 1 ml, includes shipping and necessary supplies
L8606	Injectable bulking agent, synthetic implant, urinary tract, 1 ml syringe, includes shipping and necessary supplies
Q3031	Collagen skin test

ICD-10 codes covered if selection criteria are met:

N36.42 - N36.43	Intrinsic (urethral) sphincter deficiency (ISD)
N39.3 - N39.9, R32	Urinary incontinence

ICD-10 codes not covered for indications listed in the CPB:

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Code	Code Description
N30.00 - N30.91	Cystitis
N31.9	Neuromuscular dysfunction of bladder, unspecified [Neurogenic bladder]
N34.0 - N34.2	Urethritis
N35.010 - N35.92	Urethral stricture
N39.0	Urinary tract infection, site not specified
Z92.3	Personal history of irradiation

***InterStim Continence Control Therapy/Sacral Nerve Stimulation [not covered for bilateral sacral nerve stimulation for urinary incontinence]*:**

CPT codes covered if selection criteria are met:

64561	Percutaneous implantation of neurostimulator electrode array; sacral nerve (transforaminal placement) including image guidance, if performed
64581	Incision for implantation of neurostimulator electrode array; sacral nerve (transforaminal placement)
64585	Revision or removal of peripheral neurostimulator electrode array
64590	Insertion or replacement of peripheral or gastric neurostimulator pulse generator or receiver, direct or inductive coupling
64595	Revision or removal of peripheral or gastric neurostimulator pulse generator or receiver
95970	Electronic analysis of implanted neurostimulator pulse generator system (e.g., rate, pulse amplitude and duration, configuration of wave form, battery status, electrode selectability, output modulation, cycling, impedance and patient compliance measurements); simple or complex brain, spinal cord, or peripheral (i.e., cranial nerve, peripheral nerve, sacral nerve, neuromuscular) neurostimulator pulse generator/transmitter, without reprogramming
95971	simple spinal cord, or peripheral (i.e., peripheral nerve, sacral nerve, neuromuscular) neurostimulator pulse generator/transmitter, with intraoperative or subsequent programming
95972	complex spinal cord, or peripheral (ie, peripheral nerve, sacral nerve, neuromuscular) (except cranial nerve) neurostimulator pulse generator/transmitter, with intraoperative or subsequent programming

HCPCS codes covered if selection criteria are met:

A4290	Sacral nerve stimulation test lead, each
C1767	Generator, neurostimulator (implantable), non-rechargeable
C1778	Lead, neurostimulator (implantable)
C1787	Patient programmer, neurostimulator
C1816	Receiver and/or transmitter, neurostimulator (implantable)
C1820	Generator, neurostimulator (implantable), with rechargeable battery and charging system
C1883	Adaptor/extension, pacing lead or neurostimulator lead (implantable)
C1897	Lead, neurostimulator test kit (implantable)
E0745	Neuromuscular stimulator, electronic shock unit
L8679	Implantable neurostimulator, pulse generator, any type
L8680	Implantable neurostimulator electrode, each

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Code	Code Description
L8681	Patient programmer (external) for use with implantable programmable neurostimulator pulse generator, replacement only
L8682	Implantable neurostimulator radiofrequency receiver
L8683	Radiofrequency transmitter (external) for use with implantable neurostimulator radiofrequency receiver
L8684	Radiofrequency transmitter (external) for use with implantable sacral root neurostimulator receiver for bowel and bladder management, replacement
L8685	Implantable neurostimulator pulse generator, single array, rechargeable, includes extension
L8686	Implantable neurostimulator pulse generator, single array, non-rechargeable, includes extension
L8687	Implantable neurostimulator pulse generator, dual array, rechargeable, includes extension
L8688	Implantable neurostimulator pulse generator, dual array, non-rechargeable, includes extension
L8689	External recharging system for battery (internal) for use with implantable neurostimulator, replacement only
L8695	External recharging system for battery (external) for use with implantable neurostimulator, replacement only

ICD-10 codes covered if selection criteria are met:

N39.41	Urge incontinence
R33.8 - R33.9	Other and unspecified retention of urine
R35.0	Frequency of micturition
R39.14	Feeling of incomplete bladder emptying
R39.15	Urgency of urination

ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):

E11.40 – E11.49	Type 2 diabetes mellitus with neurological complications
E75.21 - E75.29 E75.4, E75.6	Disorders of sphingolipid metabolism and other lipid storage disorders
G10 - G32.89	Systemic atrophies primarily affecting the central nervous system
G35.A - G47.9	Demyelinating diseases of CNS and episodic and paroxysmal disorders
G50.0 - G59	Disorders of the peripheral nervous system
G90.01 - G91.9	Disorders of autonomic nervous system
N13.9	Urinary obstruction
N31.0 - N31.1, N31.9	Neurogenic bladder
N32.0	Bladder neck obstruction
N35.010 - N35.92	Urethral stricture
N39.3	Stress incontinence, (female) (male)
N39.46	Mixed incontinence (female) (male)

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Code	Code Description
N40.0	Enlarged prostate without lower urinary tract symptoms

***Vaginal Cones (no specific codes)*:**

Other HCPCS codes related to the CPB:

A4335	Incontinence supply; miscellaneous

***Pessary (Bladder Neck Support Prosthesis)*:**

CPT codes covered if selection criteria are met:

57160	Fitting and insertion of pessary or other intravaginal support device

HCPCS codes covered if selection criteria are met:

A4561	Pessary, rubber, any type
A4562	Pessary, non-rubber, any type
A4564	Pessary, disposable, any type

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence
N39.46	Mixed incontinence (female) (male)
N81.0 - N81.9	Female genital prolapse

***Tension-Free Vaginal Tape Procedures (no specific codes)*:**

Other CPT codes related to the CPB:

51992	Laparoscopy, surgical; sling operation for stress incontinence (e.g., fascia or synthetic)
57288	Sling operation for stress incontinence (e.g., fascia or synthetic)

Other HCPCS codes related to the CPB:

C1771	Repair device, urinary, incontinence, with sling graft
C2631	Repair device, urinary, incontinence, without sling graft

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence
N39.46	Mixed incontinence (female) (male)

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Code	Code Description

Colposuspension and Sling Procedures **[Not covered for adjustable retropubic subureathral sling]:**

CPT codes covered if selection criteria are met:

51990	Laparoscopy, surgical; urethral suspension for stress incontinence
51992	sling operation for stress incontinence (e.g., fascia or synthetic)
53440	Sling operation for correction of male urinary incontinence (e.g., fascia or synthetic)
53442	Removal or revision of sling for male urinary incontinence (e.g., fascia or synthetic)
57287	Removal or revision of sling for stress incontinence (e.g., fascia or synthetic)
57288	Sling operation for stress incontinence (e.g., fascia or synthetic)

HCPCS codes covered if selection criteria are met:

C1771	Repair device, urinary, incontinence, with sling graft
C2631	Repair device, urinary, incontinence, without sling graft

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence
N39.46	Mixed incontinence (female) (male)

***Biofeedback*:**

CPT codes covered if selection criteria are met:

90912	Biofeedback training, perineal muscles, anorectal or urethral sphincter, including EMG and/or manometry, when performed; initial 15 minutes of one-on-one physician or other qualified health care professional contact with the patient
90913	Biofeedback training, perineal muscles, anorectal or urethral sphincter, including EMG and/or manometry, when performed; each additional 15 minutes of one-on-one physician or other qualified health care professional contact with the patient (List separately in addition to code for primary procedure)

HCPCS codes covered if selection criteria are met:

E0746	Electromyography (EMG), biofeedback device

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence
N39.41 - N39.498, R32	Urinary incontinence

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Code	Code Description

***Percutaneous Tibial Nerve Stimulation*:**

CPT codes covered if selection criteria are met:

64566	Posterior tibial neurostimulation, percutaneous needle electrode, single treatment, includes programming

HCPCS codes covered if selection criteria are met:

A4545	Supplies and accessories for external tibial nerve stimulator (e.g., socks, gel pads, electrodes, etc.), needed for one month
E0745	Neuromuscular stimulator, electronic shock unit

ICD-10 codes covered if selection criteria are met:

N32.81	Overactive bladder [idiopathic]
N39.41	Urge incontinence
R35.0	Frequency of micturition

***Transurethral Radiofrequency Therapy (Renessa Procedure), Micro-ablative radiofrequency*:**

CPT codes covered if selection criteria are met:

53860	Transurethral, radiofrequency micro-remodeling of the female bladder neck and proximal urethra for stress urinary incontinence[Not covered for Micro-ablative radiofrequency for the treatment of over-active bladder]

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence

ICD-10 codes not covered for indications listed in the CPB::

N32.81	Overactive bladder

***Urethral inserts*:**

HCPCS codes covered if selection criteria are met:

A4336	Incontinence supply, urethral insert, any type, each

ICD-10 codes covered if selection criteria are met:

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Code	Code Description
N39.3 - N39.9	Urinary incontinence

***Cunningham Clamp*:**

HCPCS codes covered if selection criteria are met:

A4356	External urethral clamp or compression device (not to be used for catheter clamp), each [Cunningham Clamp]

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9, R32	Urinary incontinence [post-prostatectomy urinary incontinence]

***Macroplastique (polydimethysiolxane)*-no specific code:**

HCPCS codes covered if selection criteria are met:

L8606	Injectable bulking agent, synthetic implant, urinary tract, 1 ml syringe, includes shipping and necessary supplies

***Neocontrol System*-no specific code:**

***Radiofrequency Micro-Remodeling with the SURs System (paraurethral or transvaginal)* - no specific code:**

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9. R32	Urinary incontinence

***Extraurethral (Non-circumferential) Retropubic Adjustable Compression Devices (ProACT Therapy System)*:**

CPT codes not covered for indications listed in the CPB:

53451	Periurethral transperineal adjustable balloon continence device; bilateral insertion, including cystourethroscopy and imaging guidance
53452	unilateral insertion, including cystourethroscopy and imaging guidance
53453	removal, each balloon
53454	percutaneous adjustment of balloon(s) fluid volume

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Code	Code Description

HCPCS codes not covered for indications listed in the CPB:

A4356	External urethral clamp or compression device (not to be used for catheter clamp), each
A4360	Disposable external urethral clamp or compression device, with pad and/or pouch, each

Subfascial tibial nerve stimulation:

CPT codes not covered for indications listed in the CPB:

0817T	Open insertion or replacement of integrated neurostimulation system for bladder dysfunction including electrode(s) (eg, array or leadless), and pulse generator or receiver, including analysis, programming, and imaging guidance, when performed, posterior tibial nerve; subfascial [BlueWind Revi]
0819T	Revision or removal of integrated neurostimulation system for bladder dysfunction, including analysis, programming, and imaging, when performed, posterior tibial nerve; subfascial [BlueWind Revi]

ICD-10 codes not covered for indications listed in the CPB:

N32.81	Overactive bladder
N39.41	Urge incontinence

Laser Therapy :

CPT codes not covered for indications listed in the CPB:

Genityte Procedure and FemiLift - no specific code:

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9, R32	Urinary incontinence

Protect PNS system:

CPT codes not covered for indications listed in the CPB:

0587T	Percutaneous implantation or replacement of integrated single device neurostimulation system for bladder dysfunction including electrode array and

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Code	Code Description
	receiver or pulse generator, including analysis, programming, and imaging guidance when performed, posterior tibial nerve
0588T	Revision or removal of percutaneously placed integrated single device neurostimulation system for bladder dysfunction including electrode array and receiver or pulse generator, including analysis, programming, and imaging guidance when performed, posterior tibial nerve
0589T	Electronic analysis with simple programming of implanted integrated neurostimulation system for bladder dysfunction (eg, electrode array and receiver), including contact group(s), amplitude, pulse width, frequency (Hz), on/off cycling, burst, dose lockout, patient-selectable parameters, responsive neurostimulation, detection algorithms, closed-loop parameters, and passive parameters, when performed by physician or other qualified health care professional, posterior tibial nerve, 1-3 parameters
0590T	Electronic analysis with complex programming of implanted integrated neurostimulation system for bladder dysfunction (eg, electrode array and receiver), including contact group(s), amplitude, pulse width, frequency (Hz), on/off cycling, burst, dose lockout, patient-selectable parameters, responsive neurostimulation, detection algorithms, closed-loop parameters, and passive parameters, when performed by physician or other qualified health care professional, posterior tibial nerve, 4 or more parameters

ICD-10 codes not covered for indications listed in the CPB:

N32.81	Overactive bladder
R35.0	Frequency of micturition
R39.15	Urgency of urination
R39.41	Urge incontinence

Pudendal nerve stimulation:

CPT codes not covered for indications listed in the CPB:

97014	Application of a modality to 1 or more areas; electrical stimulation (unattended) [Not covered for high-intensity focused electromagnetic therapy for the treatment of stress urinary incontinence]
97032	Application of a modality to 1 or more areas; electrical stimulation (manual), each 15 minutes [Not covered for high-intensity focused electromagnetic therapy for the treatment of stress urinary incontinence]

HCPCS codes not covered for indications listed in the CPB:

E0740	Incontinence treatment system, pelvic floor stimulator, monitor, sensor and/or trainer
S9002	Intra-vaginal motion sensor system, provides biofeedback for pelvic floor muscle rehabilitation device

ICD-10 codes not covered for indications listed in the CPB:

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Code	Code Description
N39.3 - N39.9, R32	Urinary incontinence

***Autologous Myoblast Transplantation*:**

***Autologous muscle-derived cell therapy*-No specific code:**

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9, R32	Urinary incontinence

***Collagen Porcine Dermis mesh*- no specific code:**

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9, R32	Urinary incontinence

Electro-acupuncture :

CPT codes not covered for indications listed in the CPB:

97813	Acupuncture, 1 or more needles; with electrical stimulation, initial 15 minutes of personal one-on-one contact with the patient
97814	Acupuncture, 1 or more needles; with electrical stimulation, each additional 15 minutes of personal one-on-one contact with the patient, with insertion of needle(s) (List separately in addition to code for primary procedure)

ICD-10 codes not covered for indications listed in the CPB:

N31.0 - N31.1, N31.9	Neurogenic bladder

Flyte System (mechanotherapy):

HCPCS codes not covered for indications listed in the CPB:

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Code	Code Description
E0715	Intravaginal device intended to strengthen pelvic floor muscles during kegel exercises
E0716	Supplies and accessories for intravaginal device intended to strengthen pelvic floor muscles during kegel exercises

ICD-10 codes not covered for indications listed in the CPB:

N39.3	Stress incontinence (female) (male)

Stem Cell Therapy:

CPT codes not covered for indications listed in the CPB:

38241	Hematopoietic progenitor cell (HPC); autologous transplantation

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9, R32	Urinary incontinence

Transobturator Tape-no specific code:

Other CPT codes related to the CPB:

51992	Laparoscopy, surgical; sling operation for stress incontinence (e.g., fascia or synthetic)
53440	Sling operation for correction of male urinary incontinence (eg, fascia or synthetic)
57288	Sling operation for stress incontinence (e.g., fascia or synthetic)

Other HCPCS codes related to the CPB:

C1771	Repair device, urinary, incontinence, with sling graft
C2631	Repair device, urinary, incontinence, without sling graft

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9	Urinary incontinence [intractable and has failed behavioral and pharmacologic treatments]

ICD-10 codes not covered for indications listed in the CPB:

N39.46	Urge incontinence

Pelvic Floor Stimulation:

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Code	Code Description

CPT codes covered if selection criteria are met:

97014	Application of a modality to 1 or more areas; electrical stimulation (unattended)
97032	Application of a modality to 1 or more areas; electrical stimulation (manual), each 15 minutes

HCPCS codes covered if selection criteria are met:

E0740	Incontinence treatment system, pelvic floor stimulator, monitor, sensor and/or trainer
G0238	Electrical stimulation (unattended), to one or more areas for indication(s) other than wound care, as part of a therapy plan of care

ICD-10 codes covered if selection criteria are met:

N39.3 - N39.9, R32	Urinary incontinence (female) (male)

Bariatric Surgery:

CPT codes not covered for indications listed in the CPB:

43644 – 43645, 43770 – 43775, 43842 – 43848, 43886 – 43888	Bariatric surgery

ICD-10 codes not covered for indications listed in the CPB:

N39.3 - N39.9, R32	Urinary incontinence

Adjustable Transobturator Male System:

CPT codes not covered for indications listed in the CPB:

Adjustable Transobturator Male System **- no specific code**

Magnetic Stimulation:

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Code	Code Description

CPT codes not covered for indications listed in the CPB:

***Magnetic Stimulation* - no specific code**

Moxibustion:

CPT codes not covered for indications listed in the CPB:

***Moxibustion* - no specific code**

ICD-10 codes not covered for indications listed in the CPB:

N39.3	Stress incontinence (female) (male)
N39.498	Other specified urinary incontinence [post-stroke urinary incontinence]

Platelet-rich plasma:

CPT codes not covered for indications listed in the CPB:

0232T	Injection(s), platelet rich plasma, any site, including image guidance, harvesting and preparation when performed

HCPCS codes not covered for indications listed in the CPB:

P9020	Platelet rich plasma, each unit

ICD-10 codes not covered for indications listed in the CPB:

N39.3	Stress incontinence (female) (male)

Pelvic Muscle Trainers:

No specific code

HCPCS codes covered for indications listed in the CPB:

E0740	Incontinence treatment system, pelvic floor stimulator, monitor, sensor and/or trainer [not covered for Athena pelvic muscle trainer]

Other HCPCS codes related to the CPB:

A4335	Incontinence supply; miscellaneous

Subcutaneous and subfascial tibial nerve stimulation:

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Code	Code Description

CPT codes not covered for indications listed in the CPB:

0988T	Open insertion or replacement of integrated neurostimulation system for bladder dysfunction including electrode(s) (eg, array or leadless), and pulse generator or receiver, including analysis, programming, and imaging guidance, when performed, posterior tibial nerve; subcutaneous and subfascial [INTIBIA tibial nerve stimulation device]

Subcutaneous Tibial Nerve Stimulation:

CPT codes not covered for indications listed in the CPB:

0816T	Open insertion or replacement of integrated neurostimulation system for bladder dysfunction including electrode(s) (eg, array or leadless), and pulse generator or receiver, including analysis, programming, and imaging guidance, when performed, posterior tibial nerve; subcutaneous [eCoin Peripheral Neurostimulator System] [Altaviva Implantable Tibial Neuromodulation device]
0818T	Revision or removal of integrated neurostimulation system for bladder dysfunction, including analysis, programming, and imaging, when performed, posterior tibial nerve; subcutaneous [eCoin Peripheral Neurostimulator System][Altaviva Implantable Tibial Neuromodulation device]

Transcutaneous electrical nerve stimulation:

HCPCS codes not covered for indications listed in the CPB:

E0720	Transcutaneous electrical nerve stimulation (TENS) device, 2 lead, localized stimulation
E0730	Transcutaneous electrical nerve stimulation (TENS) device, 4 or more leads, for multiple nerve stimulation

ICD-10 codes not covered for indications listed in the CPB:

N32.81	Overactive bladder

Transcutaneous tibial nerve stimulation:

HCPCS codes not covered for indications listed in the CPB:

A4545	Supplies and accessories for external tibial nerve stimulator (e.g., socks, gel pads, electrodes, etc.), needed for one month
E0736	Transcutaneous tibial nerve stimulator [ZIDA Wearable Neuromodulation System]
E0737	Transcutaneous tibial nerve stimulator, controlled by phone application [Vivally System]

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Code	Code Description

ICD-10 codes not covered for indications listed in the CPB:

N32.81	Overactive bladder
N39.41	Urge incontinence
R35.0	Frequency of micturition
R39.15	Urgency of urination

Background

Urinary incontinence (UI) is the inability to voluntarily control voiding of urine from the bladder. It affects people of all ages especially elderly women. Urinary incontinence is not part of the normal aging process; however, age-related changes in the functioning of the lower urinary tract make the elderly more susceptible to this malady. There are 4 prevalent types of UI in adults:

I. stress incontinence,
II. urge incontinence,
III. overflow incontinence, and
IV. mixed stress and urge incontinence.

I. stress incontinence,
II. urge incontinence,
III. overflow incontinence, and
IV. mixed stress and urge incontinence.

Stress incontinence is more common but less difficult to control than urge incontinence. Mixed incontinence is more prevalent than urge incontinence in women while the latter is more commonly seen in men. In women, stress incontinence (SI) is generally caused by an incompetent urethral mechanism which arises from damage to the urethral sphincter or weakening of the bladder neck support that typically occurred during childbirth. Some women develop SI as a consequence of multiple anti-incontinence procedures resulting in a condition known as intrinsic urethral sphincter deficiency. In men, SI is usually a consequence of operations for benign prostatic hypertrophy or prostatic carcinoma. The mechanisms of post-prostatectomy UI may involve bladder dysfunction, sphincter incompetence, and mixed. Urge incontinence occurs when one senses the urge to void, but is unable to prevent leakage of urine before reaching the bathroom. It is usually associated with an overactivity of the detrusor muscle. Overflow incontinence is the result of the bladder's inability to empty normally. It may be due to an underactive detrusor muscle or obstruction of the urethra resulting in the overdistension of the bladder and therefore overflow of urine. Multi-channel urodynamics studies are not indicated in the first-line assessment of patients with urinary incontinence/voiding dysfunctions. Guidelines from the American Urological Association (2012), the European Association of Urology (2013) and the National Institute for Health and Care Excellence (2013) are useful in determining when multi-channel urodynamics studies should be performed.

American Urological Association guidelines on adult (Winters et al, 2012) provided the following recommendations:

- Stress Urinary Incontinence (SUI) / Prolapse
  - Clinicians may perform multi-channel urodynamics in patients with both symptoms and physical findings of stress incontinence who are considering invasive, potentially morbid or irreversible treatments. (Option; Evidence Strength: Grade C)

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    - Clinicians should perform stress testing with reduction of the prolapse in women with high grade pelvic organ prolapse (POP) but without the symptom of SUI. Multi-channel urodynamics with prolapse reduction may be used to assess for occult stress incontinence and detrusor dysfunction in these women with associated lower urinary tract symptoms (LUTS). (Option; Evidence Strength: Grade C)

- Overactive Bladder (OAB), Urgency Urinary Incontinence (UUI), Mixed Incontinence

  Clinicians may perform multi-channel filling cystometry when it is important to determine if altered compliance, detrusor overactivity (DO) or other urodynamic abnormalities are present (or not) in patients with urgency incontinence in whom invasive, potentially morbid or irreversible treatments are considered. (Option; Evidence Strength: Grade C)

- LUTS (Lower Urinary Tract Symptoms)

  Clinicians may perform multi-channel filling cystometry when it is important to determine if DO or other abnormalities of bladder filling/urine storage are present in patients with LUTS, particularly when invasive, potentially morbid or irreversible treatments are considered. (Expert Opinion)

European Association of Urology guidelines on urinary incontinence (EAU, 2013) stated that "'urodynamics is generally used as a collective term for all tests of bladder and urethral function. These guidelines will review both non-invasive estimation of urine flow, i.e., uroflowmetry, and invasive tests, including multichannel cystometry, ambulatory monitoring and video-urodynamics, and different tests of urethral function, such as urethral pressure profilometry, Valsalva leak point pressure estimation and retrograde urethral resistance measurement. Multichannel cystometry, ambulatory monitoring and video-urodynamics aim to observe the effects on intra-vesical and intra-abdominal pressures while reproducing a patient's symptoms. Bladder filling may be artificial or physiological and voiding is prompted. Any incontinence observed may be categorized as SUI, detrusor overactivity (DO) incontinence, a mixture of SUI/DO incontinence, or, rarely, urethral relaxation incontinence. A test may fail to reproduce a patient's symptoms because of poor diagnostic accuracy or because the symptoms are not directly attributable to a urodynamically measurable phenomenon. Urodynamic testing is widely used as an adjunct to clinical diagnosis, to direct decisions about treatment and to provide prognostic information. When clinical diagnosis is difficult because of an unclear history or inconclusive examination, urodynamics may provide the only 'diagnosis' available. Although it is unlikely that carrying out a test, in itself, would alter the outcome of treatment, it remains possible that the test results would influence treatment decisions to such an extent that better outcomes would be achieved. This has been the rationale for using urodynamics prior to surgery."

National Institute for Health and Care Excellence guideline *Urinary Incontinence: The Management of Urinary Incontinence in Women* (NICE, 2013) provided the following recommendations regarding urodynamic testing:

- Do not perform multi-channel cystometry, ambulatory urodynamics, or video-urodynamics before starting conservative management. [2006, amended 2013]
- After undertaking a detailed clinical history and examination, perform multi-channel filling and voiding cystometry before surgery in women who have:

    - Symptoms of over-active bladder leading to a clinical suspicion of detrusor over-activity, or
    - Symptoms suggestive of voiding dysfunction or anterior compartment prolapse, or
    - Had previous surgery for stress incontinence [2006, amended 2013]

- Do not perform multi-channel filling and voiding cystometry in the small group of women where pure SUI is diagnosed based on a detailed clinical history and examination. [2006, amended 2013]
- Consider ambulatory urodynamics or videourodynamics if the diagnosis is unclear after conventional urodynamics. [2006, amended 2013]

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Treatments for UI include pelvic muscle exercises (Kegel exercise), behavioral therapies such as bladder training and/or biofeedback, pharmacotherapies (e.g., anti-cholinergic agents, musculo-tropic relaxants, calcium channel blockers, tricyclic anti-depressants, or a combination of anti-cholinergic, anti-spasmodic medications and tricyclic anti-depressants), and a variety of surgical procedures including intra-urethral injection of collagen, and implantation of an artificial urinary sphincter. Specifically, urge incontinence is more effectively managed with peripherally acting receptor agonists or antagonists while stress incontinence is better controlled by pelvic muscle exercises, behavioral therapies, or corrective surgery.

Electrical stimulation has also been employed in the treatment of UI, especially in Europe. The mechanism of action of electrostimulation is still unclear, but it probably serves to provide a kind of muscular training similar to that of pelvic floor exercise. In this regard, Green and Laycock (1990) demonstrated that interferential currents produce increases in muscle activity as indicated by pressure probes at the peri-vaginal and abdominal areas. It is also conceivable that electrical stimulation may improve re-innervation of partially damaged pelvic floor muscles by enhancing the sprouting of sensory motor axons. Additionally, repeated stimulation of the pelvic floor musculature may also help to strengthen the supporting ligaments. Examples of electrical stimulation devices include the Innova and Minnova systems (Empi, Inc., St. Paul, MN). An assessment of nonsurgical treatments for urinary incontinence prepared for the Agency for Healthcare Research and Quality (AHRQ) (Shamliyan, et al., 2012) found that intravaginal electrical stimulation increased continence rates and improved stress urinary incontinence more often than sham stimulation. The AHRQ assessment stated that a high level of evidence suggests increased continence rates and improvement in UI with electrical stimulation. This conclusion was based upon nine studies that examined intravaginal electrical stimulation. The studies included women with predominant urgency UI, clinical or urodynamic stress UI, or urodynamic mixed UI. Electrical stimulation was described with different levels of detail and had variable stimulation parameters, depending on the UI type being treated, including the use of 4 Hz, 10 Hz, 20 Hz, or 50 Hz frequency for 4 weeks, 7 to 8 weeks, 12 weeks, or 15 weeks.

Peri-urethral injection of bulking agents has been shown to be safe and effective for the treatment of UI resulting from intrinsic sphincter deficiency. One of the first bulking agents available on the market is Contigen (Bard, Canada), which is a sterile, injectable gel composed of highly purified bovine collagen that has been minimally cross-linked with 0.0075 % glutaraldehyde. All patients are required to undergo a pre-treatment skin test. Patients who exhibit local hypersensitivity will not be considered for treatment. Intra-urethral injections of glutaraldehyde crossed-linked (GAX) collagen are performed under local or general anesthesia. In men, the procedure is usually carried out trans-urethrally. The urethra and bladder are monitored by means of a 21F cystoscope, and the bladder filled with fluid cystoscopically. A percutaneous 10F or 12F suprapubic catheter is inserted into the bladder and its location verified. This catheter drains the bladder during the implantation procedure. Aliquots of GAX collagen are injected in a circumferential manner around the sphincter. In patients in whom no sphincter could be seen or in whom no sphincter existed (after radical prostatectomy), GAX collagen is injected circumferentially around the bladder neck. Once the lumen is occluded with the water running through the cystoscope, the injections are terminated, and the cystoscope is not inserted past the area of injection again.

In women, intra-urethral injections of GAX collagen are performed transurethrally or periurethrally. In the latter approach, the proximal urethra and bladder neck are visualized under direct cystoscopy. A 22G spinal needle is advanced parallel to the urethra in the peri-urethral tissue up to the bladder neck, and its position is confirmed by cystoscopy. Aliquots of collagen are then injected cystoscopically into the peri-urethral tissues to occlude the urethra. If intraluminal extravasation occurs, the injection is stopped, and another injection site is chosen. At the end of the procedure performed under local anesthetics, the patient is asked to cough or strain while in the supine position and then in the upright position. If leakage still occurs, more collagen is injected. If no leakage occurs, and there is no urinary retention, the patient is discharged from the hospital. When the procedure is performed under general anesthesia and the bladder is filled, an 8F feeding tube is inserted to empty the bladder and then removed. All patients receive perioperative antibiotics. If patients remain incontinent after treatment, re-injections of collagen are performed. Satisfactory results are usually obtained within 3 treatment sessions.

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Angioli and colleagues (2008) stated that in recent years they used a lot of bulking agents including bovine collagen, Macroplastique (polydimethylsiloxane), calcium hydroxylapatite, ethylene vinyl alcohol copolymer, dextranomer in the treatment of urinary incontinence. Urethral injection have success in 40 % to 90 %. These investigators asserted that Macroplastique is the most effective and safe on the basis of literature data and of their experience. This surgical procedure, in fact, has good percentage of success in accurately selected patients. In the authors' experience, Macroplastique can also be used in oncological patients, in elderly women, in patients with important co-morbidity and with high surgical risk with good objective and subjective results.

In a prospective, randomized, controlled trial, Ter Meulen and associates (2009) evaluated the effectiveness of Macroplastique (MPQ) Implantation System (MIS) in women with urodynamic stress UI (SUI) and urethral hypermobility after an unsuccessful conservative treatment. These subjects had no prior incontinence surgery. A total of 24 women received MPQ; 21 controls underwent a pelvic floor muscle exercises home program. Follow-up was at 3 months and the MPQ group also at 12 months. At 3 months, pad usage decreased significantly more in the MPQ group than in the control group ($p = 0.015$). According to physician and patient self-assessment, respectively, 71 % and 63 % women in the MPQ group were considered cured or markedly improved. This was significantly higher compared to controls. There was a significant higher increase of Incontinence Quality-of-Life questionnaire score in the MPQ group compared to controls ($p = 0.017$). Improvements in MPQ group at 3 months are sustained to 12 months. Adverse events were mild and transient. The authors concluded that the Macroplastique Implantation System is an acceptable option for women with SUI and urethral hypermobility.

Plotti et al (2009) prospectively investigated the effectiveness and complications of Macroplastique transurethral implantation in cervical cancer patients affected by SUI after radical hysterectomy (RH). Patients affected by de novo SUI post type 3 RH were considered for eligibility in this study. Pre-operative and post-operative assessment included a standardized urogynecological history, urogynecological and neurological physical examination, evaluation of severity of SUI symptoms, a 3-day voiding diary, urine culture and urodynamic assessment. All patients underwent transurethral implantation using the MIS. Patient follow-up was performed 6 and 12 months after surgery. A total of 24 consecutive patients were enrolled. At the 12 month follow-up SUI cure rate was 42 % (10 of 24 patients), the improvement rate was 42 % (10 of 24) and the failure rate was 16 % (4 of 24). The overall success rate was 84 % (10 patients cured and 10 improved). No intra-operative or post-operative early complications were found. The 4 patients in whom treatment was not successful had pre-operative urethral hypermobility. Subjective patient perception of SUI symptom severity showed significant improvement (mean severity of urinary loss perception 6.6 +/- 1.8 versus 2.3 +/- 3.3, $p < 0.05$). The frequency of incontinence on the 3-day voiding diary was significantly reduced at the follow-up (14.5 +/- 5.8 versus 4.3 +/- 7.9 episodes per 3 days, $p < 0.05$). The authors concluded that bulking agents urethral injection could be a valid option having no surgical complications. This therapeutic strategy is able to treat SUI and improve well being of cervical cancer patients after radical surgery.

Ghoniem et al (2009) evaluated the effectiveness and safety of Macroplastique as minimally invasive endoscopic treatment for female SUI primarily due to intrinsic sphincter deficiency. A total of 247 females with intrinsic sphincter deficiency were randomized 1:1 and treated with a transurethral injection of Macroplastique or Contigen (served as the control). Repeat treatment was allowed after the 3-month follow-up. Effectiveness was determined 12 months after the last treatment using Stamey grade, pad weight and Urinary Incontinence Quality of Life Scale scores. Safety assessment was recorded throughout the study. After 12 patients were excluded from study, 122 patients received Macroplastique injection and 125 received Contigen injection. Mean patient age was 61 years and the average history of incontinence was 11.2 years. Of the patients 24 % had undergone prior incontinence surgery. At 12 months after treatment 61.5 % of patients who received Macroplastique and 48 % of controls had improved 1 Stamey grade. In the Macroplastique group the dry/cure rate was 36.9 % compared to 24.8 % in the control group ($p < 0.05$). In the Macroplastique and control groups the 1-hour pad weight decrease was 25.4 and 22.8 ml from baseline ($p = 0.64$), and the mean improvement in Urinary Incontinence Quality of Life Scale score was 28.7 and 26.4 ($p = 0.49$), respectively. The authors concluded that Macroplastique injection was statistically more effective than Contigen for SUI primarily due to intrinsic sphincter deficiency with a 12.1 % cure rate difference.

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Available evidence indicates that intra-urethral injection of bulking agents is safe and effective for the treatment of UI, especially in women, resulting from intrinsic sphincter deficiency. Appropriate candidates should have no improvement in incontinence with conservative measures. For collagen-based products, a pre-treatment skin test for collagen should be performed, and show no evidence of local hypersensitivity. Patients whose incontinence does not improve after 3 treatment sessions are considered treatment failures. Periurethral injections of bulking agents should be avoided in the following persons: previous pelvic radiation therapy (less likely to benefit); unstable or non-compliant bladder; patients with severe allergies manifested by a history of anaphylaxis, or history or presence of multiple severe allergies; patients with an acute condition involving cystitis, urethritis, or infection; patients undergoing or planning to undergo desensitization injections to meat products (for collagen products).

The tension-free vaginal tape (TVT) procedure is an established treatment for intractable stress UI in persons who have failed behavioral and pharmacological treatments. An earlier assessment conducted by the Society of Obstetricians and Gynecologists of Canada (2003) concluded that "[t]he TVT procedure is promising but currently under evaluation in trials that will establish its efficacy and safety."

Recent randomized trials and studies with long-term follow-up have indicated that TVT procedure is safe and effective for the treatment of stress UI. In a randomized controlled study (n = 72), Paraiso et al (2004) concluded that the TVT procedure results in greater objective and subjective cure rates for urodynamic SUI than does laparoscopic Burch colposuspension. This is in agreement with the results those by Valpas et al (2004) and Ward et al (2004). In a multi-center randomized controlled trial (n = 128), Valpas and associates reported that treatment with TVT results in higher objective and subjective cure rates at 1 year than treatment by means of laparoscopic mesh colposuspension. In another multi-center randomized controlled study (n = 344), Ward and colleagues concluded that the TVT procedure appears to be as effective as colposuspension for the treatment of urodynamic stress UI at 2 years.

Based on the results of a controlled trial with a 2-year follow-up (n = 50), Meschia and colleagues (2004) stated that TVT can be recommended for patients with prolapse and occult SUI. In a comparison study (n = 61), deTayrac and co-workers (2004) concluded that trans-obturator sub-urethral tape appears to be equally efficient as TVT for surgical treatment of SUI in women, with no reduction of bladder outlet obstruction at 1-year follow-up.

In a prospective observational, multi-center study (n = 90), Nilsson et al (2004) reported that the TVT procedure for treatment of female SUI is effective over a period of 7 years. This finding extends the observation of that by Abdel-Fattah and associates (2004) who concluded that the Pelvicol pubovaginal sling is a safe procedure in the surgical management of SUI with similar success rate and patient satisfaction rate to TVT up to 3 years of follow-up. An assessment by the National Institute for Clinical Excellence (NICE, 2003) concluded that "[t]he tension-free vaginal tape (TVT) procedure is recommended as one of a range of surgical options for women with uncomplicated urodynamic stress incontinence in whom conservative management has failed." The Ontario Health Technology Advisory Committee (2004) concluded that TVT be offered as one option to treat women who are affected by SUI severely enough to warrant a surgical treatment approach.

There is evidence that percutaneous tibial nerve stimulation (PTNS) (Urgent PC Neuromodulation System, Uroplasty, Inc., Minneapolis, MN) is an effective treatment for chronic non-neurogenic urinary voiding dysfunctions (e.g., overactive bladder/urge incontinence) in persons who have failed conservative treatments. In general, 3 to 12 treatments (once weekly) with PTNS are needed for symptom relief. If a patient fails to improve after 12 PTNS treatments, further treatments are unlikely to be effective.

Percutaneous tibial nerve stimulation is regarded as an intermediate therapy between pelvic muscle exercise and sacral nerve stimulation (e.g., InterStim). Treatments are usually administered in twelve 30-min sessions. Van der Pal et al (2006a) examined the relationship between QOL and voiding variables in patients with lower urinary tract dysfunction treated with PTNS (n = 30). These investigators concluded that PTNS is useful for treating refractory urge incontinence and should at least be considered as a therapeutic alternative before resorting to aggressive surgery. Patients must have a reduction of greater than or equal to 2 pads/day before their QOL improves, and this might be the best definition of successful therapy for patients with urge

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UI. De Gennaro and colleagues (2004) assessed pain tolerability and the preliminary results of PTNS in children with unresponsive lower urinary tract symptoms (n = 23). These researchers concluded that PTNS is safe, minimally painful and feasible in children. It seems helpful for treating refractive non-neurogenic lower urinary tract symptoms. This is in agreement with the findings of Hoebeke et al (2002) who reported that PTNS has a significant effect on voiding frequency, the uroflowmetry curve and bladder capacity in children with non-neurogenic bladder sphincter dysfunction. van Balken (2007) stated that PTNS is carried out in 12 weekly sessions of 30 mins each, through a percutaneously placed needle cephalad to the medial malleolus. Success can be obtained in about 2/3 of patients.

Guidelines from the American Urologic Association (Gormley et al, 2012) have concluded: "Clinicians may offer peripheral tibial nerve stimulation (PTNS) as third-line treatment in a carefully selected patient population. Option (evidence strength grade C; balance between benefits and risks/burdens uncertain)".

There is insufficient evidence to support the use of the Neocontrol system, which uses extracorporeal magnetic innervation (ExMI), for treatment of urinary incontinence. The clinical role of this technology as a conservative incontinence therapy has not been defined, and longer follow-up than that reported is required to determine the durability of treatment results. An assessment prepared for the California Technology Assessment Forum (CTAF, 2004) concluded that pelvic floor magnetic stimulation for UI does not meet CTAF's criteria. The assessment concluded that "There is insufficient evidence from randomized clinical trials to conclude that pelvic floor magnetic stimulation is as beneficial as these alternative therapies." Since the CTAF assessment was published, an additional randomized controlled clinical trial (Culligan et al, 2005) and an uncontrolled prospective study (Voorham - van der Zalm, 2006) found extracorporeal magnetic stimulation to be ineffective.

Radiofrequency (RF) energy has been used for various clinical applications. Characteristics of RF energy allow it to be used for precisely controlled thermal therapy directed at soft tissues so as to induce such changes as collagen deposition and tissue shrinkage. These soft tissue effects are currently being examined for the treatment of genuine SUI in women. Ross et al (2002) evaluated the effectiveness of RF electrothermal energy in the treatment of genuine SUI (n = 94). The authors concluded that RF bipolar electrothermal energy appears to be a safe an efficient means of treating mild to moderate genuine SUI. It resulted in shrinkage and elevation of paravaginal connective tissue, stabilizing the urethra and bladder neck, thereby restoring continence. The authors stated, however, that long-term follow-up is necessary.

Sotomayor and Bernal (2003) studied the safety and quality of life impact of transurethral RF energy tissue micro-remodeling of the proximal urethra and bladder outlet in patients with SUI. Forty-one patients with SUI were sequentially enrolled into 4 treatment groups and then underwent rapid outpatient treatment under conscious sedation using an investigational RF energy delivery device. At 6 months, 75 to 80 % of patients in all 4 groups have demonstrated an improvement in quality of life. Two groups demonstrated statistical significance in both mean quality of life improvement and incontinence frequency reduction at 6 months. However, it is unclear if treatment resulted in clinically significant improvements in these parameters. Furthermore, long-term effectiveness of this approach is still unavailable.

Sotomayor and Bernal (2005) published longer follow-up findings of their 2003 study. They reported that significant incontinence episode frequency reduction was demonstrated by 3 of 4 treatment groups. They also noted that RF micro-remodeling demonstrated 12-month safety, quality of life improvement, and incontinence episode frequency reduction. No one treatment group demonstrated clear superiority in efficacy outcomes. Moreover, the authors stated that this pilot study had a number of limitations and weakness, namely, the trial was uncontrolled, and there were few subjects in any one treatment group. Also, diagnosis and follow-up evaluation lacked urodynamic testing.

Lenihan and colleagues (2005) examined the feasibility, safety, and patient comfort associated with RF tissue micro-remodeling in women with SUI given oral and local anesthesia. A total of 16 women with SUI and hypermobility (based on history and physical examination) with no history of previous definitive incontinence therapy were enrolled in this study. The women had a mean age of 49.7 years (range of 30 to 76 years) and a mean duration of incontinence of 7.6 years (range of 1 to 30 years). The non-surgical RF micro-remodeling treatment, which was previously shown to be of significant benefit when administered

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under intravenous (IV) sedation in an out-patient surgical center setting, was successfully completed in all 16 women. Either the treating physician or the patient had the option to convert to IV sedation during the procedure if there was too much discomfort; however, this did not occur in any of the 16 patients. Thus, neither the treating physician nor any patient determined that conversion to IV conscious sedation was needed for treatment completion. The first 6 patients received an oral sedative and oral analgesic as well as a local peri-urethral anesthetic block with 10-ml of 2 % lidocaine. The final 10 patients (63 %) received only 1 oral sedative or analgesic and a total of 10-ml lidocaine local anesthetic. Two women who received the maximum oral regimen (both oral sedation and analgesics) experienced nausea and emesis when drinking immediately after treatment, and 1 of these women also experienced urinary retention, which resolved after 24 hours of catheterization. Immediately before discharge, subjects classified their pain on a scale from 0 ("no pain") to 10 ("terrible pain"). Mean score was 1.8, and 38 % of subjects selected "0". The authors concluded that this pilot trial demonstrated the feasibility, safety, and patient comfort associated with performing a novel new successful technique of non-surgical RF of the urethra for the treatment of SUI in an office-based setting using oral plus local anesthesia. It should be noted that this study was not designed to evaluate the effectiveness of RF micro-remodeling in the treatment of SUI.

Lenihan (2005) examined the effect of menopause and hormone replacement therapy (HRT) on incontinence quality of life (I-QOL) score improvement in women with moderate-to-severe SUI after transurethral RF tissue micro-remodeling. A total of 173 women with genuine SUI with bladder outlet hypermobility were enrolled. Subjects were randomly assigned to undergo either RF micro-remodeling (n = 110) or sham treatment (n = 63). Participants were analyzed by menopausal status and HRT use for 10-point or greater I-QOL score improvement (an increase associated with subjective and objective SUI improvement). Radiofrequency micro-remodeling resulted in 81 % of subjects achieving 10-point or greater I-QOL score improvement versus 49 % of sham subjects at 12 months (p = 0.04). Outcomes did not differ statistically when pre-menopausal (85 %), post-menopausal using HRT (70 %), and post-menopausal not using HRT (71 %) groups were compared. The authors concluded that menopausal status and HRT demonstrated no impact on the quality of life improvement experienced by women with moderate-to-severe SUI who underwent RF tissue micro-remodeling. They also stated that further studies in pre-menopausal and post-menopausal women with SUI that measure additional effectiveness outcomes after RF micro-remodeling may provide further information concerning the clinical impact of menopause and HRT on this collagen-based treatment modality.

Appell and co-workers (2006) performed a prospective, randomized, controlled trial to demonstrate the 12 months safety and effectiveness of transurethral RF collagen micro-remodeling in women with SUI. Women with SUI, bladder outlet hypermobility, and leak point pressure (LPP) greater than or equal to 60 cm H(2)O were randomized to RF micro-remodeling or "sham treatment". Adverse events (AEs) were recorded. Incidence of greater than or equal to 10-point I-QOL score improvement, a magnitude of improvement with a demonstrated responsiveness to patient satisfaction with treatment and to greater than or equal to 25 % reduction in both incontinence episode frequency and stress pad weight, served as a subjective outcome measurement. Change in mean LPP served as an objective outcome measurement. The 12-month RF micro-remodeling safety profile was statistically no different than that of sham treatment (a brief bladder catheterization). Seventy-four % of women with moderate-to-severe baseline SUI experienced greater than or equal to 10-point I-QOL score improvement at 12 months (p = 0.04). Women who underwent RF micro-remodeling demonstrated LPP elevation at 12 months, while sham-treated women demonstrated LPP reduction (p = 0.02). The authors concluded that transurethral RF micro-remodeling is a safe treatment for women with SUI. In women with moderate-to-severe SUI, this novel therapy resulted in statistically significant improvement in QOL of a magnitude associated with patient satisfaction with the treatment. Women who underwent RF micro-remodeling demonstrated a statistically significant elevation in mean LPP at 12 months. While this study found statistically significant improvement in frequency and severity of incontinence episodes, the criterion of greater than or equal to 25 % reduction in both incontinence episode frequency and stress pad weight seems to be a "low bar" to clear. While RF micro-remodeling demonstrated a statistically significant elevation in mean LPP at 12 months; its clinical relevance is unclear. It is also interesting that the authors concluded that "transurethral RF micro-remodeling is a safe treatment for women with SUI" (effectiveness was not addressed). Furthermore, this study appeared to have the same group of subjects as reported by Lenihan (2005) -- 110 women underwent RF micro-remodeling and 63 underwent virtually identical "sham treatment".

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In a retrospective study, Appell and associates (2007) evaluated long-term safety and effectiveness of RF collagen denaturation for SUI in 21 patients from a 12-month, randomized controlled trial utilizing 3-day diaries and the I-QOL survey. Significant increases in overall I-QOL scores 3 years or more post-treatment was the primary end point. Secondary end points were reductions in frequency and severity of incontinence episodes. After 3 years, mean overall I-QOL score improvement was 12.7 (+/- 26); 56 % of patients achieved 50 % or more reduction in frequency. No new AEs occurred. These results indicated that RF collagen denaturation is safe and provides durable effectiveness. This was a longer follow-up (3 years) study of the previous studies reported (Lenihan, 2005; Appell et al, 2006). The authors also noted that additional studies of RF collagen denaturation are ongoing, including a study to expand the indication of RF collagen denaturation in patients who experienced suboptimal responses to a surgical intervention.

Vianello et al (2007) reviewed recent literature on mini-invasive surgical technique for the treatment of female SUI. Surgical aspects, intra-operative and peri-operative complications and objective and subjective outcomes were analyzed and compared. Studies had to investigate at least 40 women with a minimum follow-up of 12 months. A total of 38 prospective studies were found: 27 of them were on mid-urethral slings; 8 assessed urethral injections; and 3 RF treatment. Fifteen studies were randomized. Follow-ups ranged from 12 to 60 months, except for sexual function which had a 6-month follow-up. Ten out of 38 studies assessed patients who did not refer pelvic organ prolapse or detrusor over-activity and had not undergone any previous anti-incontinence procedure. The authors concluded that mid-urethral slings showed good outcomes and are safe and brief to perform and have a relatively short learning curve. Urethral injections showed discouraging results, as they have poor outcomes and repetitive treatments are frequently necessary. Injections can be used in women with contra-indications to major surgical procedures, with intrinsic sphincter deficiency as the main cause of incontinence. Radiofrequency showed worse results than mid-urethral slings, but is a valuable choice in women who refuse more invasive procedures.

Appell (2008) stated that patients who received transurethral collagen denaturation by means of non-ablative RF energy applied through a transurethral probe have shown improvements in quality of life and in Valsalva leak point pressure. This procedure presents a beneficial non-surgical treatment option for women with SUI.

An assessment by the California Technology Assessment Forum (Karliner, 2008) on RF micro-remodeling for the treatment of female SUI stated that while RF micro-remodeling (Renesss) for SUI does not show as high success rates as the gold standard approaches (Burch and TVT), it does demonstrate a good safety profile and moderate improvement in objective urinary leakage and quality of life, particularly for women with moderate-to-severe SUI. It stated that RF micro-remodeling with the Renessa System meets its criteria for safety, effectiveness and improvement in health outcomes for the treatment of moderate-to-severe female SUI in non-pregnant women who are either not able or not willing to undergo surgery for their condition.

In a continuing, prospective, 36-month, open-label, single-arm clinical trial, Elser et al (2009) evaluated the effectiveness of non-surgical transurethral collagen denaturation (Renessa) in women with SUI caused by bladder outlet hypermobility. Twelve-month results from intent-to-treat (ITT) analysis were reported. Women with SUI secondary to bladder outlet hypermobility for 12 months or longer who failed earlier conservative treatment and had not received earlier surgical or bulking agent therapy were included in the study. Subjects were treated as out-patients and received an oral antibiotic and local peri-urethral anesthesia before undergoing Renessa therapy. Voiding diaries and in-office stress pad weight tests yield objective assessments. Subjective measures include the Incontinence Quality of Life (I-QOL), Urogenital Distress Inventory (UDI-6), and Patient Global Impression of Improvement (PGI-I) instruments. A total of 136 women received treatment (ITT population). Patients experienced significant reductions versus baseline in median number of leaks caused by activity/day and activity/week ($p < 0.0026$ for both), with 50 % of patients reporting 50 % or more reduction. Pad weight tests revealed that 69 % of women had 50 % or more reduction in leakage (median reduction 15.2 g; $p < 0.0001$); 45 % were dry (29 % no leaks; 16 % less than 1-g leakage). Significant improvements occurred in median scores on the I-QOL (+9.5 [range of -66.0 to 91.0]; $p < 0.0001$) and mean scores on the UDI-6 (-14.1 +/- 24.7; $p < 0.0001$). Furthermore, 71.2 % showed I-QOL score improvement, including 50.3 % with 10-point or greater improvement, and 49.6 % reported on the PGI-I that they were "a little," "much," or "very much" better. The authors concluded that at 12 months, treatment of SUI with non-surgical transurethral collagen denaturation resulted in significant improvements in activity-related leaks and quality of life.

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It is also interesting to note that transvaginal RF bladder neck suspension procedure for SUI has not been shown to provide satisfactory results. Buchsbaum and colleagues (2007) evaluated the outcome and patient acceptance of the transvaginal RF bladder neck suspension procedure. A retrospective chart review of 18 women treated with the transvaginal RF bladder neck suspension procedure for SUI was conducted. Data on demographics, urodynamics, daily leakage episodes, complications, patient satisfaction, and further intervention were collected. The mean number of leaks per day was 5.7. There were no complications. Post-operatively, 2 patients were continent, 4 were improved, and 10 were unimproved. The mean number of daily leaks was reduced to 2.7. Five patients reported to be extremely satisfied with the procedure; 1 patient was satisfied, and 10 were not satisfied. Seven patients sought additional treatment within 1 year. Low cure rate, low patient satisfaction, and high rate of additional treatment led these researchers to abandon transvaginal RF bladder neck suspension procedure as a treatment option.

Ismail (2008) evaluated the safety and effectiveness of transvaginal RF remodeling of the endopelvic fascia as a primary procedure for SUI due to urethral hypermobility in women. It included 24 patients who had the procedure at 2 district general hospitals. Outcome measures included the pad test, urodynamic assessment, continence diary, pain scores, as well as operative and post-operative complications and assessment was made on recruitment during hospital admission and at 3, 6 and 12 months follow-up. A rising failure rate was noted as early as 3 months, leading to a cumulative cure rate of 45.8 % at 12 months follow-up. This low effectiveness could be attributed to inherent weakness of the endopelvic fascia. No major complications were encountered and pain scores were mild. In this regard, a draft assessment by the California Technology Assessment Forum (2008) on RF micro-remodeling for the treatment of female SUI stated that RF micro-remodeling with the SURx System (paraurethral or transvaginal) does not meet its criteria for safety, effectiveness and improvement in health outcomes for the treatment of female SUI.

Polypropylene meshed tape may be placed at the mid urethra or bladder neck using retropubic or trans-obturator approaches. Various types of sub-urethral tapes inserted via the trans-obturator route (TVT obturator route [TVTO] and trans-obturator tape [TOT]) have been used for the treatment of SUI. In a systematic review, Latthe and co-workers (2007) evaluated the effectiveness and complications of TOTs as treatment of SUI. Randomized controlled trials (RCTs) that compared the effectiveness of TVTO or TOT with synthetic TVT by retropubic route for the treatment of SUI in all languages were included. Two reviewers extracted data on participants' characteristics, study quality, population, intervention, cure and adverse effects independently. There were 5 RCTs that compared TVTO with TVT and 6 RCTs that compared TOT with TVT. When compared by subjective cure, TVTO and TOT at 2 to 12 months were no better than TVT (odds ratio [OR] 0.85; 95 % confidence interval [CI]: 0.60 to 1.21). Adverse events such as bladder injuries (OR 0.12; 95 % CI: 0.05 to 0.33) and voiding difficulties (OR 0.55; 95 % CI: 0.31 to 0.98) were less common, whereas groin/thigh pain (OR 8.28; 95 % CI: 2.7 to 25.4), vaginal injuries or erosion of mesh (OR 1.96; 95 % CI: 0.87 to 4.39) were more common after tape insertion by the trans-obturator route. The authors concluded that the evidence for short-term superiority of effectiveness of TOTs is currently limited. Bladder injuries and voiding difficulties are lower, but the risk of vaginal erosions and groin pain is higher with TVTO/TOT. Methodologically sound and sufficiently powered RCTs with long-term follow-up are needed, and the results of continuing trials are awaited.

In a prospective, single-blinded, multi-center RCT, Barry et al (2008) compared the safety and effectiveness of the trans-obturator tape (Monarc) with the retropubic tape (tension-free vaginal tape, TVT) for the treatment of SUI. A total of 187 women with SUI were randomly allocated to undergo surgery with either the Monarc sling (n = 80) or TVT (n = 107). Outcome measures were intra-operative complications (especially bladder injury), peri-operative complications, symptomatology, quality of life as well as urodynamic outcomes. At 3 months, data were available on 140 women, 82 (59 %) TVT and 58 (42 %) Monarc. The TVT group was significantly more likely to be complicated by bladder injury (7 TVT, 0 Monarc, $p < 0.05$). Blood loss and operative time were significantly less in the Monarc group, which was 49 mls (31) versus that of the TVT group, which was 64 mls (41) $p < 0.05$; 18.5 mins (6.5) TVT versus 14.6 mins (6) Monarc ($p < 0.001$). The subjective and objective SUI cure rates were 86.6 % (71) versus 72.4 % (42) $p = 0.77$ and 79.3 versus 84.5 %, $p = 0.51$ for the TVT and Monarc groups, respectively. Both groups reported similar improvement in incontinence impact and satisfaction with their operation, although return to activity was significantly quicker with the trans-obturator route ($p = 0.029$). The authors concluded that the transobturator tape appears to be as effective as the retro-pubic tape in the short-term, with a reduction in the

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risk of intra-operative bladder injury, shorter operating time, decreased blood loss, and quicker return to usual activities.

Barber et al (2008) compared the safety and effectiveness of the trans-obturator tape to TVT in the treatment of SUI in patients with and without concurrent pelvic organ prolapse. A total of 170 women were randomized to receive TVT or trans-obturator tape. Subjects with detrusor over-activity or previous sling surgery were excluded. The primary outcome was the presence or absence of abnormal bladder function, a composite outcome defined as the presence of any the following: incontinence symptoms of any type, a positive cough stress test, or re-treatment for SUI or post-operative urinary retention assessed 1 year following surgery. This study was a non-inferiority study design. Of 180 women who enrolled in the study, 170 underwent surgery and 168 returned for follow-up, with a mean follow-up of 18.2 +/- 6 months. Mean operating time, length of stay, and post-operative pain scores were similar between the 2 groups. Bladder perforations occurred more frequently in the TVT group (7 % compared with 0 %, p = 0.02); otherwise, the incidence of peri-operative complications was similar. Abnormal bladder function occurred in 46.6 % of TVT patients and 42.7 % of trans-obturator tape patients, with a mean absolute difference of 3.9 % favoring trans-obturator tape (95 % CI: -11.0 % to 18.6 %.). The "p" value for the 1-sided non-inferiority test was 0.006, indicating that trans-obturator tape was not inferior to TVT. The authors concluded that the trans-obturator tape is not inferior to TVT for the treatment of SUI and results in fewer bladder perforations. Moreover, they also noted that larger studies are needed to assess the relative risk of the less common but potentially severe complications that have been seen with both procedures. Furthermore, studies with longer follow-up are necessary to ascertain if the effectiveness of trans-obturator tape is durable.

Koch and Zimmern (2008) evaluated the evidence base for surgical management of SUI in women. Pubovaginal sling has a higher success rate than the Burch at the expense of a higher morbidity. A prophylactic Burch procedure at the time of an abdominal sacrocolpopexy can reduce secondary SUI and urge incontinence. Sub-urethral tapes have a higher cure rate for patients with predominant SUI and can safely be placed at the time of concomitant pelvic surgery. The TVT has a higher rate of lower urinary tract injury and voiding dysfunction when compared with trans-obturator tape. The authors concluded that the Burch and pubovaginal sling have a high success rate for treating SUI; prospective RCTs are needed to evaluate the long-term results of sub-urethral slings. This is in agreement with the observation of Rogers (2008) who stated that the use of the trans-obturator tape (one of the many newer techniques) entails the placement of polypropylene mesh through the obturator foramen rather than through the retropubic space, but large, randomized trials with adequate follow-up comparing these newer anti-incontinence procedures are limited.

A systematic evidence review by Sung et al (2007) found that the trans-obturator approach was associated with a lower risk of complications than the retropubic approach to mid-urethral slings for the treatment of stress incontinence, but there was insufficient evidence to compare the effect of surgical approaches on objective and subjective outcomes.

Guidelines on choice of surgery for SUI from the Society of Obstetricians and Gynecologists of Canada (Robert et al, 2005) concluded that there is insufficient evidence to support the use of the TOT procedure for stress urinary incontinence. Guidelines on UI from the National Collaborating Centre for Women's and Children's Health concluded that the TOT procedure is recommended as alternative treatment option for SUI if conservative management has failed, "provided women are made aware of the lack of long-term outcome data." This was a "D" recommendation, based on consensus or low quality evidence. Earlier guidance on the TOT procedure from the National Institute for Health and Clinical Excellence (NICE, 2005) was withdrawn when NICE was made aware that one of the main studies that was considered in the overview of evidence on the safety and efficacy of this procedure had been withdrawn by the journal that published it.

Tahseen and Reid (2009) estimated changes in overactive bladder (OAB) symptoms and urge UI in patients undergoing the TOT procedure for SUI and mixed UI. Telephone interviews were conducted using the International Consultation on Incontinence-Female Lower Urinary Tract Symptoms questionnaire, the International Consultation on Incontinence questionnaire-Overactive Bladder (ICIQ-OAB), and the Verbal Analogue Satisfaction (VeAS) scale. Pre-operative OAB scores were compared with post-operative scores in women with SUI only (group 1), mixed UI with predominant stress leakage (group 2), and mixed UI with

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predominant urge (group 3). Case notes were reviewed for pre-operative assessment and complications. At median follow-up of 13 months, significant improvement was noted in ICIQ-OAB scores, from a median of 10 (1 to 15) pre-operatively to a median of 3 (0 to 11) post-operatively ($p < 0.001$). Overall, urge UI was cured in 19 of 44 (43 %) patients, improved in a further 16 (36 %), and was persistent in only 9 (21 %). In group 2 (SUI predominant), urge UI was cured in 10 of 23 (43.5 %) patients, improved in 10 (43.5 %), and persistent in 3 (13 %). In group 3 (urge UI predominant), urge UI was cured in 9 of 21 (43 %) patients, improved in 6 (28.5 %), and persistent in 6 (28.5 %). Post-operative lower urinary tract symptom scores were low in all 3 groups (median 4/48 [0 to 18]). Stress incontinence was cured in 77 %, improved in a further 19 %, and unchanged in 4 %. Median VeAS score was 9 (2 to 10); 21 % (11/52) of participants had low satisfaction scores (less than 8) owing to persistent urge and slow voiding. The authors concluded that marked resolution or improvement (79 %) in urge UI after the TOT procedure was noted, and no case of de novo urge UI was identified. Moreover, the authors noted that it is unclear how to predict who will benefit and remain free of urge following the surgery. Furthermore, they stated that larger outcome studies of TOT with longer follow-up are needed, ideally using standardized, validated assessment tools, focusing on the common problem of mixed UI, with clear reporting criteria, and assessment at baseline and after surgery.

On behalf of the Agency for Healthcare Research and Quality, the Vanderbilt Evidence-based Practice Center systematically reviewed evidence on treatment of OAB, urge UI, and related symptoms. These investigators focused on prevalence and incidence, treatment outcomes, comparisons of treatments, modifiers of outcomes, and costs. They included studies published in English from January 1966 to October 2008; and excluded studies with fewer than 50 subjects, fewer than 75 % women, or lack of relevance to OAB. Of 232 included publications, 20 were good quality, 145 were fair, and 67 poor. These researchers calculated weighted averages of outcome effects and conducted a mixed-effects meta-analysis to examine outcomes of pharmacotherapies across studies.

Overactive bladder affects more than 10 to 15 % of adult women, with 5 to 10 % experiencing urge UI monthly or more often. Six available medications are effective in short-term studies: estimates from meta-analysis models suggest extended release forms (taken once-daily) reduce urge UI by 1.78 (95 % CI: 1.61, 1.94) episodes per day, and voids by 2.24 (95 % CI: 2.03, 2.46) per day. Immediate release forms (taken twice-daily or more) reduce urge UI by 1.46 (95 % CI: 1.28, 1.64), and voids by 2.17 (95 % CI: 1.81, 2.54). As context, placebo reduces urge UI episodes by 1.08 (95 % CI: 0.86, 1.30), and voids by 1.48 (95 % CI: 1.19, 1.71) per day. No one drug was definitively superior to others, including comparison of newer more selective agents to older antimuscarinics. Procedural and surgical treatments, such as sacral nerve stimulation (neuromodulation), and bladder instillation of oxybutynin or injections of botulinum toxin, were found to treat symptoms in select groups of women though more information is needed to understand safety and effectiveness. Acupuncture was the sole complementary and alternative medicine treatment, among reflexology and hypnosis, with early evidence of benefit. The strength of the evidence is insufficient to fully inform choice of these treatments. Select behavioral interventions were associated with symptom improvements comparable to medications. Limited evidence suggests no clear benefit from adding behavioral interventions at the time of initiation of pharmacotherapy. The authors concluded that OAB and associated symptoms are common; treatment effects are modest. Quality of life and treatment satisfaction measures suggest such improvements can be important to women. The amount of high quality literature available is meager for helping guide women's choices. Gaps include weak or absent data about long-term follow-up, poorly characterized and potentially concerning harms, information about best choices to minimize side effects, and study of how combinations of approaches may best be used. This is problematic since the condition is chronic and a single treatment modality is unlikely to fully resolve symptoms for most women.

Sirls et al (2002) reported the long-term results of the FemSoft urethral insert for the management of female SUI. This 5-year controlled multi-center study enrolled 150 women. Outcome measures included pad weight tests (PWT), voiding diary (VD), quality of life (QOL) and satisfaction questionnaires. Outcome measures during the baseline period were compared to evaluations during follow-up. Concurrent evaluations with and without device use were also performed. Safety evaluations included urinalysis and culture, LPP and cystoscopy. Adverse events were recorded throughout the study. One to 2 years of follow-up were collected on all study participants (mean of 15 months). Statistically significant reductions in overall daily incontinence episodes ($p < 0.001$) and PWT urine loss ($p < 0.001$) were observed with the device at all

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follow-up intervals, and 93 % of women had a negative PWT at 12 months. Women were satisfied with ease of use of the device, comfort and dryness, and significant improvements in QOL were observed ($p < 0.001$). Sub-group analysis revealed that the insert was effective, despite the presence of urgency, low LPP, failed surgery and advanced age. Adverse events included symptomatic urinary tract infection in 31.3 %, mild trauma with insertion in 6.7 %, hematuria in 3.3 %, and migration in 1.3 % of women. The results of PWT and VD demonstrated device efficacy. Women were satisfied and significant improvements in QOL were observed; AE were transient and required minimal or no treatment. The authors concluded that the urethral insert should be considered as an option for the management of SUI.

Robinson et al (2003) evaluated the safety and efficacy of an urethral device (NEAT) and compared it with the Reliance Insert. The ease of use of both devices was then evaluated. A total of 24 women with mixed or SUI were enrolled in the study. Study subjects were blinded and randomly assigned to a device group. Device efficacy was assessed by pad weighing at 0 and 4 months. Success was defined as a 50 % or greater reduction in urine loss using the formula 100 [(pad weight without device - pad weight with device)/pad weight with device]. Safety was evaluated using urinalysis and urine cultures. Ease of use assessment scales were also completed. Eleven patients were randomized to the Reliance Insert and 13 to the NEAT device. There were no significant differences between the 2 groups in age, height, weight, duration of incontinence, pad weight, leakage score, parity or QOL score. Based on the pad weight success formula, there was no significant difference in device success between the 2 groups at 4 months. Women who were post-menopausal had a trend towards a higher level of success in reduction of their pad weight. Previous treatment, diagnosis and hormone replacement therapy all had no relationship to device success. Leakage score data showed that subjects had a significant decrease in urine leakage when using either device. There was no statistically significant difference in ease of use between the 2 devices. Adverse symptoms most commonly noted were awareness of the device (62.5 %), urgency (29.2 %), and urethral discomfort or pain (20.8 %). One urinary tract infection (UTI) was observed. The most common finding on urinalysis was trace hematuria (15.8 %). The authors concluded that the NEAT device appears to be at least as effective and safe as the Reliance Insert. Both devices are effective at decreasing urine leakage in patients with SUI or mixed UI. The risk of UTI is low, but these devices may cause trace hematuria.

The Genityte procedure is a novel approach for the treatment of SUI. It entails the use of laser that works in a similar fashion to skin tightening treatments. The treatment stimulates the skin's natural production of collagen making it more supple and elastic. Genityte works to regain bladder control by tightening the tissue around the urethra. The number of treatments needed to restore the function of a woman's urethra supposedly depends largely on how much collage is still present in her skin. The clinical value of the Genityte procedure needs to be validated by well-designed studies.

In a pilot study, Groen and colleagues (2005) evaluated the results of chronic pudendal nerve neuromodulation (CPNN) on women with idiopathic refractory detrusor over-activity incontinence. A percutaneous screening test (PST) was performed in patients with urodynamically demonstrated detrusor over-activity incontinence. Such a test includes the performance of a cystometrogram without and with percutaneous pudendal nerve stimulation and is considered positive if stimulation results in a more than 50 % increase in the bladder volume at the first involuntary detrusor contraction or the maximum cystometric capacity. Patients with a positive PST qualified for the implantation of a mini-neurostimulator with an integrated electrode, a so-called Bion(R), adjacent to the pudendal nerve at Alcock's Canal. Five-day voiding-incontinence diaries were the main tool for the evaluation of therapy. A PST was performed in 14 women; 6 patients responded positively and received a Bion(R). The degree of incontinence decreased significantly in this group, which also included patients in whom sacral neuromodulation had failed. There were no severe adverse events. The authors concluded that CPNN may reduce the degree of detrusor over-activity incontinence, even in patients in whom sacral neuromodulation fails.

Spinelli et al (2005) stated that pudendal nerve stimulation has beneficial effects on numerous pelvic floor function impairments such as urinary and/or fecal incontinence, retention, and constipation. In preceding literature the implant technique required a fairly complex and invasive surgery, although recent advances with percutaneous placement of the lead through an introducer have made the procedure much less invasive. These researchers performed staged procedure similar to that of sacral neuromodulation (SNM) to place tined lead near the pudendal nerve, using neurophysiological guidance that allowed accurate pudendal nerve

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stimulation through either perineal or posterior approach. They have named this approach CPNN. A total of 15 neurogenic patients (8 males, 7 females) with symptoms of urge UI due to neurogenic over-active bladder underwent CPNN. All patients had complete neurophysiological and urodynamic evaluation at baseline and follow-up and were asked to complete voiding and bowel diary for 7 days. During screening, average number of incontinent episodes per day decreased from 7 +/- 3.3 to 2.6 +/- 3.3 ($p < 0.02$, paired t-test). Eight patients became continent, 2 improved by more than 88 % (from 9 to 1 daily incontinence episode) and 2 patients reduced the number of incontinence episodes by 50 %. The implantable pulse generator (IPG) was subsequently implanted in those 12 patients. Three patients without improvement did not continue to second stage. In implanted patients with 6 months follow-up, urodynamic evaluation showed an objective improvement in the maximum cystometric capacity which increased from 153.3 +/- 49.9 to 331.4 +/- 110.7 ml ($p < 0.01$, paired t-test). The maximum pressure decreased from 66 +/- 24.3 to 36.8 +/- 35.9 cm H2O ($p = 0.059$, paired t-test). Eight patients reported significant improvement in bowel function. The authors concluded that CPNN is feasible. Neurophysiological guidance is mandatory to place the lead near the pudendal nerve either using perineal or posterior approach. They stated that further studies must be carried out to identify the best stimulation parameters and to verify the long-term results.

Seif and associates (2005) noted that sacral neuromodulation is known to be an alternative therapeutic option for patients with anti-cholinergic resistant overactive bladder (OAB). For the same indication, a micro-stimulation system called BION is available since last year. The BION-stimulator, which only measures 2.8 x 0.3 cm, is designed for pudendal nerve stimulation. Its implantation technique as well as the first clinical results were presented and discussed. During an out-patient PST, a pudendal nerve stimulation is performed with a needle electrode in local anesthesia. A 50 % increase in the urodynamic parameters (bladder capacity, first desire to void, compliance, etc.) is an indication for a chronic implantation of the BION stimulator, which also can be placed in local anesthesia. Two patients have been treated with a BION-stimulator in the authors' clinic so far. Patient 1 suffered from an OAB with frequent UI and patient 2 had a sensory OAB with high voiding frequency. After the BION(R)-implantation, patient 1 showed a reduction in incontinence episodes by 31.5 % a day and patient 2 had lowered voiding frequencies from 12.6 to 7 a day. The post-operative urodynamic investigations confirmed these clinical results. The authors concluded that the BION-system and CPNN seem to be alternatives to sacral neuromodulation, however, patient selection is difficult as subchronic stimulation for a longer period of time is not possible so far.

Madjar et al (2001) reviewed the evolution of appliances and devices used for treating post-prostatectomy UI. These investigators used the MEDLINE to search the literature from 1966 to March 2000 and then manually searched bibliographies to identify studies that their initial search may have missed. The evolution of treatment for post-prostatectomy UI may be traced back to the 18th century. Two main schools of thoughts simultaneously evolved. The first fixed urethral compression devices were constructed to enable urethral obstruction by fixed resistance. This outlet resistance allows voiding after intra-abdominal and intra-vesical pressure is elevated but it is sufficient to prevent leakage between urinations. The other school of thought preferred creation of dynamic urethral compression in which outlet resistance is not fixed but may be decreased when voiding is desired or elevated between urinations. Therapeutic fixed and dynamic urethral compression interventions may be further divided into external or internal compressive devices or procedures. External fixed compression devices may be traced back to antiquity. A penile clamp, similar to the later Cunningham clamp, and a truss designed to compress the urethra by external perineal compression were presented in the Heister textbook of surgery, Institutiones Chirurgicae, as early as 1750. Dynamic compressive devices applied externally were developed much later, such as the first artificial urinary sphincter, described by Foley, in 1947 and the Vincent apparatus, described in 1960. The modern era of fixed urethral compression began in 1961 with Berry. Acrylic prostheses impregnated with bismuth to allow radiographic visualization were produced in various shapes and sizes, and used to compress the urethra against the urogenital diaphragm. In 1968 the UCLA group under the direction of Kaufman began to use cavernous crural cross-over to compress the bulbous urethra (Kaufman I). Later, 2 other modifications were described, including approximation of the crura in the midline using a polytetrafluoroethylene mesh tape (Kaufman II) and an implantable silicone gel prosthesis (Kaufman III). With the advent of the artificial urinary sphincter pioneered by Scott in 1973 interest in passive urethral compression disappeared in favor of the implantation of an inflatable circumferential prosthetic sphincter. Recently, there has been a trend back to passive urethral compression. The authors concluded that much creativity has been dedicated to solve the complex and challenging problem of post-prostatectomy UI. Devices used for treating this condition may be

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grouped according to the mechanism of action and how they are applied. Passive urethral compression, long abandoned in favor of dynamic implantable sphincters, has re-emerged.

Moore et al (2004) evaluated the safety, effectiveness, comfort, and patient satisfaction with 3 penile compression devices: the Cunningham clamp, C3, and U-Tex. The devices were tested in random order in a multiple-period, cross-over study design using a Latin squares configuration. The subjects had undergone radical prostatectomy 6 months or more before the study, had no neurologic or cognitive impairment, and had not undergone radiotherapy. Baseline penile Doppler ultrasonography was followed by ultrasound scanning with each device. In random order, subjects completed a 4-hour pad test, with and without each device, and the questionnaire. A total of 12 men completed the study. The mean Mini-Mental State Examination score was 29.6 (SD 1.2, range of 27 to 30). The mean urine loss at baseline was 122.8 g (SD 130.8). The mean urine loss with each device was 53.3 g (SD 65.7) with the U-Tex, 32.3 g (SD 24.3) with C3, and 17.1 g (SD 21.3) with the Cunningham clamp ($p < 0.05$). No device had an impact on the resistive index; the C3 and U-Tex allowed good cavernosal artery flow, and the Cunningham clamp significantly lowered the distal blood flow velocity (from 12.5 to 7.3 cm/s [left systolic velocity] to 9.5 cm/s [right systolic velocity]) even at the loosest setting. The Cunningham clamp was ranked positively by 10 of 12 men; 2 of 12 men rated the C3 positively; none rated the U-Tex positively. The authors concluded that the Cunningham device was the most effective and most acceptable to users, but also contributed to reduced systolic velocity in all men. None of the devices completely eliminated urine loss when applied at a comfortable pressure. Individualized instruction to cognitively capable men is necessary to ensure appropriate application, comfort, and fit.

An UpToDate review on "Urinary incontinence in men" (Clemens, 2012) states that "[a]djunctive measures include incontinence pads, indwelling catheters, external urinary catheters, and penile incontinence clamps. The treatment of urinary incontinence with an indwelling catheter is usually a poor management choice, as it is associated with urethral trauma, infection, and nephrolithiasis. Incontinence pads and indwelling catheterization are discussed elsewhere. In men, external urinary catheters (condom catheters) can be useful in managing urinary incontinence, with less associated morbidity compared to indwelling catheterization. Successful use of an external catheter requires adherence of the condom sheath to the penis. Use of external catheters may not be possible in some patients who are not able to keep catheters in place (e.g., due to skin infections) or not physically able to place catheters (e.g., obesity, neurologic impairment). In patients with neurogenic bladder dysfunction, the use of an external catheter may be associated with progressive renal damage unless it is confirmed with urodynamics that bladder storage pressures remain low .... Another option is the use of a penile incontinence clamp. A clamp is most suitable in ambulatory men with stress incontinence and good bladder storage function. Clamps are meant to be used on an intermittent basis. Their use in men with sensory abnormalities should be avoided, as tissue damage from the clamp can occur with prolonged use".

The Athena pelvic muscle trainer is an electronic device designed to strengthen pelvic muscle in women. This would appear to be similar to Kegelmaster. Per CPB 223, Aetna does not cover the Kegelmaster, Gyneflex or similar devices for the treatment of UI because these devices are considered exercise machines, and they do not meet Aetna's definition of covered durable medical equipment (DME). Furthermore, there is a clinical trial on the effectiveness of the Athena pelvic muscle trainer device in the treatment of stress, urge or mixed incontinence in women. http://clinicaltrials.gov/ct2/show/NCT01073878.

Elmi et al (2011) evaluated the effectiveness of endo-urethral autologous myoblast transplantation in the treatment of UI in children with bladder exstrophy-epispadias complex. Subjects were evaluated at 4 years of follow-up regarding the safety, efficacy and durability of the procedure, and health related quality of life. A total of 7 boys underwent autologous myoblast transplantation between May and December 2006. All patients had persistent UI after bladder neck reconstruction and bulking agent injection. Patients were followed for 4 years after autologous myoblast transplantation regarding clinical outcomes and cystometric, urodynamic, uroflowmetric and urethrocystoscopic evaluations. Health related quality of life was also measured before treatment and at final follow-up. No evidence of urinary obstruction was observed. Five children (71 %) were completely continent and 2 (29 %) were socially dry with complete daytime dryness at final follow-up. Health related quality of life was improved significantly. Urodynamic studies revealed a progressive increase in bladder capacity ($p < 0.001$). Mean detrusor leak point pressure showed a 27 cm H(2)O (158 %) increase during 4-year follow-up. Uroflowmetry parameters of voided volume and average

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maximum flow rate were improved significantly ($p < 0.001$). The authors concluded that the 4-year outcomes demonstrate that autologous myoblast transplantation for UI in children with bladder exstrophy-epispadias complex is relatively reliable, reproducible, safe and effective with minimal morbidity. This novel treatment represents a promising therapeutic approach in patients with UI. They stated that further randomized trials with larger numbers of patients and longer follow-up are needed.

According to the Interstim product labeling, the safety and effectiveness of bilateral sacral nerve stimulation has not been established (Medtronic, 2008).

In a pilot study, Marcelissen et al (2011) examined if bilateral sacral nerve stimulation can be effective to restore treatment efficacy in patients in whom unilateral sacral neuromodulation fails. Patients in whom unilateral sacral neuromodulation failed were included in analysis. The percutaneous nerve evaluation test was used to evaluate the effect of contralateral and bilateral stimulation. The stimulation electrode was placed in the contralateral S3 foramen and symptoms were self-recorded using a 3-day voiding diary. Clinical success was defined as more than 50 % improvement in at least 1 relevant voiding diary parameter versus baseline. The 15 study patients underwent test stimulation with percutaneous nerve evaluation. In 3 patients lead migration was suspected and, thus, they were not included in analysis. Four of the remaining 12 patients had a successful response to percutaneous nerve evaluation, of whom 3 were eventually implanted with a contralateral lead. After 12 months of treatment 2 of the 3 patients had a successful outcome. The authors concluded that only a select group of patients appeared to benefit from bilateral stimulation after unilateral therapy failure. They stated that further investigation is needed to determine the predictive factors and cost-effectiveness of this treatment.

Guidelines from the American Urologic Association (Gormley et al, 2012) have concluded: “Clinicians may offer sacral neuromodulation (SNS) as third-line treatment in a carefully selected patient population characterized by severe refractory OAB symptoms or patients who are not candidates for second-line therapy and are willing to undergo a surgical procedure. Recommendation (evidence strength grade C; benefits outweigh risk/burdens)”.

Woodruff et al (2008) stated that little is known about the host response to the various biologic and synthetic graft materials used as substitutes for autologous fascia. These researchers investigated the host response to sling graft materials in humans. A total of 24 women undergoing sling revision had a portion of the graft material removed for comparative analysis. At exploration, the degree of graft preservation (integrity), encapsulation, infection, and fibrosis was quantified. A histopathologic analysis was performed by systematically examining each specimen for the inflammatory response, neovascularity, and host fibroblast infiltration. A total of 24 grafts were explanted at 2 to 34 months after implantation. The indications for removal were a lack of sling efficacy in 2, urinary retention in 9, and sling obstruction in 13. The types of graft material were polypropylene mesh (PPM) in 10, autologous fascia in 5, porcine dermis in 4, cadaveric dermis in 3, and cadaveric fascia in 2. No graft degradation had occurred in PPM material. Autologous and cadaveric fascia had the most demonstrable graft degradation. No encapsulation had occurred with autologous fascia or PPM. The porcine dermis was the most encapsulated. No host infiltration had occurred with the encapsulated porcine grafts, and only peripheral infiltration of fibroblasts had occurred in the cadaveric grafts. The PPM grafts had the greatest number of fibroblasts throughout the entire graft. Neovascularity was the most prevalent in mesh and was also present in the autologous fascia. Giant cells were seen in 2 mesh and 2 porcine grafts. The authors concluded that the results of this study have shown that porcine dermis has the potential to encapsulate. The degree of host tissue infiltration was greatest with PPM, and no degradation of the mesh material had occurred with time.

An UpToDate review on “Treatment of urinary incontinence” (DuBeau, 2012) does not mention the use of collagen porcine dermis mesh as a therapeutic option. Furthermore, an UpToDate review on “Overview of transvaginal placement of reconstructive materials (surgical mesh or biograft) for treatment of pelvic organ prolapse or stress urinary incontinence” (Trabuco and Gebhart, 2012) states that “Midurethral slings, using macroporous polypropylene mesh, are the most common procedures for treatment of SUI [11]. A sling made of microporous material (ObTape) for mid-urethral slings was associated with high complication rates and was removed from the market”. It does not mention the use of collagen porcine dermis.

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In an open, prospective, single-center study, Cornu et al (2011) evaluated the safety of intrasphincteric injections of autologous muscular cells in patients with post-prostatectomy incontinence (PPI; n = 12). Patients underwent intrasphincteric injections of autologous muscular cells isolated from a biopsy of deltoid muscle. The primary endpoint was the Q(max) variation at the 3-month visit in order to assess potential bladder outlet obstruction. Secondary endpoints assessed side effects and efficacy parameters based on symptoms, quality of life score, voiding diary, pad-test, and urethral pressure profile at 1, 2, 3, 6 and 12 months after injection. No immediate complication occurred and no significant variation was noted on Q(max). The only side effects possibly product-related were 3 cases of urinary tract infection treated by antibiotics. An acceptable safety and tolerability of the procedure whatever the injected dose of muscular cells was demonstrated. Results on efficacy after 1 year were heterogeneous, with 4/12 patients describing reduced urine leakage episodes, 1/12 patient presenting increased maximal closure pressure, and 8/12 patients showing improvement on pad-test. The authors concluded that cell therapy consisting of intrasphincteric injections of autologous muscular cells in patients with PPI was a feasible and safe procedure. They stated that these findings pointed out that some subjects may positively respond to this procedure, but clinical efficacy remains to be confirmed.

In a prospective, dose ranging, feasibility study, Carr et al (2013) evaluated the 12-month safety and potential efficacy of autologous muscle derived cells (Cook MyoSite Incorporated, Pittsburgh, PA) as therapy for SUI. A total of 38 women in whom SUI had not improved with conservative therapy for 12 or more months underwent intra-sphincter injection of low-doses (1, 2, 4, 8 or 16 × 10(6)) or high-doses (32, 64 or 128 × 10(6)) of autologous muscle derived cells, which were derived from biopsies of their quadriceps femoris. All patients could elect a second treatment of the same dose after 3-month follow-up. Assessments were made at 1, 3, 6 and 12 months after the last treatment. The primary end point was the incidence and severity of adverse events. In addition, changes in SUI severity were evaluated by pad test, diary of incontinence episodes and quality of life surveys. Of the 38 patients, 33 completed the study. Treatment-related complications were limited to minor events such as pain/bruising at the biopsy and injection sites. Of patients who received 2 treatments of autologous muscle derived cells who were eligible for analysis, a higher percentage of those in the high-dose versus the low-dose group experienced a 50 % or greater reduction in pad weight (88.9 %, 8 of 9 versus 61.5 %, 8 of 13), had a 50 % or greater reduction in diary reported stress leaks (77.8 %, 7 of 9 versus 53.3 %, 8 of 15) and had 0 to 1 leaks during 3 days (88.9 %, 8 of 9 versus 33.3 %, 5 of 15) at final follow-up. The authors concluded that injection of autologous muscle derived cells in a wide range of doses appears safe with no major treatment-related adverse events reported. They stated that treatment with autologous muscle derived cells shows promise for relieving SUI symptoms and improving quality of life. Moreover, they noted that the most effective dose of cells has yet to be determined, and a placebo-controlled study powered to determine treatment efficacy is necessary. Two ongoing studies have been designed to address these issues.

Phe and colleagues (2014) described the minimally invasive adjustable continence therapy (ACT) balloon placement surgical technique and analyzed the results of ACT balloon in the treatment for female SUI. A review of the literature was performed by searching the PubMed database using the following search terms: ACT balloons, female urinary incontinence, and female continence. A total of 8 studies were published between 2007 and 2013. The mean follow-up of these studies was 1 to 6 years. The mean age of the patients ranged between 62 and 73 years; 40 to 100 % of patients had already been treated surgically for their SUI. A significant reduction in the number of pads used per day was observed after ACT balloon placement, with improvement of short pad tests from 49.6 to 77.3 g pre-operatively to 11.2 to 25.7 g after ACT balloon placement; 15 to 44 % of patients considered that their SUI had been cured and 66 to 78.4 % were satisfied with the result. The explantation rate ranged between 18.7 and 30.8 %. Quality of life was significantly improved, and no major complication was reported. The authors concluded that ACT balloons constitute a reasonable, minimally invasive alternative for the treatment for female SUI due to intrinsic sphincter disorder, especially in patients who have already experienced failure of standard surgical treatment and in clinical settings incompatible with invasive surgical placement of an artificial urinary sphincter (especially women over the age of 80 years). Moreover, they stated that long-term results are essential to evaluate the effectiveness of this treatment.

Test Stimulation of the InterStim

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The InterStim product labeling states that, in clinical studies, subjects underwent anywhere from 1 to 6 test stimulation procedures before implantation of InterStim.

The Medtronic InterStim test stimulation lead kit manual stated that "Of the 260 patients (45.0 %) who qualified for implantation, 169 (65.0 %) had a successful result (minimum of 50 % improvement in dysfunctional voiding symptoms) during their first test stimulation procedure. Of the remaining 91 patients, 56 (21.5 %) obtained a successful result during a second test stimulation, and 35 (13.5 %) obtained a successful result during three or more test stimulations. Reasons for repeat test stimulation procedures included inadequate responses to test stimulation or technical problems …. The safety and effectiveness of this therapy has not been established for pediatric use (patients under the age of 16), patients with neurological disease origins, such as multiple sclerosis or diabetes, and bilateral stimulation".

Axonics Neuromodulation System for the Treatment of Urinary Incontinence

Benson et al (2020) noted that sacral neuromodulation (SNM) is a guideline-recommended treatment for voiding dysfunction including urgency, urge incontinence (UI), and non-obstructive retention as well as fecal incontinence. The Axonics System is a miniaturized, rechargeable SNM system designed to provide therapy for at least 15 years, which is expected to significantly reduce revision surgeries as it will not require replacement as frequently as the non-rechargeable SNM system. The ARTISAN-SNM study is a pivotal study designed to treat patients with urgency UI (UUI). These researchers presented clinical results at 1 year. A total of 129 eligible UUI patients were treated. All subjects were implanted with a quadripolar tined lead and neurostimulator in a single procedure. Effectiveness data were collected using a 3-day bladder diary, a validated quality of life questionnaire (ICIQ-OABqol), and a participant satisfaction questionnaire. Therapy responders were defined as participants with greater than or equal to 50 % reduction in UUI episodes compared to baseline. Data were analyzed on all 129 subjects. At 1 year, 89 % of the subjects were therapy responders. The average UUI episodes per day reduced from 5.6 ± 0.3 at baseline to 1.4 ± 0.2. Subjects experienced an overall clinically meaningful improvement of 34 points on the ICIQ-OABqol questionnaire. All study participants (100 %) were able to recharge their device at 1 year, and 96 % of participants reported that the frequency and duration of recharging was acceptable. There were no serious device-related adverse events (AEs). The authors concluded that the Axonics System was safe and effective at 1 year, with 89 % of participants experiencing clinically and statistically significant improvements in UUI symptoms.

Geynisman-Tan et al (2021) described factors associated with satisfaction with the Axonics SNM System at 1 year. This was a secondary analysis of data collected in the ARTISAN-SNM study -- a prospective, single-arm, multi-center trial of the Axonics r-SNM System. ARTISAN-SNM recruited subjects with UUI to undergo a single, non-staged implant of the lead and rechargeable neurostimulator. Participants were considered therapy responders if they had greater than or equal to 50 % reduction in UUI episodes in a 3-day period at 1-month post-implant. Bladder diaries and satisfaction (7-point Likert scale) were evaluated at 1 year. A total of 124 participants (110 "responders" and 14 "non-responders") had complete data at baseline, 1 month and 1 year following implant. Most subjects were satisfied with Axonics at 1 year: 68.5 % were "very satisfied", 25.8 % were "moderately satisfied", and 2.4 % were "slightly satisfied". At 1 year, treatment effectiveness, as measured by electronic bladder diaries, was significantly associated with satisfaction. Participants who were "very satisfied" had a larger reduction in voids per day ($p = 0.01$), leaks per day ($p = 0.004$), urgent leaks per day ($p = 0.04$), and voids in which the urgency was desperate per day ($p = 0.03$) compared to those less satisfied; 12 of the 14 "non-responders" continued to see improvements in symptom reduction from 1 month to 1 year; 9/14 (64 %) were "responders" at 1 year with 6 reporting being "very satisfied" and 1 reporting being "moderately satisfied". The authors concluded that satisfaction 1 year after implantation of Axonics SNM was extremely high and correlated with the degree of symptom improvement, which increased over time.

Pezzella et al (2021) stated that SNM is a guideline-recommended treatment with proven therapeutic benefit for UUI patients. The Axonics System is the 1st Food and Drug Administration (FDA)-approved rechargeable SNM system and is designed to deliver therapy for a minimum of 15 years. The ARTISAN-SNM study was designed to evaluate UUI participants treated with the Axonics System. These researchers presented 2-year follow-up results. A total of 129 UUI participants underwent implantation with the Axonics

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System. Therapeutic response rate, participant quality of life (QOL), and satisfaction were determined using 3-day voiding diaries, ICIQ-OABqol, and satisfaction questionnaires. Participants were considered responders if they had a 50 % or greater reduction in UUI episodes post-treatment. At 2 years, 93 % of the participants (n = 121 completers at 2 years) were therapy responders, of which 82 % achieved greater than or equal to 75 % reduction in UUI episodes and 37 % were dry (100 % reduction). Daily UUI episodes reduced from 5.6 ± 0.3 at baseline to 1.0 ± 0.2 at 2 years. Statistically significant improvements in ICIQ-OABqol were reported. All participants were able to recharge their device and 94 % of participants reported that the recharging frequency and duration were acceptable. Participant demographics nor condition severity were correlated with clinical outcomes or recharging experience. No unanticipated or serious device-related AEs occurred. The authors concluded that at 2 years, participants treated with the Axonics System demonstrated sustained safety and effectiveness, high levels of satisfaction with therapy and recharging. Participant-related factors were not associated with effectiveness or recharging outcomes, indicating the reported results were applicable to a diverse population.

Wang et al (2021) noted that over-active bladder (OAB) and UUI affect millions of women and men and results in billions of dollars in health-care expenses. First- and 2nd-line therapy includes behavioral modifications and/or pharmacotherapies; however, many patients' symptoms remain or worsen on these treatments. There has been concern regarding the detrimental side effects of the most widely prescribed medications for these bladder symptom management. As a result, there has been increased interest in continuous sacral neuromodulation, an FDA-approved therapy for refractory UUI. These investigators reviewed current research on the effectiveness and patient/provider satisfaction and safety profile of the Axonics System. Furthermore, they addressed the current state SNM and potential future direction and applicability. The authors concluded that the Axonics system is a safe and effective device for the treatment of OAB and UUI. In additional, it affords patient's the convenience of a rechargeable, compact, MRI safe system. It should be noted that the rechargeable system, while allowing for approximately 15 years of battery and lead life, may have its challenges in terms of charge burden. Furthermore, this system is easily adapted for experienced implanters of sacral neuromodulating devices.

Furthermore, an UpToDate review on “Urgency urinary incontinence/overactive bladder (OAB) in females: Treatment” (Lukacz, 2022) states that “Sacral neuromodulation -- SNM is a minimally invasive surgical electrical stimulation option to treat OAB symptoms that is offered to patients whose symptoms do not respond to initial interventions and pharmacotherapy. Several devices are available, including InterStim micro system, InterStim II, and Axonics, which include MRI-compatible options. InterStim micro and Axonics have rechargeable implanted programmable device options, which can increase battery life to 15 years or more and may be more cost-effective than the non-rechargeable option. These devices require the patient to have the cognitive ability and desire to manage the technology, perform a testing procedure; monitor the impact of stimulation on urinary incontinence episodes, urgency, and pad usage for a week or two; and manage the recharging process should they select this option. The InterStim II device has a non-rechargeable battery that requires replacement every 3 to 5 years”.

Polyacrylamide Hydrogel (Bulkamid)

Kasi et al (2016) performed a systematic review on the effectiveness of polyacrylamide hydrogel (PAHG, Bulkamid) in the treatment of female patients with SUI with regard to reproducibility, feasibility, safety and clinical outcome. These investigators searched MEDLINE (1966 to 2015), Scopus (2004 to 2015), POPLINE (1974 to 2015) and ClinicalTrials.gov (2008 to 2015) along with reference lists of electronically retrieved studies. Observational studies, prospective, retrospective and RCTs were included. Two reviewers independently selected studies, assessed the risk of bias and tabulated data to structured forms. These researchers included 8 studies, which enrolled a total of 767 patients who received treatment with PAHG. They found that 186 of 767 women (24.3 %, range of 12 to 35 %) required re-injection in order to achieve adequate effectiveness. The most frequent adverse effects were pain at the site of injection (4 to 14 %) and urinary tract infections (3 to 7 %). Both the number of incontinence episodes/24 hours and the number of ml/24 hours were significantly reduced 1 year following treatment and the quality of life of patients was significantly improved. The authors concluded that PAHG is a safe intervention for treating women with

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SUI, but repeat injections are often required. They stated that further research is needed to compare its effectiveness with other bulking agents.

In a prospective, observational study, Altman and colleagues (2017) examined the effects of transurethral polyacrylamide hydrogel injection in patients considered ineligible for mid-urethral sling surgery. A total of 81 patients received treatment with transurethral polyacrylamide hydrogel injection. Subjects were considered ineligible for placement of a mid-urethral sling based on significant co-morbidity (48 %), 1 or more previously failed invasive treatments (16 %), mixed UI (27 %), continuous incontinence (5 %), or previous pelvic radiation therapy (4 %). Longitudinal assessment of subjective treatment outcomes was performed using the validated UDI and the pelvic floor impact questionnaire (PFIQ) at baseline, 2, and 6 months. To deal with repeated measurements, mixed linear models were used to examine changes in the outcomes over time. There was a significant improvement in the overall UDI score from baseline to month 2 follow-up (FU) ($p < 0.001$). No major differences between the month 2 and 6 FUs were observed. The largest difference in effect was observed for the irritative and stress subscales; 25 patients (33 %) requested a 2nd injection at the month-2 FU visit. At month-6 FU, the UDI scores for patients having had only 1 injection were largely unchanged, whereas all UDI domains worsened further for patients having had a 2nd injection at the month-2 visit. After the injection, there were 3 minor AEs (3.7 %) and no serious AEs. The authors concluded that in patients considered ineligible for mid-urethral sling surgery, transurethral injection with polyacrylamide hydrogel may alleviate UI symptoms. Repeat injections did not improve outcomes in this complicated group of patients.

Mohr and associates (2017) noted that mixed UI (MUI), defined as mixed symptoms of SUI and OAB, is a difficult entity if conservative treatment has failed. Cure rates are low compared with SUI, particularly the OAB component, may deteriorate after sling insertion. Bulking agents pose an appealing alternative for the treatment of MUI. They have shown beneficial effect in small case studies, but larger series are lacking. In a prospective study, these investigators examined the safety and efficacy profile of Bulkamid in female patients with MUI. A total of 154 women with MUI symptoms (components of SUI/OAB within the limits of 60 to 40 % either way) received bulking therapy with Bulkamid. Patients were followed-up 3 months post-operatively. Primary outcome was the domain Incontinence impact on the KHQ; secondary outcomes were the other KHQ domains, visual analog scale (VAS), and International Continence Society (ICS) standardized pad weight test as objective measurement of incontinence. Statistically significant improvements were found for all KHQ domains, pad weight test, and the VAS before and after bulking; overall complication rate was 13 %. The authors concluded that the findings of this study showed improvement in MUI after bulking therapy according to both subjective and objective outcomes. These investigators could advocate bulking therapy for treating MUI, as it is simple and safe and showed both objective and subjective improvement and relief. They stated that long-term results (up to 1 year) are awaited.

Hussain and Bray (2019) noted that urethral bulking agents (UBAs) to treat SUI was first described in the 1930s, when paraffin was used to increase urethral resistance. Since then, several agents have been introduced to the market, with varying degrees of safety, efficacy, and durability. The agents currently available include calcium hydroxyl apatite (Coaptite), carbon coated zirconium (Durasphere; polydimethylsiloxane elastomer (Macroplastique); and polyacrylamide hydrogel (Bulkamid). The latest product, PDMS-U (Urolastic) is a silicone gel that polymerizes when injected. The short-term efficacy of UBAs are generally encouraging; however, longer follow-up results show the success rates are reduced and many women will require a repeat treatment.

In a controlled clinical trial, Itkonen Freitas and co-workers (2020) examined if polyacrylamide hydrogel is non-inferior to tension-free vaginal tape in the treatment of women with primary SUI. Subjects were randomized to tension-free vaginal tape or polyacrylamide hydrogel treatment. The primary outcome was patient satisfaction and secondary outcomes were effectiveness in reducing urinary leakage and complications at 1-year follow-up. For statistical power, significance was considered at 5 %, power was set at 80 % and the non-inferiority limit was 20 % with a 10 % expected drop-out rate. A total of 224 women with primary SUI entered the study between September 28, 2015 and March 1, 2017. Of the women 111 were randomized to tension-free vaginal tape and 113 were randomized to polyacrylamide hydrogel. At 1 year a satisfaction score of 80 or greater on a VAS of 0 to 100 was reached in 95.0 % and 59.8 % of patients treated with tension-free vaginal tape and polyacrylamide hydrogel, respectively. Therefore, polyacrylamide

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hydrogel did not meet the non-inferiority criteria set in this study. As secondary outcomes, the cough stress test was negative in 95.0 % of tension-free vaginal tape cases versus 66.4 % of polyacrylamide hydrogel cases (difference 28.6 %, 95 % CI: 18.4 to 38.5). However, most peri-operative complications, including those in 19 tension-free vaginal tape cases versus 3 polyacrylamide hydrogel cases (difference 16.0 %, 95 % CI: 7.8 to 24.9), and all 6 re-operations due to complications (difference 5.9 %, 95 % CI: 1.2 to 12.4) were associated with tension-free vaginal tape. The authors concluded that mid-urethral tension-free vaginal tape slings were associated with better satisfaction and cure rates than polyacrylamide hydrogel in women with primary SUI. However, complications were mainly associated with tension-free vaginal tape. Therefore, tension-free vaginal tape should be offered as 1st-line treatment in women who expect to be completely cured by the initial treatment and are willing to accept the complication risks. Since polyacrylamide hydrogel treatment also provides high satisfaction and cure rates, women with primary SUI can be offered polyacrylamide hydrogel as an alternative treatment.

Furthermore, an UpToDate review on "Stress urinary incontinence in women: Persistent/recurrent symptoms after surgical treatment" (Morgan, 2020) states that "Several materials are available for peri-urethral injection therapy. Products that are now commonly used include: Polyacrylamide hydrogel (Bulkamid): a homogeneous, stable hydrophilic polymer gel composed of 2.5 % cross-linked dextranomer polyacrylamide and 97.5 % water, available in Europe".

On January 28, 2020, the Bulkamid Urethral Bulking System was approved by the FDA. It is indicated for urethral injection for the treatment of SUI due to ISD in adult women who have SUI or stress predominant mixed incontinence.

Adjustable Retropubic Sub-Urethral Sling for Stress Urinary Incontinence

In a single-center, prospective study, Leizour and associates (2016) evaluated the safety and effectiveness of the adjustable sub-urethral sling Remeex in the treatment of male SUI. Participants were patients treated for SUI after radical prostatectomy (RP) or transurethral resection of prostate. The severity of incontinence was evaluated by the number of pads used per day. Success rate, complications and number of adjustments were studied. From February 2011 to May 2015, Remeex was implanted in 25 patients. The average pre-operative number of pads used per day was 3.8 (± 1.8). Sling tension has been adjusted the day after surgery in all patients. Mean follow-up was 31 months (± 15). During follow-up, 6 patients did not need any re-adjustment (24 %) and 15 patients (60 %) had to be re-adjusted. One Remeex system had to be completely removed because of a sub-occlusive syndrome; 3 patients had early infection requiring partial system removal (Varitensor). At the end of follow-up, 9 patients were cured (36 %), 9 patients (36 %) were significantly improved and 7 patients (28 %) were not improved; 5 patients were waiting for a new re-adjustment. The authors concluded that in this short series of patients who had prostatic surgery, at mid-term follow-up, the placement of an adjustable sub-urethral sling was associated with an improvement or cure of UI symptoms in 2/3 of cases.

Magnetically Controlled Endo-Urethral Artificial Urinary Sphincter

Mazzocchi and colleagues (2016) stated that UI is a widespread dysfunction that affects more than 300 million people worldwide. At present, no technological solutions are able to restore continence in a minimally invasive and effective way. These researchers described the design, fabrication, and testing of a novel artificial endo-urethral urinary sphincter that attempts to fully restore continence. The device can be inserted/retracted in a minimally invasive fashion without hospital admission, does not alter the body scheme and can be applied to both women and men. The device core is a uni-directional polymeric valve and a magnetically activated system, which is able to modulate its opening pressure. Bench tests and ex-vivo tests on a human cadaver demonstrated that the device was able to fully restore continence and allowed urination when desired. The authors concluded that the proposed system showed a high potential as a technological solution that may restore a normal daily life in patients affected by UI. These preliminary findings need to be validated by well-designed studies.

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Transcutaneous Electrical Nerve Stimulation (TENS) in the Treatment of Overactive Bladder

Sharma and colleagues (2016) stated that OAB accounts for 40 to 70 % cases of incontinence. The etiology is unknown though detrusor instability is found in urodynamic evaluation of almost all cases. Detrusor instability or hyperreflexia can be inhibited by direct inhibition of impulses in the pre-ganglionic afferent neuron or by inhibition of bladder pre-ganglionic neurons of the efferent limb of micturition reflex. Transcutaneous electrical nerve stimulation (TENS) is based on the gate control theory of abolishing the local micturition reflex arc. In a prospective experimental study, these researchers evaluated the safety and effectiveness of TENS in idiopathic OAB. They evaluated the effectiveness of TENS versus placebo in reducing OAB symptoms (n1 = 20, n2 = 20); 10 treatment sessions (5 sessions/week) of 30 minutes were conducted. There was a significant improvement in Overactive Bladder Symptom Scores (OABSS) in TENS group and 2 patients were completely dry following TENS therapy. The authors concluded that in elderly women, patients with OAB where other co-medications had their own anti-cholinergic side effects and impairment of cognition is a concern, TENS can be a useful intervention. The author also noted that future advancements will likely emphasis on the exact placement site of electrodes with less collateral stimulation. The main drawbacks of this study were; (i) small sample size (n = 20) for the TENS-treated group), and (ii) all patients were uniformly treated with alternate high- and low-frequency TENS with burst therapy intermittently, to prevent the development of tolerance. High-intensity was used to achieve maximum effect, and (iii) the ideal stimulation protocol needs to be worked out.

In a randomized, double-blind, placebo-controlled study, Borch and associates (2017) evaluated the immediate effect on natural fill urodynamic parameters and bladder function during TENS in children with OAB and daytime UI (DUI). A total of 24 children with severe OAB and DUI (mean age of 8.5 ± 1.2 years) underwent 48-hour natural fill urodynamics. After 24 hours of baseline investigation, the children were randomized to either active continuous TENS (n = 12) or placebo TENS (n = 12) over the sacral S2-S3 outflow. The urodynamic recordings were analyzed manually for 3 different bladder contraction patterns resulting in a void. The number of bladder contractions not leading to a void was also calculated. Maximum voided volume (MVV) and average voided volume (AVV) were identified for both the baseline and the intervention day. It was found that TENS had no immediate objective effect on bladder capacity. The difference (before minus after treatment) in MVV/EBC in the active TENS group = 0.03 ± 0.23 versus placebo TENS group = -0.01 ± 0.10 (p = 0.61). Furthermore, there was no significant difference in the proportion of different bladder contraction types between the 2 groups; TENS did not significantly influence the number of bladder contractions not leading to a void. Results were presented as mean ± SD. The authors concluded that there was no immediate objective effect of TENS on bladder activity assessed by natural fill urodynamics in children with OAB and DUI.

Transperineal Implantation of Permanent Adjustable Balloon Continence Device

There is insufficient evidence to support the use of extra-urethral (non-circumferential retropubic adjustable compression devices (ProACT Therapy System) for treatment of UI. The ProACT Therapy System (Uromedica, Inc.), an adjustable continence therapy, is a minimally invasive urological device designed to treat persons with SUI. A Horizon Scanning Report of the ProACT by the Australian Safety and Efficacy Register of New Interventional Procedures - Surgical (ASERNIPS, 2006) raised questions about the safety of the device. The report concluded that current available literature on the ProACT system suggested that the device is safe for implantation. However as the studies presented suggest there are recurring safety issues with the device, namely post-operative complications such as the migration of the device and erosion of the urethra or the bladder. Although these complications were able to be corrected through removal and later re-implantation of the device in most cases, this presents an added risk to the patient as a result of the re-implantation procedure. Intra-operatively, implantation of the device is not reported as overly difficult and successful implantation may increase as surgeons familiarize themselves with the procedure. Further studies investigating the long-term (more than 2 years) effects of the ProACT Therapy System are needed to ascertain any long-term advantage of the ProACT Therapy System over other treatment options. Furthermore, randomized controlled trials or comparative studies are needed to compare differences in rates

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of complications between the ProACT Therapy System and other treatment options. The National Institute for Health and Clinical Excellence (2006) has concluded that current evidence on the safety and efficacy of insertion of extra-urethral (non-circumferential) retropubic adjustable compression devices for SUI does not appear adequate for this procedure to be used without special arrangements for consent and for audit and research

Aboseif et al (2009) examined the safety, effectiveness, adjustability and technical feasibility of the adjustable continence therapy device (Uromedica, Plymouth, MN) for the treatment of recurrent female SUI. Female patients with recurrent SUI were enrolled in the study and a defined set of exclusionary criteria were followed. Baseline and regular follow-up tests to determine eligibility, and to measure subjective and objective improvement were performed. A trocar was passed fluoroscopically and with digital vaginal guidance to the urethro-vesical junction through small incisions between the labia majora and minora. The adjustable continence therapy device was delivered and the balloons were filled with isotonic contrast. The injection ports for balloon inflation were placed in a subcutaneous pocket in each labia majora. Device adjustments were performed percutaneously in the clinic post-operatively. An approved investigational device exemption FDA protocol was followed to record all adverse events. A total of 162 subjects underwent implantation with 1 year of data available on 140. Mean Stamey score improved by 1 grade or more in 76.4 % (107 of 140) of subjects. Improvement in the mean incontinence quality of life questionnaire score was noted at 36.5 to 70.7 ($p < 0.001$). Reductions in mean Urogenital Distress Inventory (60.3 to 33.4) and Incontinence Impact Questionnaire (54.4 to 23.4) scores also occurred ($p < 0.001$). Mean provocative pad weight decreased from 49.6 to 11.2 gm ($p < 0.001$). Of the patients 52 % (67 of 130) were dry at 1 year (less than 2 gm on provocative pad weight testing) and 80 % (102 of 126) were improved (greater than 50 % reduction on provocative pad weight testing). Complications occurred in 24.4 % (38 of 156) of patients. Explantation was required in 18.3 % (28 of 153) of the patients during 1 year. In terms of the complications 96.0 % were considered to be mild or moderate. The authors concluded that the Uromedica adjustable continence therapy device is an effective, simple, safe and minimally invasive treatment for recurrent female SUI. It can be easily adjusted percutaneously to enhance efficacy and complications are usually easily manageable. Explantation does not preclude later repeat implantation. Moreover, the authors stated that additional studies are needed to determine the long-term durability of the device.

In an editorial that accompanied the afore-mentioned study, Gilling (2009) stated that the results appear superior to those of bulking agents, although comparison of these heterogenous groups is difficult.

Kocjancic et al (2010) evaluated the implantation procedure and assessed patient outcomes of adjustable continence therapy for severe intrinsic sphincter deficiency and recurrent female SUI. The adjustable continence device consists of 2 silicone balloons on either side of the proximal urethra under the bladder neck, each attached to a titanium port buried in the labia to allow post-operative titration. Urodynamic assessment was done in 57 female patients in whom previous pelvic surgery had failed. Pad use and an incontinence quality of life questionnaire were evaluated before ACT implantation, post-operatively at 1, 3, 6 and 12 months, and annually thereafter. Patients recorded the overall impression and percent of improvement post-operatively based on the Patient Global Impression Index and a visual analog scale. Mean follow-up was 72 months (range of 12 to 84). At 6-year follow-up in 29 patients, mean pad use improved from 5.6 daily at baseline to 0.41 and intrinsic sphincter deficiency improved from 27.2 to 78.6 ($p < 0.001$). As measured on the visual analog scale, 68 % of patients considered themselves dry. On the Patient Global Impression Index questionnaire 64 % were very much improved, 23 % were much improved and 13 % were only minimally improved or unchanged. No patients considered themselves worse after the procedure. Complications necessitating device removal developed in 21.1 % of patients. The authors concluded that relative ease of insertion and the ability to tailor this therapy to individual needs makes this an attractive option for the challenging treatment for recurrent SUI due to intrinsic sphincter deficiency. Furthermore, they noted that these findings are encouraging, especially in terms of patient subjective outcomes, but their study was limited by the number of patients treated, the modification in procedural technique during the study, and the lack of more objective data. More studies are needed to establish the actual ACT mechanism of action in previously failed surgical cases and more closely monitor objective outcomes in the light of procedural and post-operative management.

Dynamometry for Quantification of Pelvic For Muscle Strength

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Deegan and colleagues (2018) stated that there remains no gold standard for quantification of voluntary pelvic floor muscle (PFM) strength, despite international guidelines that recommend PFM assessment in females with UI. In this study, methods currently reported for quantification of skeletal muscle strength across disciplines were systematically reviewed and their relevance for clinical and academic use related to the pelvic floor were described. These investigators performed a systematic review via Medline, PubMed, CINHAL, and the Cochrane database using key terms for pelvic floor anatomy and function were cross-referenced with skeletal muscle strength quantification from 1946 to 2016. Full text peer-reviewed articles in English having female subjects with UI were identified. Each study was analyzed for use of controls, type of methodology as direct or indirect measures, benefits, and limitations of the technique. A total of 1,586 articles were identified of which 50 met the inclusion criteria; 9 methodologies of determining PFM strength were described including: digital palpation, perineometer, dynamometry, electromyography (EMG), vaginal cones, ultrasonography, magnetic resonance imaging (MRI), urine stream interruption test, and the Colpexin pull test; 32 % lacked a control group. The authors concluded that technical refinements in both direct and indirect instrumentation for PFM strength measurement are allowing for sensitivity. However, the most common methods of quantification remain digital palpation and perineometry; techniques that pose limitations and yield subjective or indirect measures of muscular strength. Moreover , they stated that dynamometry has potential as an accurate and sensitive tool, but is limited by inability to assess PFM strength during dynamic movements.

The Leva Pelvic Floor Trainer

The Leva pelvic floor trainer is intended for the purpose of rehabilitation and training of weak pelvic floor muscles for the treatment of stress, mixed and mild moderate urge incontinence in women. This device interacts with the user via smart phone technology. There is lack of evidence that the use of this device provides better outcomes than conventional Kegel exercises.

Oliveira and colleagues (2017) noted that strengthening exercises for PFM (SEPFM) are considered the 1st approach in the treatment of SUI. Nevertheless, there is no evidence about training parameters. These researchers identified the protocol and/or most effective training parameters in the treatment of female SUI. A literature research was conducted in the PubMed, Cochrane Library, PEDro, Web of Science and Lilacs databases, with publishing dates ranging from January 1992 to March 2014. The articles included consisted of English-speaking experimental studies in which SEPFM were compared with placebo treatment (usual or untreated). The sample had a diagnosis of SUI and their age ranged between 18 and 65 years. The assessment of methodological quality was performed based on the PEDro scale. A total of 7 high methodological quality articles were included in this review. The sample consisted of 331 women, mean age of 44.4 ± 5.51 years, average duration of urinary loss of 64 ± 5.66 months and severity of SUI ranging from mild to severe. SEPFM programs included different training parameters concerning the PFM. Some studies have applied abdominal training and adjuvant techniques. Urine leakage cure rates varied from 28.6 to 80 %, while the strength increase of PFM varied from 15.6 to 161.7 %. The authors concluded that the most effective training protocol consists of SEPFM by digital palpation combined with biofeedback monitoring and vaginal cones, including 12-week training parameters, and 10 repetitions per series in different positions compared with SEPFM alone or a lack of treatment.

In a randomized controlled trial (RCT), Weinstein et al (2022) examined if pelvic floor muscle training (PFMT) using a motion-based digital intra-vaginal device is more effective than home PFMT for treatment of stress urinary incontinence (SUI) or stress-predominant mixed UI (MUI). In a remote, virtually executed 8-week prospective, superiority study, women with SUI or stress-predominant MUI were randomized to PFMT using a motion-based digital therapeutic device or a home training program using written and narrated instructions. Primary outcomes were change in UDI-6 (Urogenital Distress Inventory, Short Form) score and SUI episodes on a 3-day bladder diary. A sample size of 139 per group (n = 278) was planned to meet the power analysis requirements for the UDI-6 score (n = 278) and the bladder diary (n = 78). Pre-specified secondary outcomes included QOL surveys and adherence reporting. From September 2020 to March 2021, a total of 5,353 participants were screened, and 363 were randomized: 182 in the intervention, and 181 in the control group. There were no baseline clinicodemographic differences between groups. The mean change in UDI-6 score was significantly greater for the intervention group compared with the control group (18.8

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versus 14.7, p = 0.01). The median (inter-quartile range [IQR]) number of SUI episodes on the 3-day bladder diary was significantly reduced from 5 (3 to 8) and 5 (3 to 8) episodes to 1 (0 to 3) and 2 (1 to 4) (p = 0.005) in the intervention group compared with control group, respectively. A significantly greater number of participants in the intervention group than in the control group reported they were "much improved" or "very much improved" on the PGI-I (Patient Global Impression of Improvement) (63/143 [44.1 % versus 45/156 [28.8 %], OR of 1.94, 95 % CI: 1.21 to 3.15). There were no device-related severe AEs. The authors concluded that in this all-remote, virtually-conducted study, PFMT guided by a motion-based digital therapeutic device resulted in significantly improved UI symptoms and reduction of UI episodes compared with a home training program.

The authors stated that a drawback of this trial was the inability to carry out a physical examination before enrollment. For example, pelvic organ prolapse beyond the introitus was an exclusion criterion, and subjects were asked about “seeing or feeling a bulge”, a question that has been used in other epidemiologic studies. Baseline pelvic floor muscle strength assessment may have added value to a study of 1st-line UI treatment, although research and expert consensus supported digital health in the remote context, including initiation of PFMT in the absence of a physical examination. Moreover, these researchers stated that longer-term follow-up is underway to better understand the durability of the treatment regimen and examine the need for maintenance exercises to sustain the benefits of therapy.

Weinstein et al (2023) examined the long-term effectiveness of an 8-week regimen of PFMT guided by a motion-based digital therapeutic device compared with a standard home program in the treatment of SUI and stress-predominant MUI. The primary virtual trial was carried out from October 2020 to March 2021; a total of 363 women with SUI or stress-predominant MUI were randomized to complete PFMT using the device (intervention group) or a standard home pelvic floor muscle training program (control group) for 8 weeks. Primary outcomes included change in UDI-6 score and SUI episodes on a 3-day bladder diary. The PGI-I was also assessed, with "much better" and "very much better" responses considered as improvement. In this planned secondary analysis, symptom and adherence data were collected in follow-up at 6 and 12 months. A modified intention-to-treat (ITT) analysis was carried out using Student's t tests and $\chi 2$ tests as appropriate. Of 299 subjects analyzed at 8 weeks, 286 (95.7 %) returned 6- and 12-month data (151 in the control group, 135 in the intervention group). Mean age was 51.9 ± 12.8 years, and mean BMI was 31.8 ± 7.4; 84.6 % of subjects were parous, and 54.9 % were post-menopausal. Mean change in UDI-6 score from baseline to 6 and 12 months was significantly greater in the intervention group than in the control group (20.2 ± 20.9 versus 14.8 ± 19.5, p = 0.03 and 22.7 ± 23.3 versus 15.9 ± 20.3, p = 0.01, respectively). Subjects in the intervention group had more than twice the odds of reporting improvement on the PGI-I compared with subjects in the control group (OR 2.45, 95 % CI: 1.49 to 4.00). The authors concluded that the use of this technology may facilitate remote access to PFMT for women with UI and represented an effective modality for scaling conservative 1st-line care above PFMT home programs. For women choosing 1st-line care at home for SUI or MUI, a motion-based digital therapeutic device may be considered to optimize durable treatment results.

The authors concluded that drawbacks of this trial included the lack of physical examination and other objective measures of pelvic floor muscle performance at baseline and follow-up. Furthermore, bladder diaries were not collected at 6 or 12 months to enable comparison of number of UI episodes reported during the active study period. In addition, although these investigators were able to collect information regarding continued use for subjects in the intervention group due to reporting from the device, they were unable to collect parallel information for subjects in the control group. Although this limited their ability to understand the presence or absence of continued PFMT in the control group, it was inherent in the design of the control group and typical for the use of home PFMT.

In a retrospective, cohort study, Keyser et al (2023) examined the effectiveness of a prescription digital therapeutic (pDTx) in reducing UI symptoms in real-world users. This trial examined data from users of a pDTx designed to guide PFMT between July 1, 2020 and December 31, 2021. The primary outcome was UI symptom change as reported via in-app UDI-6. Included subjects were female, 18 years of age or older with a diagnosis of SUI, urgency UI, or MUI who completed the UDI-6 at baseline and 8 weeks. Demographic, symptom, and adherence data were summarized. Paired t-test and Wilcoxon signed rank test were used to analyze change in outcomes from baseline to 8 weeks across adherence and UI diagnosis groups. Of 532

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women with UI, 265 (50 %) met criteria and were included in the analysis. Mean age was 51.2 ± 11.5 years (range of22 to 84, n = 265). Mean BMI was 27.3 ± 6.2 kg/m2 (range of 15.2 to 46.9, n = 147). Most subjects had SUI (59 %) followed by MUI (22 %), urgency UI/OAB (11 %), and unspecified UI (8 %). UDI-6 scores improved by 13.90 ± 15.53 ($p \leq 0.001$); 62 % met or exceeded minimum clinical important difference (MCID). Device-reported PFMT adherence was 72 % at 4 weeks and 66 % at 8 weeks (100 % = 14 uses/week). Subjects in each diagnosis category reported significant improvement on UDI-6 score from baseline to 8 weeks. No association between UDI-6 score improvement and adherence category, age, BMI, or UI subtype was identified. The authors concluded that the findings of this study showed the effectiveness of a pDTx in reducing UI symptoms among this cohort of users in a real-world setting. Users with stress, mixed, and urgency UI achieved statistically and clinically significant symptom improvement over an 8-week period. These researchers stated that enhanced data collection of relevant demographic and clinical information will further add to the value and applicability of the data. They stated that these findings may inform additional research and development, including efforts to improve in-app data collection and promote adherence. These investigators stated that given the opportunity of pDTx, further investigation designed to present larger patient cohorts is planned, including application of machine learning to expanded data sets. Clinically validated pDTx designed to treat UI in women may help to scale treatment and management of this significant, yet undertreated health condition.

Weinstein et al (2024) noted that there are sparse data regarding the long-term effectiveness of PFMT for the treatment of UI. These researchers examined the impact of an 8-week PFMT program guided by a motion-based intra-vaginal device versus a standard home program over 24 months. Between October 2020 and March 2021, a total of 363 women with stress or M UI were randomized and completed an 8-week PFMT program using a motion-based intra-vaginal device (intervention group) or a home program following written/video instructions (control group). Participants were not asked to continue training after the 8-week program. At 18 and 24 months' follow-up, the UDI-6 and PGI-I were collected. In the original trial, a total of 139 subjects in each arm were needed to detect a 0.3 effect size (alpha = 0.05, power 0.8, 1-tailed t test) in the difference in UDI-6 scores. A total of 231 subjects returned 24-month data. Mean age at 24 months was 51.7 ± 14.5 years, and mean BMI was 31.8 ± 7.4 kg/m2. Mean change in UDI-6 scores from baseline to 24 months was greater in the intervention group than the control group (−21.1 ± 24.5 versus −14.8 ± 19.4, p = 0.04). Reported improvement using PGI-I was greater in the intervention group than in the control group at 24 months (35 % versus 22 %, p = 0.03, OR 1.95 (95 % CI: 1.08 to 3.57). The authors concluded that PFMT guided by a motion-based prescription intra-vaginal device yielded durable and significantly greater UI symptom improvement than a standard home program, even in the absence of continued therapy.

The authors stated that drawbacks of this study included the lack of bladder diaries at these time-points, and the limitations inherent to a remotely conducted study including the absence of a physical examination. These researchers noted that although a more robust, in-person follow-up containing these additional data points may be ideal, ease of access to remotely obtained surveys may have resulted in a larger percentage of participants who engaged in follow-up. They stated that the balance between the ease of access for research participants using remote or app-based data collection tools, and the value of in-person evaluation is an ongoing discussion among researchers.

Vaginal Laser Therapy

Pergialiotis and co-workers (2017) presented available evidence related to vaginal laser therapy as a treatment option for SUI. These investigators searched the Medline (1966 to 2017), Scopus (2004 to 2017), Clinicaltrials.gov (2008 to 2017) and Cochrane Central Register of Controlled Trials (CENTRAL) (1999 to 2017) databases for relevant studies in this field. They included all observational studies (prospective and retrospective, randomized and non-randomized) that reported outcomes on vaginal laser therapy as a therapeutic option for SUI. A total of 13 studies were included that recruited 818 patients who underwent laser therapy for SUI. The methodological quality of most included studies was low, as they were either individual case-control studies, case series or poor-quality cohorts (Oxford Level of Evidence 3b and 4). According to the existing evidence, laser therapy may be a useful, minimally invasive approach for treating SUI. However, the methodological limitations of included studies rendered them prone to significant bias, limiting their scientific integrity. The authors concluded that as the demand for minimally invasive

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approaches for treating SUI increases, it is expected that more patients will seek alternative treatments over current standards (mid-urethral slings). They stated that given the limitations of the existing studies, it appeared that conducting future trials is needed to elucidate this field.

Alsulihem and Corcos (2021) examined the available literature to evaluate the safety, efficacy, and outcomes of lasers in the treatment of female SUI and OAB. PubMed search was performed up to May 2020, including observational and investigational human studies that documented the effects on laser treatment in SUI and OAB. A total of 27 studies, recording subjective or objective measures in SUI or OAB were included. Lasers used included Er:YAG and Fractional CO2 lasers. The overall quality of studies was poor, and 23/27 studies were case-series studies (Level of evidence [LOE] = 4). Er:YAG laser showed a modest reduction in mild SUI cases, with benefits lasting a maximum of 13 to 16 months. Er:YAG laser for OAB showed conflicting results, with a trend to improve OAB symptoms for up to 12 months. Fractional CO2 laser showed an improvement of mild SUI in few studies; however, no long-term data are available. For OAB symptoms, studies showed minimal improvement that was examined in short-term studies. When reported, AEs were insignificant, however, they were not reported systematically. Several drawbacks have been noticed in the current literature of vaginal lasers, including large variation in laser settings and protocols, short-term follow-up, lack of urodynamic evaluation, and appropriate objective measures. The authors concluded that based on the available literature, lasers cannot be recommended as a therapeutic option at this time. These researchers stated that future better-quality studies are needed to document the exact mechanism of action, longevity, safety and its eventual place into the current treatment algorithms of SUI and OAB.

Vibratory Perineal Stimulation

Rodrigues and associates (2018) noted that PFM play an important part in the urinary continence mechanism. Changes in their structure and functionality may lead to a predisposition to pelvic floor dysfunction such as UI. Some techniques for conservative treatment of UI are already well documented. However, new approaches have been found that require scientific proof of their effectiveness, such as vibratory stimulation (VS). These researchers performed a systematic review of studies that investigated the use of perineal VS (PVS) for the treatment of SUI. This study followed the recommendations of the Cochrane Collaboration for systematic reviews. Studies that used PVS for the treatment of female UI were eligible. A total of 56 studies were found, of which ten were duplicates and were excluded. Analysis of the titles and abstracts led to the exclusion of 30 studies, leaving 16 for detailed analysis. Of these, only 3 were included as they fulfilled all the eligibility criteria previously established for the present study. In spite of the heterogeneity of the protocols, all the studies had the goal of assessing the effects of vibration on the PFM, and the stimulation was found to be effective in reducing urinary leakage, improving muscle strength and consequently the patients' QOL. The authors concluded that because of the heterogeneity and the small number of studies, it is not possible to draw a conclusion as to the effectiveness of PVS for the treatment of SUI, and further studies are needed to provide scientific support for its use.

Stem Cell Therapy

Goldman et al (2012) reviewed the current state of research in the use of stem cells (SCs) for SUI and assessed the likelihood of this becoming a relevant treatment option. The peer-reviewed literature consisting of relevant clinical and animal studies on the topic of SUI was surveyed and reviewed. Animal studies have demonstrated the potential utility of SCs in promoting functional recovery of the urethra after simulated childbirth injury. Research in animals suggests similar urethral recovery after injection of bone marrow derived mesenchymal SC secretions as after injection of the SCs themselves. Therefore, whether the improvements result from the injection of the SCs themselves or from their secretion of specific proteins is unclear. Early clinical trials have demonstrated the feasibility and short-term safety of injecting muscle-derived SCs into the urethra to treat SUI. The authors concluded that larger and longer-term clinical trials are needed

Burdzinska and associates (2018) noted that cell therapy constitutes an attractive alternative to treat SUI. Although promising results have been demonstrated in this field, the procedure requires further optimization.

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The most commonly proposed cell types for intra-urethral injections are muscle derived cells (MDCs) and mesenchymal stem/stromal cell (MSCs). These investigators evaluated the effects of MDC-MSC co-transplantation into the urethra. Autologous transplantation of labeled MDCs, bone marrow MSCs or co-transplantation of MDC-MSC were performed in aged multi-parous female goats (n = 6 in each group). The mean number of cells injected per animal was 29.6 × 106(± 4.3 × 106); phosphate-buffered saline (PBS)-injected animals constituted the control group (n = 5). Each animal underwent urethral pressure profile (UPP) measurements before and after the injection procedure. The maximal urethral closure pressure (MUCP) and functional area (FA) of UPPs were calculated. The urethras were collected at the 28th or the 84th day after transplantation. The marker fluorochrome (DID) was visualized and quantified using in-vivo imaging system in whole explants. Myogenic differentiation of the graft was immunohistochemically evaluated. The grafted cells were identified in all urethras collected at day 28 regardless of injected cell type. At this time-point the strongest DID-derived signal (normalized to the number of injected cells) was noted in the co-transplanted group. There was a distinct decline in signal intensity between day 28 and day 84 in all types of transplantation. Both MSCs and MDCs contributed to striated muscle formation if transplanted directly to the external urethral sphincter. In the MSC group those events were rare. If cells were injected into the submucosal region they remained undifferentiated usually packed in clearly distinguishable depots. The mean increase in MUCP after transplantation in comparison to the pre-transplantation state in the MDC, MSC and MDC-MSC groups was 12.3 % (± 11.2 %, not significant (ns)), 8.2 % (± 9.6 %, ns) and 24.1 % (± 3.1 %, p = 0.02), respectively. The mean increase in FA after transplantation in the MDC, MSC and MDC-MSC groups amounted to 17.8 % (± 15.4 %, ns), 15.2 % (± 12.9 %, ns) and 17.8 % (± 2.5%, p = 0.04), respectively. The authors concluded that the findings of this study suggested that MDC-MSC co-transplantation provided a greater chance of improvement in urethral closure than transplantation of each population alone.

Fazeli and colleagues (2019) stated that in recent years, the administration of stem cells has been considered a new therapeutic option for UI. These researchers examined the efficacy of MSC transplantation in the treatment of UI. Combinations of the key words “mesenchymal stem cells”, “MSCs”, “urinary incontinence”, “urethral sphincter” and “involuntary urination” were searched in PubMed and Science Direct databases. Following application of exclusion criteria to the 1,946 papers obtained and review and duplicate articles were removed, 23 articles were considered further. The search was limited to the animal model studies. The data obtained from the evaluation of different studies indicated that the injected MSCs play an important role in the neovascularization and the recovery of muscle cells in UI models through the paracrine process. The authors concluded that the available evidence suggested that further trials are needed to be focused on clinical phase of MSC therapy on the patients affected by UI.

In a systematic review and meta-analysis, Mariotti et al (2023) examined clinical trials on the use of autologous stem cell (SC) injection for the treatment of SUI. These investigators analyzed the effect in terms of UI improvement and continence recovery following SC treatment. They carried out a literature search following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Entry into the analysis was restricted to data collected from prospective studies on humans, including female and male patients with SUI. These researchers performed a cumulative meta-analysis to examine the trend in the effect size across different groups at follow-up. Available data were compared in terms of event rate (ER) for the percentage of pad-free patients. A total of 12 studies were enclosed in the analysis. Sample size of patients with SUI ranged from 5 to 123 cases, mainly female cases. Autologous muscle-derived SC (MDSC) was used in 9 and adipocyte-derived SC (ADSC) in 3 trials. Considering a random effect model, ER of continence recovery was 0.41 (95 % CI: 0.29 to 0.54), with similar results between the ADSC (ER, 0.40; 95 % CI: 0.12 to 0.69] and the MDSC group (ER 0.41; 95 % CI: 0.27 to 0.55) [I2 = 84.69 %; Q 104.69 – p < 0.01] (Test of group differences; p = 0.96). The authors concluded that autologous MDSC or ADSC injection for the treatment of SUI showed to be a safe procedure and a 41 % mean rate of continence recovery was described. These researchers stated that a higher effort should be produced to design better clinical trials, objectively examining either modifications inside the urethral sphincter or long-term functional results in terms of pad test and UI questionnaires.

Artificial Urinary Sphincter (AMS 800)

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The artificial urinary sphincter (AUS) has been shown to be effective for UI due to intrinsic urethral sphincter deficiency (IUSD), and is a useful alternative when conservative interventions have failed. Implantation of an AUS is a commonly used surgical option for the management of male urethral deficiency especially, following prostatectomy. A clinical practice guideline for UI in adults by the Agency for Healthcare Policy and Research (1992) recommends that post-prostatectomy patients wait at least 6 to 12 months before AUS placement, and try behavioral and pharmacologic therapies first. To be considered for AUS implantation, the patient must be motivated and have enough dexterity and ability to operate the device.

The AUS (AS 800, American Medical Systems, Minnetonka, MN) is an externally controlled urethral occlusion device. The transfer of fluid within the device is controlled by a pressure regulating balloon placed extra-peritoneally in the patient's pelvis or abdominal cavity and a control pump placed in a subcutaneous pocket in the scrotum or labium. Squeezing of the pump allows the fluid within the closed-loop system to be transferred from the cuff to the balloon. It takes a few minutes before the cuff re-inflates automatically to the preset level, thus allowing the urethra to remain opened for voiding. The AS 800 has the option of primary deactivation. Primary deactivation is performed to limit the cuff compression during the early post-operative healing period, thus minimizing the risk of cuff erosion and infection. In the male, the preferred site of cuff placement is around the bladder neck because erosion is less likely. When implantation of the device at the bladder neck is precluded by previous surgery, the cuff is placed around the bulbous urethra. In the female, the AUS cuff is placed around the bladder neck. The device is implanted abdominally or vaginally.

Aetna's selection criteria for AUS are consistent with the AHCPR clinical practice guidelines for urinary incontinence in adults. Potential candidates for AUS implantation should be evaluated preoperatively to exclude severe detrusor instability as well as to ensure adequate bladder stability and compliance prior to implantation of the AUS. Appropriate candidates for implantation of an AUS must have adequate motivation and sufficient manual dexterity to operate the device. Post-prostatectomy patients should wait 6 to 12 months and attempt behavioral and pharmacologic therapies first. AUS may also be indicated in: patients with epispadias-exstrophy in whom bladder neck reconstruction has failed; women in whom behavioral or pharmacologic therapies, or other surgical options have failed; and children with intractable UI who are refractory to pharmacologic therapies or unsuitable for other types of operation.

Peyronnet and associates (2019a) performed a systematic review of studies reporting the outcomes of AMS-800 AUS implantation in women with SUI resulting from intrinsic sphincter deficiency (ISD). A systematic literature search of the Medline and Embase databases was performed in June 2018 in accordance with the PRISMA statement; no time limit was used. Study selection and data extraction were performed by 2 independent reviewers. Of 886 records screened, 17 were included. All were retrospective or prospective non-comparative case series; 1 study reported on vaginal AUS implantation, 11 on open AUS implantation, 2 on laparoscopic AUS implantation, 2 on robot-assisted AUS implantation, and 1 compared open and robot-assisted implantations. The vast majority of patients had undergone at least 1 anti-incontinence surgical procedure prior to AUS implantation (69.1 to 100 %). The intra-operative bladder neck injury rates ranged from 0 % to 43.8 % and the intra-operative vaginal injury rates ranged from 0 % to 25 %. After mean follow-up periods ranging from 5 to 204 months, the complete continence rates ranged from 61.1 % to 100 %. The rates of explantation, erosion and mechanical failure varied from 0 % to 45.3 %, 0 % to 22.2 % and 0 % to 44.1 %, respectively. The authors concluded that AMS-800 AUS could provide excellent functional outcomes in women with SUI resulting from ISD but at the cost of a relatively high morbidity. These researchers stated that high level of evidence studies are needed to help better define the role of AUS in SUI armamentarium in women.

Reus and colleagues (2020) stated that the use of the AUS for non-neurogenic severe SUI in women due to sphincter deficiency is either not specifically registered and/or reimbursed in some countries worldwide, as opposed to severe SUI in men, in whom it is considered the gold standard. With waning popularity of synthetic mid-urethral slings for the treatment of SUI, evidence-based assessment of AUS performance and safety is mandatory for patient counselling. These investigators carried out a systematic review of studies evaluating short- to long-term AUS performance and safety outcomes in non-neurogenic women with severe SUI. PubMed/Medline, Embase, and the Cochrane Central Register of Controlled Trials were searched, from 1987 to 2018, without language restriction. Included studies had to report outcomes after AUS implantation in at least 5 women with non-neurogenic SUI, after a minimum follow-up of 6 months. A total of 12 articles

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collecting data from 886 patients were identified, no study being randomized or prospective. The reported zero pad rates ranged from 42 % to 86 %, revision rates from 6 % to 44 %, and mechanical failure rates from 2 % to 41 %. Procedure serious AE rates ranged from 2 % to 54 % and rates of serious adverse device effects such as explantation ranged from 2 % to 27 %. The authors concluded that the level of evidence supporting the use of an AUS for non-neurogenic SUI in women is very low; AUS outcome assessments necessitate well-designed randomized trials, in accordance with current evidence-based medicine requirements.

Peyronnet and co-workers (2019b) stated that widespread adoption of the AMS-800 AUS in female patients has been hampered by the surgical morbidity of its implantation through an open approach. These researchers described a standardized technique of robotic bladder neck AUS implantation in female patients, and reported the peri-operative and functional outcomes obtained by multiple surgeons with this technique. They retrospectively reviewed the charts of all female patients who underwent robotic AUS implantation for UI due to ISD between March 2012 and March 2017 in 5 institutions. Most of the 10 surgeons involved were not highly experienced in female AUS implantation and/or in robotic surgery. The AUS was implanted at the bladder neck through a trans-peritoneal robotic approach. The finger placed by the assistant surgeon in the vagina is paramount to expose the vesico-vaginal space and guided the robotic surgeon throughout the bladder neck dissection. The primary end-point was the incontinence categorized as complete continence(i.e., no pads used), improved incontinence, or unchanged incontinence. A total of 49 women underwent a robotic AUS implantation. There were 8 intra-operative complications (16.3 %): 5 bladder neck injuries and 3 vaginal injuries; 9 patients experienced post-operative complications (18.3 %), but only 2 were Clavien greater than or equal to 3 (4.1 %). After a median follow-up of 18.5 months, 1 explantation (vaginal erosion, 2.1 %) and 3 revisions (1 mechanical and 2 non-mechanical failure, 6.1 %) were needed. At last follow-up, 40 patients were fully continent (81.6 %), 6 had improved incontinence (12.2 %), and 3 had unchanged incontinence (6.1 %). The authors concluded that in this first multi-center series of robot-assisted AUS implantation, this technique appeared feasible, safe, and reproducible with peri-operative and functional outcomes in the early learning curve not inferior to those reported in large series of open AUS implantation from tertiary referral centers. These investigators stated that the findings of this study suggested that this technique is feasible and reproducible by surgeons with various levels of surgical expertise; however, further data are needed to confirm the findings of the present report.

The authors stated that this study had several drawbacks. First, it had numerous biases inherent to its retrospective design. The lack of a control group did not allow a proper assessment of the value of robotic female AUS implantation compared with the open or laparoscopic approaches or other therapeutic options (e.g., pubovaginal sling, bulking agents, etc.). The relatively small sample size (n = 49) was another drawback of the present series. Last, opening of the bladder dome performed in challenging cases introduced significant, although isolated, heterogeneity in the technique used, but was part of a “patient's first” policy and had certainly contributed to the safe development of this new surgical technique. Opening of the bladder dome was felt less and less necessary with increasing experience.

Screening for Urinary Incontinence in Women

In a systematic review, Nelson and colleagues (2018) examined if screening for UI in women not previously diagnosed would improve outcomes (symptoms, QOL, and function) and assessed the accuracy of screening methods and potential harms of screening. English-language searches of Ovid Medline, Cochrane Central Register of Controlled Trials, and Cochrane Database of Systematic Reviews (January 1, 1996 to March 30, 2018); ClinicalTrials.gov (April 2018), and reference lists of studies and reviews were carried out. Randomized trials, cohort studies, systematic reviews of studies that enrolled non-pregnant women without previously diagnosed UI and compared clinical outcomes and adverse effects between women who were and were not screened, and diagnostic accuracy studies that reported performance measures of screening tests. No studies evaluated the overall effectiveness or harms of screening. A total of 17 studies evaluated the diagnostic accuracy of 18 screening questionnaires against a clinical diagnosis or results of diagnostic tests. Of these, 14 poor-quality studies were based in referral clinics, enrolled only symptomatic women, or had other limitations; 1 good-quality and 2 fair-quality studies (evaluating 4 methods) enrolled women not recruited on the basis of symptoms. Areas under the receiver-operating characteristic curve for stress, urge,

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and any type of incontinence in these studies were 0.79, 0.88, and 0.88 for the Michigan Incontinence Symptom Index; 0.85, 0.83, and 0.87 for the Bladder Control Self-Assessment Questionnaire; and 0.68, 0.82, and 0.75 for the Overactive Bladder Awareness Tool. The Incontinence Screening Questionnaire had a sensitivity of 66 % and specificity of 80 % for any type of incontinence. The authors concluded that available evidence is insufficient on the overall effectiveness and harms of screening for UI in women; and limited evidence in general populations suggested fairly high accuracy for some screening methods. The main drawbacks of this review were studies enrolled few participants, often from symptomatic referral populations; used various reference standards; and infrequently reported CIs.

Despite the lack of studies determining the benefits and harms of UI screening, the Women's Preventive Services Initiative (WPSI) recommended that doctors screen women of all ages, including adolescents, for UI yearly by using a questionnaire. The WPSI recommended referring women with UI for further evaluation if it affects their activities and QOL. These recommendations were based on indirect evidence that UI is common, treatment may be effective, and the harms of screening are unlikely to be serious. The recommendations might change if studies directly evaluating the benefits and harms of screening for UI become available. There are no data to support that the correct frequency of screening is yearly (no authors listed, 2018).

Bariatric Surgery for the Treatment of Urinary Incontinence in Obese Women

Shimonov and colleagues (2017) examined the effect of bariatric surgery on female pelvic floor disorders (PFDs). A total of 80 consecutive obese women who underwent a laparoscopic sleeve gastrectomy were prospectively enrolled. Four validated questionnaires (ICIQ-UI, BFLUTS-SF, PFDI-20, PISQ-12) were used to evaluate pelvic floor symptoms before and 6 months after surgery. Outcome results were analyzed according to the presence of pre-operative UI, defined as a positive answer to the question "how often do you leak urine?" on the ICIQ-UI questionnaire. A total of 77 women (aged 41.3 ± 11.5 years; parity 1.9 ± 1.6) completed all pre- and post-operative questionnaires. Mean body mass index (BMI) before and 6 months after surgery was 42 ± 4.7 and 33 ± 4.7, respectively. Pre-operatively, 29 (37.7 %) women (mean age of 45.6 ± 11, mean BMI 42.3 ± 5.2) had UI, 17 (59 %) of whom had SUI. Surgically induced weight loss was associated with statistically significant improvement in UI and filling symptoms, pelvic organ prolapse and colorectal-anal scores, and condition-specific sexual function and QOL parameters. Specifically, the total score of the ICIQ-UI questionnaire decreased from 9.28 ± 3.6 pre-operatively to 2.9 ± 3.8 post-operatively ($p < 0.001$), and the urinary score of the PFDI-20 questionnaire decreased from 31.4 ± 17.9 pre-operatively to 9.3 ± 12.3 post-operatively ($p < 0.001$). Furthermore, 15 (51.7 %) women reported complete resolution of UI following weight loss. The authors concluded that surgically induced weight loss resulted in resolution of UI in up to 52 % of pre-operatively incontinent women and subsequent improvement in other pelvic floor symptoms. Moreover, they stated that larger studies with longer follow-up are required to examine the possible impacts of bariatric surgery on various aspects of pelvic floor function.

In a meta-analysis, Lian and associates (2017) evaluated the effects of bariatric surgery on PFD in obese women. These researchers carried out a systematic search of PubMed, Cochrane Library, CNKI and CBM databases up to October 2016, and studies reporting pre-operative and post-operative outcomes in obese women undergoing bariatric surgery were included. The Pelvic Floor Distress Inventory (PFDI-20), the Pelvic Floor Incontinence Questionnaire (PFIQ-7), the Pelvic Organ Prolapse/Urinary Incontinence Sexual Questionnaire, Female Sexual Function Index and the International Consultation on Incontinence Questionnaire-Urinary Incontinence short form score were used for evaluating pelvic floor dysfunction after bariatric surgery. A total of 11 cohort studies were finally included. Pooled results revealed that bariatric surgery was associated with a significant improvement in PFD for obese women on the whole [PFDI-20: standard mean difference [SMD] = 0.89, 95 % CI: 0.44 to 1.34), $p < 0.001$; PFIQ-7: SMD = 1.23, 95 % CI: 0.17 to 2.29), $p = 0.023$]. In the subscale analysis, there was significant improvement in UI and pelvic organ prolapse. However, no significant improvement was found in fecal incontinence and sexual function. The authors concluded that bariatric surgery is associated with significant improvement in UI, and has a benefit on pelvic organ prolapse for obese women. However, there is no significant improvement in fecal incontinence and sexual function. These investigators stated that further multi-center, large-scale and longer-term RCTs are needed to confirm these findings.

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In a meta-analysis, Zhang and colleagues (2020) examined the effectiveness of bariatric surgery in obese women with UI. Searches of PubMed, the Cochrane Library, and Embase databases were performed using "weight loss surgery/bariatric surgery/gastric bypass surgery" and "incontinentia urinae / uracratia / urinary incontinence / uroclepsia" in the title/abstract before January 2018. Then, meta-analysis was analyzed by Review Manager 5.3 (Cochrane Collaboration, Oxford, United Kingdom). The SMD and OR were used to describe results of continuous variables and dichotomous variables, respectively. Pooled data showed that bariatric surgery reduced the incidence of UI in obese women at the follow-up of 6 months (OR, 3.27; 95 % CI: 2.55 to 4.21; $p < 0.00001$) and 12 months (OR, 4.04; 95 % CI: 2.62 to 6.22; $p < 0.00001$) and significantly reduced the BMI at 6 months (SMD, 1.86; 95 % CI: 1.19 to 2.53; $p < 0.00001$) and 12 months (SMD, 2.04; 95 % CI: 1.44 to 2.64; $p < 0.00001$). In addition, bariatric surgery could also significantly increase the QOL (SMD, 0.53; 95 % CI: 0.27 to 0.80; $p < 0.00001$) and improve the function of pelvic floor disorders (SMD, 0.55; 95 % CI: 0.38 to 0.72; $p < 0.00001$) based on QOL questionnaires and Pelvic Floor Distress Inventory 20, respectively. The authors concluded that this meta-analysis demonstrated that bariatric surgery is an effective choice for obese women with UI; however, more RCTs are needed to confirm these findings.

Furthermore, UpToDate reviews on “Treatment of urinary incontinence in women” (Lukacz, 2018a) and “Treatment of urgency incontinence/overactive bladder in women” (Lukacz, 2018b) do not mention bariatric surgery as a therapeutic option.

FemiLift (CO2 Laser) for the Treatment of Urinary Incontinence in Women

In an observational study, Pitsouni et al (2016) examined the effect of the micro-ablative fractional carbon dioxide (CO2)-laser therapy on vaginal pathophysiology and the symptoms of the genitourinary syndrome of menopause (GSM). Post-menopausal women with moderate-to-severe symptoms of GSM underwent 3 sessions of CO2-laser therapy at monthly intervals. Participants were evaluated at baseline and 4 weeks after the last treatment. The primary outcomes were Vaginal Maturation Value (VMV) and Vaginal Health Index Score (VHIS). Secondary outcomes included symptoms of GSM, Female Sexual Function Index (FSFI), International Consultation on Incontinence Questionnaire of Female Urinary Tract Symptoms (ICIQ-FLUTS) and Urinary Incontinence Short Form (ICIQ-UI SF), Urogenital Distress Inventory (UDI-6) and King's Health Questionnaire (KHQ). A total of 53 post-menopausal women completed this study; VMV, VHIS and FSFI increased significantly. Dyspareunia, dryness, burning, itching, dysuria, frequency, urgency, urgency incontinence, stress incontinence and scores on the ICIQ-FLUTS, ICIQ-UI SF, UDI-6 and KHQ decreased significantly. Factors predicting for which women the CO2-laser therapy was more effective were not identified. The authors concluded that the findings of this study suggested that intravaginal CO2-laser therapy for post-menopausal women with clinical signs and symptoms of GSM may be effective in improving both vaginal pathophysiology and reported symptoms. This was an observational study, with a relatively small ($n = 53$) sample size and short-term follow-up (4 weeks). These preliminary findings need to be validated in well-designed studies.

Gonzalez et al (2018) examined the long-term effect of thermo-ablative fractional CO2 laser (TACO2L) as an alternative treatment for early stages of stress urinary incontinence (SUI) in post-menopausal women with genitourinary syndrome of menopause. A total of 161 post-menopausal patients (age of 53.38 ± 5.1 years, range of 45 to 65 years) with a clinical diagnosis of mild SUI were prospectively enrolled in the study. Patients received 1 treatment with TACO2L every 30 to 45 days, each treatment comprising 4 sessions, followed in all patients by a yearly treatment session at 12, 24 and 36 months. SUI was evaluated using the International Continence Society 1-hour pad test and the ICIQ-UI SF before and after TACO2L treatment. TACO2L treatment was associated with a significant improvement in ICIQ-UI SF scores and 1-hour pad weight test at 12 months (both $p < 0.001$), 24 months (both $p < 0.001$) and 36 months (both $p < 0.001$). Improvements were maintained for up to 36 months without the need for any further intervention. The results were confirmed by significant histological changes related to trophic restoration of the vagina, responsible for extrinsic and intrinsic mechanisms involved in urinary continence. The authors concluded that these findings suggested that TACO2L was a safe and efficient novel treatment strategy in patients with mild SUI. These researchers stated that further investigation to confirm the long-term results presented here is still needed. These researchers stated that this was a prospective, non-randomized study of an

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observational nature with no control group. Since all patients enrolled were relatively young (45 to 65 years), the results of this particular study could not be translated into older populations.

Lin and associates (2018) noted that female pelvic floor disorders, including female SUI or sexual dysfunction are notorious for affecting the quality of life (QOL) in women. It has been reported that laser therapy might result in collagen re-modeling and improvement in tissue firmness. These investigators evaluated the short-term outcome of female pelvic floor disorders treated by laser therapy. Women with self-reported symptoms of female pelvic floor disorders (limited to SUI and sexual dysfunction) were included in the study. Participants were treated with the Er:YAG laser or the fractional micro-ablative CO2 laser. The therapeutic effect was focused on SUI symptoms and sexual dysfunction. There were 31 women who underwent laser treatment, including 21 patients treated with Erbium:YAG laser and 10 treated with CO2 laser. In the Erbium:YAG laser group, ICIQ- SF scores were dropped from 8.25 ± 5.66 to 5.00 ± 3.99 (p = 0.007); and in the CO2 laser group, scores were dropped from 11.11 ± 6.85 to 6.44 ± 4.25 (p = 0.035), contributing to the drop of ICI-Q-SF scores from 9.14 ± 6.08 to 5.45 ± 4.05 for all enrolled patients (p = 0.001). However, objective measure using pad test did not show a statistically significant difference between before and after treatment (from 3.20 ± 5.84 g to 1.54 ± 3.18 g, p = 0.224). Sexual dysfunction was improved in 13 patients (44.83 %), but Female Sexual Function Index (FSFI) scores were not different before and after laser treatment (44.22 ± 23.36 versus 44.09 ± 24.51, p = 0.389). The authors concluded that laser therapy either by Erbium:YAG laser or CO2 laser appeared to be useful for female pelvic floor disorders, especially on improvement of SUI symptoms; however, the effectiveness needs further confirmation in large prospective and randomized studies. This was a small (n = 10 in the CO2 laser group), non-randomization study with short-term follow-up (2 months).

Furthermore, an UpToDate review on “Treatment of urinary incontinence in women” (Lukacz, 2019) states that “There is insufficient evidence to support the use of vaginal laser (CO2 or erbium) for the treatment of urinary incontinence in the setting of genitourinary syndrome of menopause”.

Hafidh et al (2023) stated that CO2 laser therapy is an emerging treatment for women with SUI. In a systematic review and meta-analysis, these investigators examined the effectiveness of CO2 laser therapy for management of SUI-related symptoms in women. A total of 4 databases were screened until January 2023. All effectiveness continuous endpoints were examined via subtraction of the post-treatment from pre-treatment values; data were summarized as mean difference (MD) with 95 % CI using the random-effects model. A total of 15 studies with 700 patients were analyzed. CO2 laser therapy significantly decreased the 1-hour pad weights at 3 months (n = 5 studies, MD = -3.656 g, 95 % CI: -5.198 to -2.113, p < 0.001), 6 months (n = 6 studies, MD = -6.583 g, 95 % CI: -11.158 to -2.008, p = 0.005), and 12 months (n = 6 studies, MD = -3.726 g, 95 % CI: -6.347 to -1.106, p = 0.005). Moreover, CO2 laser therapy significantly decreased the ICIQ-Urinary Incontinence Short Form Scores at 3 months (n = 10 studies, MD = -4.805, 95 % CI: -5.985 to -3.626, p < 0.001), and 12-months (n = 6 studies, MD = -3.726, 95 % CI: -6.347 to -1.106, p = 0.005). Furthermore, CO2 laser therapy significantly decreased the PFIQ scores at 6 months (n = 2 studies, MD = -11.268, 95 % CI: -18.671 to -3.865, p = 0.002), and 12 months (n = 2 studies, MD = -10.624, 95 % CI: -18.145 to -3.103, p = 0.006). In addition, CO2 laser therapy significantly decreased the UDI-6 scores at 3 months (n = 2 studies, MD = -21.997, 95 % CI: -32.294 to -11.699, p < 0.001), but not at 6 months (n = 3 studies, MD = -3.034, 95 % CI: -7.357 to 1.259, p = 0.169). Finally, CO2 laser therapy significantly increased the VHIS at 6 months (n = 2 studies, MD = 2.826, 95 % CI: 0.013 to 5.638, p = 0.047), and 12 months (MD = 1.553, 95 % CI: 0.173 to 2.934, p = 0.027). The authors concluded that CO2 laser therapy improved the SUI-related symptoms in women. Moreover, these researchers stated that to obtain solid conclusions, well-designed studies (large-sized, randomized, double-blinded, controlled trials) with standardized settings, consistent therapeutic protocols, and long-term follow-up periods are needed. Furthermore, outcome evaluations should be carried out in a uniform manner to ensure the reliability of the results.

The authors stated that this study had several drawbacks. First, the small number of studies and corresponding sample size. Indeed, there was a reduced number of studies that have examined the safety and effectiveness of CO2 laser treatment (specifically for SUI) compared to those examining its application for genitourinary syndrome of menopause (GSM). In certain studies, urinary symptoms, including SUI, were considered as components of the broader symptom profile associated with GSM, and these researchers

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decided to exclude them to maintain a uniform and strict inclusion criterion. Additionally, some outcomes did not report the long-term effectiveness of the intervention. Second, these investigators also observed heterogeneity across several endpoints, which could have impacted the validity of the meta-analyzed results. Moreover, some studies were single-arm studies without randomization, which could have potentially subjected the results to selection bias. Third, the lack of uniformity of CO2 laser application. Fourth, some studies did not provide adequate information on the technique and related parameters. Fifth, the lack of retrospectively recording the research protocol in the International Prospective Register of Systematic Reviews (PROSPERO); thus, potential bias could not be certainly excluded.

Gene Testing for Stress Urinary Incontinence

In a systematic review, Isali and colleagues (2020) provided insight into the genetic pathogenesis of SUI by gathering and synthesizing the available data from studies evaluating differential gene expression in SUI patients and identified possible novel therapeutic targets and leads. A systematic literature search was conducted through September 2017 for the concepts of genetics and SUI. Gene net-working connections and gene-set functional analyses of the identified genes as differentially expressed in SUI were performed using GeneMANIA software. Of 3,019 studies, 4 were included in the final analysis. A total of 13 genes were identified as being differentially expressed in SUI patients; 11 genes were over-expressed: skin-derived antileukoproteinase (SKALP/elafin), collagen type XVII alpha 1 chain (COL17A1), plakophilin 1 (PKP1), keratin 16 (KRT16), decorin (DCN), biglycan (BGN), protein bicaudal D homolog 2 (BICD2), growth factor receptor-bound protein 2 (GRB2), signal transducer and activator of transcription 3 (STAT3), apolipoprotein E (APOE), and Golgi SNAP receptor complex member 1 (GOSR1), while 2 genes were under-expressed: fibromodulin (FMOD) and glucocerebrosidase (GBA). GeneMANIA revealed that these genes are involved in intermediate filament cytoskeleton and extra-cellular matrix organization. The authors concluded that many genes are involved in the pathogenesis of SUI. Furthermore, whole-genome studies are needed to identify these genetic connections. These researchers stated that this study laid the groundwork for future research and the development of novel therapies and SUI biomarkers in clinical practice.

An UpToDate review on “Evaluation of women with urinary incontinence” (Lukacz, 2019) states that “The risk of urinary incontinence, particularly urgency incontinence, may be higher in patients with a family history. One study found that the risk of incontinence was increased for both daughters (relative risk [RR] 1.3, 95 % CI 1.2-1.4) and sisters (RR 1.6, 95 % CI 1.3-1.9) of women with incontinence. Twin studies attribute a 35 to 55 % genetic contribution to urgency incontinence/overactive bladder but only 1.5 % for stress incontinence”.

Furthermore, an UpToDate review on “Urinary incontinence in men” (Clemans, 2019) does not mention genetic testing as a management option.

The Adjustable Trans-Obturator Male System for the Treatment of Stress Urinary Incontinence

In a systematic review and meta-analysis, Esquinas and Angulo (2019) examined the effectiveness of the Adjustable Trans-Obturator Male System (ATOMS) device to treat male SUI. Two independent reviewers identified studies eligible for a systematic review and meta-analysis of various sources written in English, German and Spanish, using the databases PubMed, Embase and Web of Science. They excluded studies on female incontinence. These researchers employed the DerSimonian and Laird method for defining heterogeneity, calculating the grouped SMD. The primary objective of this review was the evaluation of clinical efficacy based on the achievement of dryness following device adjustment, defined as use of no pad or 1 safety pad per day (PPD). The secondary objective was focused on analyzing improvement of incontinence with the device. Magnitude of effect was calculated by analyzing decrease in PPD and/or in 24-hour pad test. Number and severity of complications according to Clavien-Dindo classification were also reviewed. The pooled data of 1,393 patients from 20 studies (13 retrospective and 7 prospective) showed that treatment with ATOMS resulted in a mean 67 % dryness rate and 90 % improvement after adjustment. Mean total number of system fillings per patient was 2.4. Mean pad count and 24-hour pad test decrease were - 4.14 PPD and - 443 cc, respectively. There was significant heterogeneity of the sample analyzed,

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mainly based on variable baseline severity of incontinence, proportion of patients treated with irradiation and different generation devices. Proportion of irradiated patients affected dryness rate ($p = 0.0014$), together with baseline severity of incontinence ($p = 0.0035$) and different generation device used ($p < 0.0001$). Standardized mean follow-up was 20.9 months, with complications occurring in 16.4 % (major complications 3.0%) and explantation in 5.75 %. No randomized study has been developed so far to compare ATOMS to other devices for treating male SUI. The authors concluded that despite the evidence being exclusively based on descriptive studies and limited follow-up, ATOMS has proven to be a safe alternative to treat different degrees of male SUI following prostate surgery; better results were observed for patients with less than 6 PPD before implantation, non-irradiated patients and use of 3rd-generation device with silicone-covered pre-attached scrotal port. These researchers stated that ATOMS appeared to be a safe and effective procedure, with pooled data showing high objective effectiveness and low rate of complications in the short- and medium-term. They noted that it would be of great interest to develop comparative prospective studies in the future among ATOMS and other devices, not only regarding effectiveness but also including patient-reported outcomes.

The authors stated that the main drawbacks of this systematic review and meta-analysis lied in the scant level of evidence provided by the design and nature of the non-controlled, and mainly retrospective, studies available, and in their relatively short follow-up. The variable nature and severity of SUI and the different proportion of patients receiving radiation likely explained the high heterogeneity observed. Combining the results of individual studies increased the total number of participants and more participants imply more statistical power. However, combining studies with differences among participants could also reduce statistical power and make real effects more difficult to identify.

In a systematic review and meta-analysis, Angulo and colleagues (2019) examined the safety and efficacy of ATOMS compared to ProACT for male SUI according to literature findings; studies on female or neurogenic incontinence were excluded. Differences between ATOMS and ProACT in primary objective: dryness status (no-pad or 1 safety pad/day) after initial device adjustment, and in secondary objectives: improvement, satisfaction, complications and device durability, were estimated using random-effect model. Statistical heterogeneity among studies included in the meta-analysis was assessed using tau2, Higgins´s I2 statistics and Cochran´s Q test. Combined data of 41 observational studies with 3,059 patients showed higher dryness (68 % versus 55 %; $p = 0.01$) and improvement (91 % versus 80 %; $p = 0.007$) rate for ATOMS than ProACT. Mean pad-count (-4 versus -2.5 pads/day; $p = 0.005$) and pad-test decrease (-425.7 versus -211.4 cc; $p < 0.0001$) were also significantly lower. Satisfaction was higher for ATOMS (87 % versus 56 %; $p = 0.002$) and explant rate was higher for ProACT (5 % versus 24 %; $p < 0.0001$). Complication rate for ProACT was also higher, but not statistically significant (17 % versus 26 %; $p = 0.07$). Mean follow-up was 25.7 months, lower for ATOMS than ProACT (20.8 months versus 30.6 months; $p = 0.02$). The rate of working devices favored ATOMS at 1-year (92 % versus 76 %; $p < 0.0001$), 2-years (85 versus 61 %; $p = 0.0008$) and 3-years (81 % versus 58 %; $p = 0.0001$). Significant heterogeneity was evidenced, due to variable incontinence severity baseline, difficulties for a common reporting of complications, different number of adjustments and time of follow-up and absence of randomized studies. The authors concluded that despite the limitations that studies available were exclusively descriptive and the follow-up was limited, literature findings confirmed ATOMS was more effective, with higher patient satisfaction and better durability than ProACT to treat male SUI.

The authors stated that the main drawbacks of this meta-analysis included the short-term follow-up (mean of 25.7 months), especially in the ATOMS-arm, and in the very high heterogeneity observed between studies; probably reflecting a variable severity of sphincteric damage included and the absence of RCTs. Furthermore, the criteria to report complications appeared variable between the studies analyzed. The drawbacks highlighted were in consonance with the publication bias identified according to Egger's linear regression. It should also be noted that the ATOMS studies had shorter follow-up than the ProACT studies (20.8 months versus 30.6 months).

Furthermore, an UpToDate review on "Urinary incontinence in men" (Clemens, 2019) does not mention Adjustable Trans-obturator Male System as a therapeutic option.

Magnetic Stimulation for Women with Stress Urinary Incontinence

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In a meta-analysis of studies with short-term follow-up, Peng and colleagues (2019) examined the efficacy of magnetic stimulation (MS) in female patients with SUI by investigating peer-reviewed RCTs. PubMed, Embase, and Cochrane library were retrieved for any peer-reviewed original articles in English. Databases were searched up to July 2018. Included studies examined effects of MS on SUI. The data were analyzed by review manager 5.3 software (Cochrane Collaboration, Oxford, UK). A total of 4 studies involving 232 patients were identified and included in present meta-analysis. Compared with the sham stimulation, the MS group had statistically significantly fewer leaks/3 days (mean difference [MD] =-1.42; 95 % CI: -2.42 to -0.59; p=0.007), less urine loss on pad test (g/24hours) (MD=-4.99; 95 % CI: -8.46 to -1.53; p=0.005), higher QOL scores (MD=0.42; 95 % CI: 0.02 to 0.82; p=0.009), and lower ICIQ scores (MD=-4.60; 95 % CI: -5.02 to -4.19; p<0.001). MS presented higher cure or improvement rate, with a statistically significant improvement in UDI and IIQ-SF scores compared to sham stimulation. No MS-related AEs were reported in study. The authors concluded that MS led to an improvement in SUI without any reported safety concerns and an improvement in patient QOL; however, the long-term outcome of this technique remains unclear and is the subject of ongoing research.

The authors stated that the drawbacks of this present study were: First, its small sample size and insufficient statistical power. Second, the stimulation parameters and duration of the studies were not consistent, which made these investigators doubted whether meta-analysis could be performed. However, the results based on RCTs were excellent despite inconsistent variables. Third, when analyzing the data of Pad test, a hug heterogeneity that most likely caused by incomplete experimental design was recorded if these researchers added the study by Manganotti et al (2007) to the analysis. Therefore, this study data were finally excluded by performing sensitivity analysis. These researchers stated that further well-designed RCTs with a long-term follow-up with a large sample size are needed.

Lukanovic et al (2021) noted that UI is becoming an increasingly common health problem; and its treatment can be conservative or surgical. In a systematic review, these researchers examined the effectiveness of MS in the treatment of UI. They compared results with findings from their clinical study – a prospective, non-randomized study conducted at the Ljubljana University Medical Center's Gynecology Division. It included 82 randomly selected female patients, irrespective of their UI type. The success rate of using MS in treating UI was based on standardized ICIQ-UI SF questionnaires. Subjects completed 10 therapy sessions on MS, and follow-up was carried out 3 months after the last therapy session. UI improved following treatment with MS. The ICIQ-UI SF score improved in patients regardless of the type of UI; however, the greatest decrease in post-treatment assessment ICIQ-UI SF scores was observed in subjects with SUI. The authors concluded that MS is a successful non-invasive conservative method for the treatment of patients with UI. Moreover, these researchers stated that future studies are needed, all of which should include a large sample size, a control group, an optimal research protocol, pre-treatment analyses, standardization, and longer follow-ups.

The authors stated that this review had several drawbacks. First, and perhaps most important, the sample was non-randomized. Although this non-probability sampling method was the most applicable and widely used method in clinical research, the sampling method did not guarantee equal chances for each subject in the target, it was less representative of the target population, and it decreased the ability to draw completely impartial conclusions regarding the effectiveness of MS. Second, the power of this study was low, as well as the power of most studies in this systematic review. An ideal study is one that has high power, which means that the study has a high chance of detecting a difference between groups if it exists, and consequently, if the study demonstrates no difference between groups, the researcher can be reasonably confident in concluding that none exists. According to the literature review, the ideal power for any study is considered to be 80 %. For this study to achieve a significance level of 95 % and a power of 80 %, the sample size should equal 189; in this study, the sample size of 76 accounted for a power of 57 %. This meant that this study had low power, and studies with lower power increase the likelihood that a statistically significant finding represents a false positive result. These investigators stated that future studies may address all of the above drawbacks and test the robustness of these findings on an extended environment. One drawback could also be that this study included only the ICIQ-UI SF as the tool for measuring the effectiveness of MS in the treatment of UI; however, this questionnaire was the only available validated questionnaire in Slovenian. These researchers were convinced that patient-reported outcomes were the most appropriate when describing treatment success or failure. As these investigators also concluded in the systematic review, they were aware that outcome measurements to generate comparable data should be standardized.

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The Neocontrol™System

In a systematic review and meta-analysis, Ho and colleagues (2020) examined the effectiveness of extracorporeal magnetic stimulation for treatment of SUI. Data sources included 4 electronic databases from inception to May 18, 2019. Two authors independently carried out the search, evaluated the methodological quality, and extracted data. The final studies included in the analysis were selected after reaching consensus with the 3rd author. A total of 20 studies were included in the systematic review and 12 of these in the meta-analysis. Quality assessment indicated that only 8 of 17 RCTs had low risk in overall risk of bias, whereas all controlled trials had serious risk of bias. The weighted mean effect size of magnetic stimulation on QOL, number of leakages, pad test outcomes, and number of incontinence events was 1.045 (95 % CI: 0.409 to 1.681), -0.411 (95 % CI: 0.178 to 0.643), -0.290 (95 % CI: 0.025 to 0.556), and -0.747 (95 % CI: -1.122 to -0.372), respectively. Subgroup analysis revealed a significant difference in the type of QOL measurement used. Sensitivity analyses revealed that a high degree of heterogeneity persisted even after omitting studies individually. The authors concluded that extracorporeal magnetic stimulation may be effective in treating UI and improving QOL without major safety concerns; however, because of a high degree of heterogeneity among studies, inferences from the results must be made with caution. These investigators encouraged researchers to conduct further qualitative and quantitative studies to develop consistent content and dosage for the intervention.

In a systematic review, Stroje et al (2023) examined the available evidence on the effectiveness of ExMI in the treatment of female patients with UI. These investigators carried out an analysis using the following electronic databases: Medline, PubMed, ScienceDirect, and the Cochrane Library (data published between 2008 and 2023). Searches of the afore-mentioned databases were performed in April 2023. Only RCTs in English studies were eligible for inclusion into this review, and evaluated with the Downs and Black checklist. A total of 11 studies met the inclusion criteria. Among these, 2 trials examined the use of ExMI and PMFT (pelvic floor muscle training); 3 studies compared active ExMI versus sham ExMI; and 4 studies examined only ExMI. Moreover, there was no control group in 2 of these studies. One study compared the effects of Kegel exercises with ExMI, while another study compared electro-stimulation with ExMI. The reviewed studies exhibited significant differences in interventions, populations, and outcome measures. The authors concluded that ExMI has shown promise as a treatment for female UI. These researchers stated that whether used alone or as a component of combination therapy, ExMI has the potential to enhance patients' QOL without significant safety concerns. Moreover, these researchers stated that more in-depth investigations are needed to examine the long-term effectiveness of this promising treatment for UI.

Moxibustion

Li and colleagues (2021) noted that UI is a frequently identified complication among stroke survivors. Moxibustion is commonly used to treat post-stroke UI in Asian countries. In a systematic review and meta-analysis, these researchers examined the evidence of using moxibustion for post-stroke UI management. A total of 12 databases were searched to identify RCTs using moxibustion to improve post-stroke UI management; 4 Chinese journals were also manually screened for potentially eligible articles. A total of 10 studies with 719 subjects and 1 completed trial without published results were included. Compared with "routine methods of treatment and/or care", the meta-analyses revealed that moxibustion had superior effects in improving UI symptoms and alleviating the severity of UI. The authors concluded that this systematic review identified preliminary research evidence that moxibustion may be effective in managing the symptoms of post-stroke UI; these investigators stated that more rigorously designed, large-scale RCTs are needed to provide more robust evidence in this area.

Zhou et al (2024) noted that SUI significantly impacts women's health and imposes substantial mental and socio-economic burdens. These researchers examined the effectiveness of various treatments for women with SUI using network meta-analysis (NMA). They systematically searched databases up until June 30, 2022, to identify relevant RCTs focusing on SUI in women. Subsequently, the quality of the included studies was assessed; and NMA was carried out using STATA 14.0 software. A total of 31 RCTs involving 2,922 participants were included in the analysis. A total of 18 treatment plans were identified. The treatment plan consisting of moxibustion + PFMT + electromyographic biofeedback (EB) showed the most significant

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reduction in ICIQ-UI-SF. Due to lack of consistency across studies, a NMA was not carried out for the outcomes of effectiveness and the 1-hour pad test. The authors concluded that the combined intervention of moxibustion + PFMT + EB appeared to be the most effective in reducing patients' reported symptoms and improving their QOL. Moreover, these investigators stated that due to the drawbacks of the included studies, more high-quality, large sample-size RCTs are needed to reinforce the current evidence.

Cervico-Sacropexy or Vagino-Sacropexy for the Treatment of Urinary Incontinence and Apical Prolapse

Page et al (2022) noted that several anatomic theories suggested that lax utero-sacral ligaments may result in apical prolapse and UI; thus, prolapse repair such as cervico-sacropexy (CESA) or vagino-sacropexy (VASA), may resolve UI. Shortcomings in current therapeutic options endorse further exploration of the potential benefit of a surgical alternative. In a systematically review, these investigators examined the evidence on the safety and effectiveness of CESA and VASA as alternative surgical therapeutic options for urge and/or mixed UI and apical prolapse. The PRISMA 2020 statement was followed. Studies from inception to September 2021 were systematically reviewed and included. Data collection, risk of bias, and certainty of evidence were evaluated using the standard Cochrane methods. The primary outcome measures were improvements in prolapse and urinary symptoms. Secondary outcomes included surgical safety and re-intervention rates for complications and recurrence. The included studies showed a moderate-to-high risk of bias and low certainty of evidence. Owing to their heterogeneity, no meta-analysis was carried out. Cure rates for mixed and urge UI and apical prolapse were 47.5 % (95 % CI: 42.4 to 52.6), 73.8 % (95 % CI: 61.9 to 85.7) and 97 % to 100 %, respectively, at a mean follow-up of 9.7 ± 7.3 months. Additional incontinence surgery in 38.9 % (216/555) of women with initial UI, and concomitant or subsequent surgery for prolapse was performed in 4.4 % (13/299). The authors concluded that CESA or VASA may remedy symptoms of urge and mixed UI and appeared to correct apical prolapse in the short-term; moreover, the overall level of evidence was low. These researchers stated that further clinical trials, integrated in the IDEAL framework, targeting better-identified patient selection, and using validated outcome measures are needed to guide further research and implementation in practice.

Trans-Perineal Ultrasound for the Diagnosis of Stress Urinary Incontinence

In a case-control, single-center study, Keshavarz et al (2021) examined the use of trans-perineal ultrasonography (TPUS) for the diagnosis of SUI. This trial entailed married women who were referred to the gynecologic and US wards with negative urinalysis and culture results. Patients with positive cough signs based on the urodynamic testing data were considered cases, whereas control women showed no cough symptoms and were recruited from the same ward. There was a significant difference ($p < 0.001$) in bladder neck descent (BND; mean ± SD, 10.89 ± 5.51 versus 7.08 ± 2.60 mm, respectively; $p = 0.0001$) and the retro-vesical (β) angle with the Valsalva maneuver (144.22° ± 19.63° versus 111.81° ± 24.47°; $p < 0.001$) between the case and control groups. Furthermore, the β angle without the Valsalva maneuver was higher in the case group (112.35° ± 23.10°) than the control group (120.17° ± 25.16°; $p = 0.001$). There was no case of a urinary leak, urethral diverticulitis, a bladder stone or mass, and cysto-urethrocele in the patients of each group. The results of multi-variate logistic regression with a backward method showed that BND (OR, 1.24; 95 % CI: 1.09 to 1.40), the β angles with and without the Valsalva maneuver (OR, 1.1; 95 % CI: 1.06 to 1.13; and OR, 1.04; 95 % CI: 1.01 to 1.06) were the predictors of SUI. A β angle higher than 127° with the Valsalva maneuver, with an area under the curve (AUC) of 0.89 (95 % CI: 0.75 to 0.96), could very well predict the SUI response. This finding showed that TPUS could be very well used to distinguish between normal and non-normal responses, with 89 % sensitivity and 79 % specificity. The authors concluded that the β angle with the Valsalva maneuver could very well predict the SUI response. Moreover, these researchers stated that since TPUS may also be an appropriate tool for post-surgical assessments of patients with SUI, it is suggested that more focus on the diagnostic accuracy of TPUS for recovery of SUI after surgery should be conducted in further studies.

The authors stated that this study had several drawbacks. First, it is recognized that findings on the Valsalva maneuver will vary with the quality of the maneuver. However, standardization of pressure would probably require invasive monitoring, since efforts at non-invasive standardization have been largely unsuccessful.

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Second, the SUI diagnoses and TPUS findings were not connected to the urodynamic findings such as maximal urethral closure pressure and urodynamic stress incontinence. Moreover, US examinations carried out on different bladder volumes might have different results. Third, the limited sample size prohibited stratification of subjects into several groups that incorporated certain variables, such as BMI, parity, and type of delivery, as these variables may affect the threshold value of the main variables examined. Fourth, this study was a single-center study and only included Iranian women. These investigators stated that further studies with different methods in different age groups and parities are needed to validate these findings.

In a prospective, observational study, Turkoglu et al (2022) examined the use of TPUS while diagnosing SUI by comparing the urethral angle (α), posterior urethra-vesical angle (β), and BND during rest and Valsalva maneuver in continent women and women with SUI. This trial was carried out with 50 women with SUI and 50 continent women. TPUS was conducted at rest and during Valsalva maneuver; and Q-tip test was performed. During the Valsalva maneuver, both α and β angles were significantly higher in women with SUI ($p < 0.001$). The difference between Valsalva and rest measurements of α and β angles (R α, R β) were also significantly higher in women with SUI ($p < 0.001$). The cut-off point determined for the R α in the diagnosis of SUI was 16° (80 % sensitivity, 98 % specificity). A statistically significant strong correlation was found between Q-tip test angle and R α value ($p = 0.000$; $r = 0.890$). Q-tip VAS pain scores were significantly higher than US VAS pain scores ($p < 0.001$). In relation to the BND comparison between the 2 groups showed that BND was significantly higher in SUI group ($p < 0.001$). The cut-off point determined for BND in the diagnosis of SUI was > 11 mm (90 % sensitivity, 98 % specificity). The authors concluded that TPUS was a practical, reliable, non-invasive and comfortable method for evaluation of SUI. It has the advantage of dynamic evaluation during the Valsalva maneuver. Rotation angles and BND exhibited high sensitivity and specificity for detection of SUI. Moreover, these researchers noted that the change in α angle with Valsalva (Rα) could be used as an alternative to Q-tip test.

The authors stated that this study had several drawbacks. First, the lack of urodynamic proof for SUI. SUI patients were all surgery candidates with observed SUI with cough test. These investigators were strict about not including patients with pelvic organ prolapse to both groups. They also excluded complicated SUI patients who require urodynamic study according to ACOG guideline such as those with prior pelvic surgery, urgency, post-void residual volume of greater than 150 cc, Q-tip test of less than 30. Second, the lack of standardization of the Valsalva maneuver, which was the case in majority of the studies since measurement of intra-abdominal pressure is not easily done and can require rectal probes. Third, these researchers did not examine levator ani muscle (LAM) injury that is better observed with three dimensional (3D) US. LAM injury is important in the pathogenesis of bladder neck mobility. Fourth, the usage of the vaginal probe instead of the convex probe since majority of the related literature used the latter. The authors chose to use the vaginal probe with curved array tip since they observed the exact same anatomical structures in the same plane with a clearer view.

In a systematic review and meta-analysis, Chen et al (2023) examined the diagnostic value of TPUS in patients with SUI using evidence-based methods. These investigators carried out a comprehensive search of the studies on the diagnosis of SUI by TPUS in PubMed, Embase, Medline, Cochrane library, Medicine, Web of Science, and clinicaltrials.gov databases on August 1, 2022. Studies were included if they met the inclusion criteria and were evaluated by different quality evaluation methods according to study types. Various US parameters were collected and counted to analyze and judge the diagnostic value of TPUS in patients with SUI. A total of 13 studies with 1,563 subjects were finally included. The combined statistics showed no significant difference in age and parity among the included patients, and the BMI of the SUI group was slightly higher than that of the normal population (MD 1.20, 95 % CI: 0.68 to 1.72). The results indicated that compared with the normal population, the α angle (MD 15.56, 95 % CI: 9.93 to 21.90), β angle (at rest: MD 10.02 mm, 95 % CI: 1.95 to 18.09; at Valsalva: MD 22.40 mm, 95 % CI: 13.79 to 31.01), BND (MD 6.82 mm, 95 % CI: 4.49 to 9.14), area of hiatus (MD 2.83 cm2, 95 % CI: 0.71 to 4.94) and bladder neck funneling (RR 4.71, 95 % CI: 1.08 to 20.62) of SUI patients were significantly different, which showed the potential value of TPUS in the diagnosis of SUI. The authors concluded that evidence-based medicine was employed to statistically analyze published studies on the diagnostic value of TPUS in patients with SUI. The results suggested that TPUS had application value in the diagnosis of SUI and had the potential to become a routine examination method to aid in clinical decision-making.

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Platelet-Rich Plasma for the Treatment of Stress Urinary Incontinence

Dankova et al (2023) stated that there is no clear clinical evidence that platelet-rich plasma (PRP) injections improve female sexual dysfunction (FSD) and SUI. In a systematic review, these researchers examined the safety and effectiveness of PRP injections in women with FSD and SUI; and explored the optimal dosing, frequency, as well as area of injections, and duration of treatment. They carried out a systematic search on PubMed, Embase, and the Cochrane Library database, as well as sources of grey literature from the date of database or source creation to January 2023. After title/abstract and full-text screening, clinical studies on humans examining the effectiveness of PRP in gynecological disorders using standardized tools were included. Risk of bias was undertaken with RoB-2 for RCT and the Newcastle-Ottawa Scale (NOS) for observational studies. A total of 4 prospective and 1 retrospective study examined FSD, while 6 prospective and 1 RCT examined SUI. A total of 327 women with a mean age of 51 ± 12 years were included. For FSD, PRP significantly improved the Female Sexual Function Index (FSFI), the Vaginal Health Index (VHI), and the Female Sexual Distress score (FSDS). For SUI, PRP resulted in a significant improvement in the ICIQ-Short Form (ICIQ-SF) and the UDI-6. The identified RCT reported a significantly higher mean score of ICIQ-SF ($p < 0.05$) and UDI-6 ($p < 0.01$) in the mid-urethral sling group compared to the PRP injections group. Regarding the risk of bias, the RCT was characterized by high risk, whereas the observational studies were of moderate risk. The protocol for PRP injections for FSD was the injection of 2-ml of PRP into the distal anterior vaginal wall once-monthly for 3 months. For SUI, 5- to 6-ml of PRP should be injected into the peri-urethral area once-monthly for 3 months. The authors concluded that despite the promising initial results of PRP injections, the level of evidence for all outcomes was deemed low due to the methodological concerns raised in most of the included studies.. These investigators stated that there is an emerging need for high-quality RCTs examining PRP injections for the treatment of FSD and SUI.

The authors stated that this study had 2 main drawbacks. First, the included studies displayed significant heterogeneity in terms of PRP preparation technique, dose, injected area, and duration of treatment; thus, a meta-analysis could not be carried out. Second, most included studies raised methodological concerns. This problem predominantly stemmed from the small number of included subjects, the suboptimal methods of reporting randomization, the relatively short follow-up, the restricted number of events, and the implementation of non-recognized scales. Accordingly, some important parameters, such as PRP preparation technique, dose of PRP, injected area, as well as percentage of patients with improvement in underlying disease symptoms, remained unreported in some of the included studies.

Subcutaneous Tibial Nerve Stimulation (eCoin)

Short-term data on subcutaneous nerve stimulation have shown promising results for the treatment of OAB and UI. The eCoin is a small, coni-shaped device that can be implanted adjacent to the tibial nerve that provides pre-programmed stimulation.

In a prospective, single-arm, open-label, multi-center study, MacDiarmid et al (2019) examined the safety and effectiveness of a fully implanted, primary battery-powered, nickel sized and shaped neuromodulation device known as the eCoin for tibial nerve stimulation for the treatment of refractory UUI. This feasibility trial included 46 subjects with refractory UUI. It was carried out at multiple sites in the U.S. and New Zealand. The device was implanted in the lower leg over the tibial nerve and activated after 4 weeks. Bladder diary data and validated QOL instruments were collected 3 and 6 months after activation and compared to baseline values. The mean ± SD age of participants was 63.4 ± 11.5 years, and 45 (98 %) were women. Episodes of UUI were reduced a relative median of 71 % after 3 months of treatment (4.2 versus 1.7 daily episodes at 3 months, $p = 0.001$). A 50 % or greater decrease in reported episodes of UUI was observed in 32 of 46 participants (69.6 %) at 3 months with more than 20 % dry at 3 and 6 months. I-QOL (Incontinence Quality of Life) scores improved an average of 25.9 points and 33 of 46 patients (72 %) indicated improvement in symptoms. A single serious AE secondary to wound care resolved with intravenous (IV) antibiotics. The authors concluded that the implantable neuromodulation device was a safe and effective treatment of UUI associated with OAB syndrome with a significant reduction or resolution of symptoms and no significant safety concerns.

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Rogers and Sen (2021) stated that the treatment of OAB with UUI symptoms follows an algorithmic pathway. Patients who fail 1st- and 2nd-line treatments may be offered PTNS, onabotulinumtoxinA injections (BOTOX) or sacral neuromodulation as a 3rd-line treatment. An implantable tibial nerve stimulator may present a more convenient and effective treatment than these options. The coin-sized neurostimulator is subcutaneously implanted in a single visit using only local anesthesia. These researchers presented an instructional video showing the brief placement of the eCoin device for the treatment of OAB with UUI. The eCoin placement technique was demonstrated on a patient in an ambulatory surgery center procedure room setting. A custom marking template was provided to indicate the location of the incision and final eCoin placement. Once the markings were made, the patient was prepped for the procedure with local anesthesia. The lower leg was then sterilized and draped. The custom marking tool was used again to remark the incision site and eCoin placement location. Once the incision was made, a custom sizing blunt dissection tool was used to create a pocket for device placement. The eCoin was then easily inserted into the pocket, located above the tibial nerve. A layered closure technique was completed. The patient was then fitted with an ankle support to provide gentle compression for 4 weeks. During this period, the patient was instructed to comply with provided after-care instructions and materials in order to prevent infection or eCoin device migration. After the 4-week healing duration, the eCoin device was activated. A total of 133 patients across 15 study sites were implanted with the eCoin device in a clinical trial. The mean implant time from incision to closure was 20.77 mins (SD 9.08). The median implant time was 18 mins. All of the patients were evaluated for wound healing approximately 2 weeks post-implant. There was 1 related severe AE, an infection resulting in uncomplicated explant at a hospital setting. At the time of this writing, patients in the study have had the device implanted for an average of 56.9 weeks. The treatment was effective and sustainable as described in other abstract submissions. The authors showed the use of a safe method of subcutaneous tibial nerve stimulation implant placement that was performed in the office under local anesthesia. The procedure time was relatively brief resulting in minimal AEs in a large cohort.

Rogers et al (2021) reported on a prospective, open-label, single-arm trial of the eCoin device carried out at 15 U.S. medical centers involving 137 subjects with refractory UUI. After implantation in the lower leg above the fascia over the tibial nerve, eCoin delivered automated stimulation sessions for the duration of the study. The primary effectiveness measure was the proportion of subjects who achieved a 50 % or greater reduction from baseline in UUI episodes after 48 weeks of therapy. The primary safety measure was device-related AEs at the same time-point. Of 137 subjects enrolled, 133 were implanted with eCoin, and 132 were included in the ITT population. Of those 132 subjects, 98 % were female, mean ± SD age was 63.9 ± 10.9 years, and baseline daily UUI episodes were 4.3 ± 3.1. The primary effectiveness analysis showed 68 % (95 % CI: 60 % to 76 %) of subjects experienced at least a 50 % reduction in UUI episodes at 48 weeks post-activation; 16 % of implanted subjects experienced device-related AEs through 52 weeks post-implantation.

Kaaki et al (2022) reported on a prospective, single-arm, open-label study, including 23 participants with refractory UUI who were previously participants in the eCoin clinical feasibility trial. This follow-on study was conducted at 7 sites in the U.S. and New Zealand. Participants were re-implanted with a new eCoin device and activated after 4 weeks. Bladder diary data and validated QOL instruments, collected at 12 weeks and 24 weeks post-activation, were compared with baseline. Participants of the study were considered responders if they reported a 50 % or greater reduction from baseline in episodes of UUI on a 3-day voiding diary. At 12 weeks of treatment, 74 % (95 % CI: 52 % to 90 %) of participants were considered responders. At 24 weeks of treatment, 82 % (95 % CI: 60 % to 95 %) of participants were considered responders with 36 % (95 % CI: 20 % to 57 %) of participants achieving complete continence. There were no device-related serious AEs reported during the study.

Gilling et al (2022) reported on a study to evaluate the safety and effectiveness of the eCoin. A feasibility clinical trial was conducted and the results after 1 year of treatment with the eCoin were presented. A total of 46 participants with refractory UUI were included in this prospective, single-arm, open-label study. This study was conducted at 7 sites in the U.S. and New Zealand. Participants in this study were implanted with the eCoin in the lower leg over the tibial nerve and activated after 4 weeks. Bladder diary data and validated QOL instruments, collected at 3, 6, and 12 months post-activation, were compared to baseline values. Responders were defined as those who had a 50 % or greater reduction in reported episodes of UUI. At 12 months, 65 % of participants were considered responders with 26 % of participants achieving complete continence. The median number of UUI episodes per day decreased from 4.2 at baseline to 1.7 at 12 months;

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70 % of participants reported feeling "better", "much better", or "very much better" on the Likert 7-point maximum scale. One participant experienced a related serious AE.

Smith (2022) stated that implantable tibial nerve stimulation (ITNS) is promising, with variable number of days and duration of stimulation possible and creating less burden on the patient. The author noted that long-term safety, effectiveness and tolerability are unknown at this time, but expected to be acceptable. With all the interest conveyed by the biotech companies, there is sure to be more to come on these technologies.

Al-Danakh et al (2022) stated that PTNS techniques have dramatically grown after approval to manage OAB. This review focused on the most current data on PTNS types (percutaneous, transcutaneous, and implant) and their mechanism of action, safety, effectiveness, advantages, drawbacks, limitation, and clinical applications. These investigators described the recent studies that addressed the tibial nerve stimulation role in OAB management. BlueWind RENOVA system, Bioness StimRouter, and eCoin are examples of emerging technologies that have evolved from interval sessions (percutaneous PTNS and transcutaneous PTNS) to continuous stimulation (implants). These can be efficiently managed at home by patients with minimum burden on the health system and fewer visits, especially in the COVID-19 pandemic. The authors concluded that the tibial nerve stimulation advancements in OAB treatment have been rapidly increasing over the recent years. It is minimally invasive and effective, similar to sacral nerve stimulation (SNM), but less aggressive. Implantable PTNS has been promised in terms of safety, effectiveness and high acceptance rate; however, evidence is still limited to short-term trials, and tolerability, method, and drawbacks remain challenges.

Bessington et al (2023) reviewed the literature on eCoin implantation, from proof-of-concept to mid-term data, with the longest period of follow-up being 12 months. The authors found that the eCoin device showed promising early data for effectiveness in managing OAB symptoms. Complication rates remained low and were mostly related to wound healing following the initial placement of the device. Research into the continued improvement and modification of the device appeared optimistic; however, longer-term data still need to be obtained. Indeed, implantable PTNS have a role in the future management of OAB, and devices such as eCoin will still need to prove a long-term benefit to be a mainstay of management.

In a prospective, single-arm, multi-center study, Lucente et al (2024) examined the continued safety and effectiveness of the eCoin ITNS for urgency urinary incontinence (UUI) in patients with OAB. The 1-year pivotal study was extended through 2 years. The ITNS is a novel and recently FDA-approved therapy. This trial was carried out on 137 subjects with refractory UUI to evaluate eCoin ITNS therapy. A 3-day voiding diary was collected along with the OAB questionnaire, Patient Global Impression of Improvement, and a custom Likert scale on subject satisfaction. The primary effectiveness measure was the proportion of subjects who achieved at least 50 % reduction from baseline in number of UUI episodes. The primary safety measure was device-related AEs. A total of 72 subjects completed the 96-week evaluation. Around 78 % (95 % CI: 67 % to 87 %) experienced at least 50 % reduction in UUI episodes; 48 % (95 % CI: 36 % to 60 %) experienced at least 75 % reduction, and 22 % (95 % CI: 13 % to 33 %) were dry on a 3-day diary. Subjects reported a decrease from baseline in their UUI episodes/day of 2.61 (SD 2.97) and 2.97 (SD 2.64) at 48 weeks and 96 weeks, respectively. Approximately 91.3 % did not require additional medications for OAB. No serious or unanticipated AEs were reported in this extension phase. The authors concluded that eCoin ITNS showed consistent continuing safety and effectiveness in treating OAB patients with UUI. The findings supported it as an excellent therapeutic option for refractory patients.

The authors stated that the study faced common challenges in neuromodulation research, precluding the implementation of a blinded, comparative trial. Furthermore, patient withdrawals and loss to follow-up were influenced by the COVID-19 pandemic, potentially impacting data reliability.

Lucente et al (2025) stated that UUI, an important subset of OAB, manifests with symptoms such as urgency, frequency, and incontinence, severely impacting quality of life (QOL). Neuromodulation therapies, including sacral nerve stimulation (SNM), percutaneous tibial nerve stimulation (PTNS), and implanted tibial nerve stimulation (ITNS), are FDA-approved for the treatment of UUI. Traditional neuromodulation entails sensory and motor response evoking amplitudes; however, emerging evidence suggested that sensory and sub-sensory settings might enhance treatment outcomes by influencing brain activation. These investigators

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examined the effectiveness of sensory and sub-sensory programming of the eCoin ITNS in reducing UUI episodes. The eCoin ITNS is a fully implantable device providing low-duty cycle tibial nerve stimulation. The ESSENCE (Effectiveness of Sensory and Sub-sensory Stimulation Amplitudes Using eCoin Implantable Tibial Nerve Stimulation in Reducing Urgency Urinary Incontinence Episodes) Trial was carried out as a double-blind RCT, and 36 subjects with UUI across 5 U.S. centers were enrolled, aiming to assess changes in UUI episodes and QOL over 3 months. Subjects were randomized to sensory or sub-sensory stimulation groups, with the sensory group activated to the amplitude at which stimulation was first felt and the sub-sensory group set 25 % below this threshold. UUI episodes were recorded using 3-day voiding diaries, and QOL was assessed via the Overactive Bladder Symptom Quality of Life Questionnaire (OABq) to evaluate the primary end-point of reduction from baseline in the number of UUI episodes per day on the 3-day voiding diary. Results showed a mean reduction in UUI episodes of 2.1 per day for the sensory group and 2.73 per day for the sub-sensory group from a pooled baseline of 5.53. Both groups reported improvements in HR-RQL and patient satisfaction. These findings aligned with previous studies on SNM, showing both sub-sensory and sensory settings were effective, potentially enhancing patient comfort and device longevity. The authors concluded that the ESSENCE Trial demonstrated that both sensory and sub-sensory amplitude settings of eCoin ITNS showed a reduction in UUI episodes, improved QOL, and increased patient satisfaction, offering the potential for optimizing neuromodulation therapies. Moreover, these researchers noted that while the study design lacked sufficient power to demonstrate statistical significance and establish the non-inferiority of the sub-sensory group compared to the sensory group, the observed data trends support the potential benefits of sub-sensory programming. These researchers stated that further investigations are needed to solidify these findings and confirm the non-inferiority of sub-sensory settings compared to sensory programming. Larger-scale studies with sufficient power for comparative analysis could better outline the differences between these 2 approaches. Nevertheless, the results highlighted the promising potential of sub-sensory stimulation as a refined strategy in ITNS therapy, balancing effective UUI management with long-term patient-centered care.

The authors stated that a drawback of this trial was the lack of a statistically significant comparison between groups. Although both groups showed a reduction in UUI, improvement in QOL, and patient satisfaction, it is unclear if the effects observed in the sub-sensory group were non-inferior to the sensory group.

Amundsen et al (2025) stated that implantable tibial neuromodulation (iTNM) systems have recently become commercially available in the U.S., and offer a new method of neurostimulation for the treatment of OAB. In the absence of head-to-head studies, this meta-analysis indirectly compared safety and effectiveness of SNM and iTNM for the treatment of OAB. These investigators carried out a comprehensive search using terms for OAB and neuromodulation. Primary effectiveness measures included a 50 % or greater reduction in (UUI episodes, urinary frequency, and/or OAB symptoms. Primary safety measures included the rate of device-related AEs. A total of 20 studies met selection criteria, encompassing 1,416 patients treated with SNM and 350 patients treated with iTNM. No comparative or placebo-controlled studies for SNM and iTNM were identified; thus, the analysis was completed using single-arm results. Weighted averages showed that the UUI responder rate was similar for both SNM and iTNM (71.8 % and 71.3 %, respectively). Similarly, weighted averages of OAB responder rates were 73.9 % for SNM and 79.4 % for iTNM. Similar rates of device-related AEs were also observed. The authors concluded that this meta-analysis found similar safety and effectiveness of SNM and iTNM for the treatment of OAB and UUI, including UUI and OAB symptom response rates, reduction in UUI episodes, significant improvements in QOL, and low rates of procedure and device-related AEs. Notably, this comparable effectiveness was observed without the use of a trial phase of neuromodulation in the iTNM studies versus SNM studies. Moreover, these researchers stated that current results suggest that iTNM may also have lower surgical re-intervention rates; however, additional follow-up time is needed to confirm if this trend will continue.

The authors stated that this meta-analysis had several drawbacks. First, while this meta-analysis featured a large number of studies and corresponding patients, 11 studies were considered to have a moderate risk of potential bias, most often related to challenges with randomization and blinding due to patient sensation of neurostimulation, along with the retrospective and single-center design of several of these studies. Differences in study populations, geography, study methods, effectiveness definitions, and stage of device development (i.e., pilot study versus pivotal study versus study completed post-marketization) may also impact generalizability of the results. Patient level data were unavailable for analysis; thus, data for all end-

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points were not always available for each study even if they had been collected. These elements, though possible with any meta-analysis and not specific to this review, should be considered when interpreting the presented results. Second, the expected difference between the length of follow-up observed in the selected iTNM and SNM studies (weighted average of 13.0 and 39.2 months, respectively). While SNM devices have been available for commercial use in the U.S. for almost 30 years (since 1996), iTNM was first FDA-approved less than 2 years ago (December 2022); accordingly, the average length of follow-up data available for SNM was much longer than iTNM. More importantly, most device- and procedure-related AEs notably occurred within the first several months following implantation for both iTNM and SNM devices. Further comparability between later revisions and/or explant rates will certainly need to be assessed. Clearly, devices designed with implanted batteries -- both iTNM and SNM -- will require eventual surgical replacement. Both the original implant and surgical revisions result in costs to medical systems and patients; published data comparing cost between iTNM and SNM are not yet available. Third, challenges in comparing frequency of urination were also identified in this analysis, primarily because of consistent differences in urinary frequency reported in the studies completed in Asia, which ultimately may describe a more severe population of patients with OAB. While the reason for this difference is unclear, the frequency was reported with range of 10 to 14.9 in iTNM and SNM studies completed in North America, Europe, and the Middle East, compared with 21.6 to 29.2 in SNM studies in Asia, resulting in challenges in comparison of results related to urinary frequency. The impact of iTNM on urinary frequency needs to be further examined in future studies. Fourth, none of the studies identified was head-to-head trials, limiting the ability for direct comparisons. Future RCTs are needed to provide additional clarity regarding comparative differences in safety and effectiveness, and to provide further guidance regarding ideal patient selection for both methods of neuromodulation, tibial versus sacral.

Subcutaneous Tibial Nerve Stimulation (Altaviva)

The Altaviva Implantable Tibial Neuromodulation (ITNM) device was developed as a leadless implantable neuromodulation device placed above the fascia in the lower leg that delivers electrical stimulation to the tibial nerve. This design may reduce the need for recurring office visits for therapy maintenance compared to percutaneous tibial neuromodulation. On September 18, 2025, the FDA approved our Altaviva rechargeable Implantable Tibial Neuromodulation device for the treatment of urge urinary incontinence (UUI).

In 2021, the FDA granted Investigational Device Exemption (IDE) approval to begin a clinical trial (TITAN 1) exploring a novel ITNM device designed to relieve symptoms of bladder incontinence. TITAN 1, a feasibility study completed in 2022, was a prospective, multicenter study that enrolled 24 subjects from 7 sites, 20 of whom were implanted with an investigational ITNM device (Lee, et al., 2023). The primary objective of the TITAN 1 study was to characterize procedural learnings through 14-days post-implant using a series of questions about the implant procedure. Subject responses to stimulation, including the lowest stimulation at which a motor response could be observed (motor threshold), the lowest stimulation amplitude at which the subject reported sensation of stimulation (sensory threshold), type of sensation (e.g., tapping), and type of motor response (e.g., great toe flexion) were collected at implant and follow-ups. Stimulation was delivered for 60 min every other day at 20 Hz, 200 µs with a stimulation amplitude set to a comfortable level above the motor and/or sensory threshold. All 20 subjects completed the 14-day visit and all 8 implanting physicians from 7 sites completed the procedural learnings questionnaire. All respondents reported the training conducted prior to the first implant was adequate. For each procedural step (identifying landmarks, dissecting to the level of the fascia, creating the subcutaneous pocket, inserting the implantable neurostimulator, and pocket closure), 90–95% of respondents rated these “easy” or “somewhat easy.” Safety data in this early phase of the study (time between study start and last subject's 14-day follow-up visit) are as follows: 5 device-, procedure-, and/or therapy-related adverse events (AEs) were reported in three subjects including 1 serious adverse event (wound infection) which resulted in device explant 6 weeks after implant. During this timeframe, no AEs were reported for suspected device migration. Motor and sensory threshold testing data were collected in the TITAN 1 study to understand responses to nerve stimulation. All twenty study subjects demonstrated motor and/or sensory response to stimulation at two implant timepoints (intra-operative and post-operative) and at the 7-day follow-up visit. Eighteen of nineteen subjects demonstrated motor and/or sensory response at the 14-day follow-up visit; for one subject, the threshold assessment was not performed.

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The Evaluation of Implantable Tibial Neuromodulation (TITAN 2) pivotal study (NCT05226286), which began in 2022, is a prospective multicenter investigational device exemption (IDE) study conducted in the US with 126 subjects evaluating the safety and effectiveness of the Medtronic Altaviva ITNM system for the treatment of OAB symptoms. Early learnings from the TITAN 1 feasibility study were used to inform the pivotal study protocol (Lee, et al., 2023).

Subcutaneous and Subfascial Tibial Nerve Stimulation (INTIBIA)

The INTIBIA system (Coloplast,(Humlebaek, Denmark)) is an investigational implantable tibial nerve stimulator designed to treat urgency urinary incontinence (UUI). The device consists of a lead implanted near the ankle and a wireless ankle stimulator to deliver electrical pulses to the tibial nerve to reduce UUI symptoms. A pivotal clinical trial of the INTIBIA system is ongoing (NCT05250908).

ZIDA Wearable Neuromodulation System (Transcutaneous Electrical Nerve Stimulation) for the Treatment of Idiopathic Non-Obstructive Urinary Retention

Coolen et al (2021) noted that TENS and PTNS provide minimally invasive ways for the treatment of idiopathic non-obstructive urinary retention (NOUR). In a systematic review, these investigators examined the effectiveness of TENS and PTNS for the treatment of idiopathic NOUR. They carried out a systematic review in accordance with the PRISMA guidelines. Embase, Medline, Web of Science Core Collection, and the Cochrane CENTRAL register of trials were searched for all relevant publications until April 2020. A total of 3,307 records were screened based on the title and abstract; 8 studies met the inclusion criteria and none of the exclusion criteria. A total of 5 studies, all from the same group, reported the effectiveness of PTNS and 2 that of TENS in adults with idiopathic NOUR. One study reported the effectiveness of TENS in children with idiopathic NOUR. Objective success was defined as a 50 % or larger decrease in the number of catheterizations per 24 hours or in the total catheterized volume in 24 hours. The objective success rate of PTNS ranged from 25 % to 41 %. Subjective success was defined as the patient's request for continued chronic treatment with PTNS, and ranged from 46.7 % to 59 %; 80 % of women who underwent trans-vaginal stimulation reported an improvement such as a stronger stream when voiding. TENS in children reduced post-void residual (PVR) volume and UTIs. The authors concluded that the effectiveness of TENS and PTNS in the treatment of idiopathic NOUR was limited and should be verified in larger randomized studies before its use in clinical practice.

Smith et al (2022) stated that bladder symptoms are common in Parkinson's disease (PD), affecting 50 % of all patients. These have significant impact on QOL as well as implications for morbidity, contributing to falls and hospital admission. The treatment of bladder symptoms can be complicated by the tendency to side-effects in patients with PD including cognitive impairment and gait instability with anti-muscarinic agents; thus, the development of new, better treatments is needed. Tibial nerve stimulation is a form of neuromodulation demonstrated to improve OAB symptoms in non-neurogenic cohorts. Previously requiring hospital attendance, these researchers examined the use of this intervention employing a simple device that can be used by patients at home. The STRIPE Trial is a phase-II RCT of TNS delivered by the Geko device, a small, self-adhesive neuromuscular stimulation device currently used for thrombo-embolism prophylaxis post-surgery. Active TNS will be compared to sham stimulation, with subjects blinded to treatment allocation and undertaking outcome assessment while still blinded. Subjects will be asked to self-administer stimulation at home twice-weekly, for 30 mins per session, over the course of 3 months. Primary outcome measure will be the International Consultation on Incontinence Overactive Bladder Questionnaire at week 12. Secondary outcomes will include pre- and post-intervention bladder diary (frequency, urgency episodes, nocturia), patient perception of global change, bowel function as well as bladder-related QOL. Subjects will be recruited from the Proactive Integrated Management and Empowerment (PRIME) cross-sectional trial in which subjects have been screened for bladder symptoms and invited to take part, as well as clinician referral from around the region. The authors concluded that this trial entails a RCT of a novel and easy to use method of delivering TNS for the treatment of PD-related bladder symptoms in the patient's own home. This may potentially have huge benefit, avoiding the problems with side effects that can be associated with anti-muscarinic agents, and providing a new potential modality of treatment.

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Bapir et al (2022) noted that OAB symptoms of frequency, urgency and UI are often associated with known neurological diseases like MS, spinal cord injury (SCI), PD, and stroke. In a systematic review, these investigators examined the effectiveness of pharmacological and non-pharmacological treatments for the treatment of neurogenic OAB. They searched 2 electronic databases (PubMed and Embase) for RCTs focusing on pharmacological and non-pharmacological treatments for OAB symptoms associated with neurological diseases published up to April 30, 2022. A total of 157 studies were retrieved; 94 were selected by title and abstract screening; after removal of 17 duplicates, 77 records were evaluated by full-text examination. A total of 62 studies were finally selected. The studies selected for review focused on the following interventions: anti-cholinergics (n = 9), mirabegron (n = 5), comparison of different drugs (n = 3), cannabinoids (n = 2), intravesical instillations (n = 3), botulinum toxin (n = 16), transcutaneous TNS (TTNS) (n = 6), acupuncture (n = 2), TENS (n = 4), PFM training (PFMT) (n = 10), others (n = 2). Anti-cholinergics were more effective than placebo in decreasing the number of daily voids in patients with PD (mean difference [MD]- 1.16, 95 % CI: - 1.80 to - 0.52, 2 trials, 86 patients, $p < 0.004$), but no significant difference from baseline was found for incontinence episodes and nocturia. Mirabegron was more effective than placebo in increasing the cystometric capacity in patients with MS (MD 89.89 mL, 95 % CI: 29.76 to 150.01, 2 trials, 98 patients, $p < 0.003$) but no significant difference was observed for symptom scores and bladder diary parameters. TTNS was more effective than its sham-control in decreasing the number of nocturia episodes (MD -1.40, 95 % CI: -2.39 to -0.42, 2 trials, 53 patients, $p < 0.005$) but no significant changes of OAB symptom scores were reported. PFMT was more effective than conservative advice in decreasing the ICIQ symptom score (MD, -1.12, 95 % CI: -2.13 to -0.11, 2 trials, 91 patients, $p = 0.03$), although the number of incontinence episodes was not significantly different between groups. The authors concluded that the findings of this meta-analysis revealed a moderate effectiveness of all considered treatments without proving the superiority of one therapy over the others. Combination treatment using different pharmacological and non-pharmacological therapies could achieve the best clinical effectiveness due to the favorable combination of the different mechanisms of action. This could be associated with fewer side effects due to drug dosage reduction. Moreover, these researchers stated that these findings were only provisional and should be considered with caution, due to the few studies included in meta-analysis, and to the small number of patients.

Ghavidel-Sardsahra et al (2022) stated that PTNS and TTNS showed a promising effect on OAB and interstitial cystitis/painful bladder syndrome. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of these therapeutic methods. They searched studies available on PubMed, Embase, Cochrane, Scopus, Web of Science, and ProQuest on March 31, 2021, to find both published and unpublished studies. The retrieved studies were screened by 2 independent researchers, and then the selected studies were critically appraised by Cochrane risk-of-bias tool for randomized trials, and Joanna Briggs Institute's check-list for quasi-experimental studies. The results of studies were synthesized using Review Manager (RevMan) 5.4 statistical software when the data were homogenous. The meta-analysis was carried out by calculating the effect size (MD) and their 95 % CIs. Of a total 3,194 publications, 68 studies were included in the qualitative evaluation, and 9 studies (11 trials) in the quantitative stage. When TTNS or PTNS were compared to sham, placebo, no treatment, or conservative management, a decrease in frequency of urination was observed in both TTNS (MD: -3.18, 95 % CI: -4.42 to -1.94, and $p < 0.00001$), and PTNS (MD: -2.84, 95 % CI: -4.22 to -1.45, and $p < 0.00001$), and overall TTNS or PTNS (MD: -2.95, 95 % CI: -4.01 to -1.88, and $p < 0.00001$). Significant improvements in mean voiding volume (MVV) and decreasing nocturia were also observed. The authors concluded that nerve stimulations with either PTNS or TTNS appeared to be effective interventions in the treatment of refractory idiopathic OAB in terms of daily voiding frequency, MVV, urgency episodes, and night-time voiding frequency. Moreover, these researchers noted that these findings did not show any improvement in terms of urinary incontinence, PVR volume, or UI, and maximum cystometric capacity that emphasized the effectiveness of these modalities on dry-OAB rather than wet-OAB.

In a scoping review, Sayner et al (2022) examined the feasibility and outcomes of TTNS as a 1st-line therapeutic option for OAB. These investigators searched 6 electronic databases to identify full-text studies from 2015 that examined the impact of TTNS on OAB and bladder dysfunction in individuals aged 18 years older. A total of 15 studies met the inclusion criteria. TTNS was compared with sham treatment, para-sacral stimulation, PFMT, anti-cholinergic medication, and PTNS. Heterogeneity in treatment application and parameters existed, with variations in treatment duration, frequency of use, and treatment settings such as

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pulse width (μs) and frequency (Hz). Results indicated that TTNS has effectiveness equal to PFMT and PTNS in the management of OAB; however, it was not as effective as anti-cholinergic medication. The authors concluded that TTNS is a promising 1st-line therapeutic option for individuals with OAB, especially in the older population and for those with neurogenic bladder. It could provide symptomatic relief from urinary incontinence, frequency, urgency, and nocturia, while avoiding the bothersome side effects of more invasive or pharmaceutical therapies. Moreover, these researchers stated that heterogeneity in treatment parameters limited generalizability and translation of the most appropriate clinical application and should be considered in future trials.

In a systematic review and meta-analysis, Tahmasb e al (2023) examined the available evidence on the effects of TTNS and PTNS on MS-induced neurogenic lower urinary tract dysfunction. Medical databases including PubMed, Scopus, Embase, and Web of Science were systematically searched from inception to September 2022. Meta-analysis was performed using the comprehensive meta-analysis tool. The inclusion criteria were fulfilled by 12 studies examining the effects of PTNS/TTNS on MS-induced neurogenic lower urinary tract dysfunction. Comparing the post-intervention results to the baseline showed that the rate of frequency was decreased in both PTNS and TTNS groups after intervention. The overall mean change of TNS on frequency was -2.623 (95 % CI: -3.58 to -1.66; $p < 0.001$, I2 : 87.04) among 6 eligible studies. The post-void residual (PVR) was decreased after treatment in both methods of TNS, with an overall MD of -31.13 ml (95 % CI: -50.62 to -11.63; $p = 0.002$, I2 : 71.81). The other urinary parameters, including urgency (MD: -4.69; 95 % CI: -7.64 to -1.74; $p < 0.001$, I2 : 92.16), maximum cystometric capacity (MD: 70.95; 95 % CI: 44.69 to 97.21; $p < 0.001$, I2 : 89.04), and nocturia (MD: -1.41; 95 % CI: -2.22 to 0.60; $p < 0.001$, I2 : 95.15), were improved after intervention, too. However, the results of subgroup analysis showed no effect of TTNS on urinary incontinence (MD: -2.00; 95 % CI: -4.06 to 0.06; $p = 0.057$, I2 : 95.22) and nocturia (MD: -0.39; 95 % CI: -1.15 to 0.37; $p = 0.315$, I2 : 84.01). In terms of mean voided volume, the evidence was related to only PTNS with a mean change of 75.01 ml (95 % CI: -39.40 to 110.61; $p < 0.001$, I2 : 85.04). The authors concluded that although the available evidence suggested that TNS might be an effective method for the treatment of neurogenic lower urinary tract dysfunction; however, the evidence base was poor, and derived from small, mostly non-randomized studies with a high-risk of bias and confounding. Moreover, these researchers stated that the long-term effects of TNS therapy and its cost-effectiveness need to be addressed by future high-quality controlled trials.

Chen et al (2023) compared and ranked the safety and effectiveness of oral medications, 3 doses of onabotulinumtoxinA, and TTNS on improving urodynamic outcomes in patient-reported outcomes and safety outcomes in patients with detrusor overactivity (NDO). These investigators searched PubMed, Embase, Medline, Cochrane Library, Medicine, and clinicaltrials.gov, from their inception to October 2022, and included RCTs on the drug, onabotulinumtoxinA, and TTNS for the treatment of patients with NDO. Outcomes included urodynamic parameters, voiding diary, QOL changes, AE rate, and PVR. A total of 26 studies and 2,938 patients were included in the statistics. Regarding effectiveness, all interventions except TTNS and alpha-blockers were statistically different for the placebo group. The urodynamic outcome and patient-reported outcome suggested that onabotulinumtoxinA injection (urodynamic outcome: onabotulinumtoxinA 200 U, the mean surface under the cumulative ranking curve (SUCRA): 87.4; patient-reported outcome: onabotulinumtoxinA 100 U, mean SUCRA: 89.8) was the most effective treatment, and the safety outcome suggested that TTNS (SUCRA: 83.3) was the safest. Cluster analysis found that anti-muscarinic agents and β3-adrenoceptor-agonists possessed good safety and effectiveness. The authors concluded that onabotulinumtoxinA injection was probably the most effective way for the treatment of patients with NDO, with increasing effectiveness but decreasing safety as the dose rises. The effectiveness of alpha-blockers and TTNS was not statistically different from the placebo group. Anti-muscarinic agents and β3-adrenoceptor-agonists exhibited good safety and effectiveness.

Yildiz and Sonmez (2023) examined the effectiveness of transcutaneous medial plantar nerve stimulation (T-MPNS) on QOL and clinical parameters associated with incontinence in women with idiopathic OAB. A total of 21 women were included in this study; all subjects received T-MPNS. Two self-adhesive surface electrodes were positioned with the negative electrode near the metatarsal-phalangeal joint of the great toe on the medial aspect of the foot and the positive electrode 2 cm inferior-posterior of the medial malleolus (in front of the medio-malleolar-calcaneal axis). T-MPNS was carried out 2 days a week, 30 mins a day, for a total of 12 sessions for 6 weeks. Subjects were evaluated for incontinence severity (24-hour pad test), 3-day

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voiding diary, symptom severity (Overactive Bladder Questionnaire [OAB-V8]), QOL (Quality of Life-Incontinence Impact Questionnaire [IIQ-7]), positive response and cure-improvement rates, as well as treatment satisfaction at baseline and at the 6th week. Statistically significant improvement was found in the severity of incontinence, frequency of voiding, incontinence episodes, nocturia, number of pads, symptom severity, and QOL parameters at the 6th week compared with baseline. Treatment satisfaction, treatment success, and cure or improvement rates were found to be high at the 6th week. The authors concluded that T-MPNS was effective on both clinical parameters and QOL associated with incontinence in women with idiopathic OAB. Moreover, these researchers stated that randomized, placebo/sham-controlled, multi-center studies are needed to validate the effectiveness of T-MPNS. These investigators stated that this trial had several drawbacks. First, this was a single-center study. Second, the study was not controlled; thus, the findings could be influenced by a placebo effect. Third, the lack of data on urodynamics and long-term follow-up of women.

In a 12-week, open-label, single-arm, multi-center study, Goudelocke et al (2024) examined the safety and effectiveness of a novel, wearable, neuromodulation system incorporating embedded EMG evaluation, representing the 1st closed-loop wearable therapy for bladder control. This trial of patients with OAB examined response of bladder diary parameters and QOL metrics. Subjects used the TTNS system, either once- or thrice-weekly, with evaluations at weeks 1, 4, 8, and 12. Enrolled subjects (n = 96) were assessed for changes in urinary frequency, urgency, and urgency urinary incontinence episodes, and QOL changes using various questionnaires. In the ITT population (n = 96, mean age of 60.8 ± 13.0years, 88.5 % women), significant reductions in 3-day diary parameters were observed for daily voids, incontinence, as well as urgency episodes at 12 weeks. QOL improvements exceeded the minimal clinically important difference (MCID) for all QOL questionnaires. Long-term results remained robust at 12 months. Device-related AEs were mild, and there were no device-related serious AEs. Mean therapy compliance at 12 weeks was 88.5 %; and high satisfaction rates were reported for the device overall. The authors concluded that the Avation device showed promising effectiveness in the treatment of adults with OAB and urge UI. At 12 weeks, both diary parameters and QOL indicators demonstrated significant improvement and remained robust at 12 months. The device had a favorable safety profile with high compliance and patient satisfaction. These investigators stated that this novel, closed-loop wearable TNS system represents a significant advancement in bladder control therapy, offering a non-invasive, patient-centered alternative with improved accessibility and ease of use.

Parodi et al (2024) stated that OAB affects millions of patients globally; and its treatment is challenging but improves the patient's QOL. Besides standard techniques for neuromodulation (sacral and pudendal neuromodulation, and PTNS), several new techniques have been examined for the treatment of symptoms of refractory OAB. These investigators described the state of the art of new neuromodulation techniques for lower urinary tract dysfunction (LUTD). They carried out a comprehensive Medline, Embase, and Scopus search In March 2023 (keywords: LUTD, new technologies, neuromodulation, LUTS, OAB, painful bladder syndromes, etc.). Studies were included according to inclusion (RCTs, prospective trials, large retrospective studies) and exclusion (case reports, outcomes not clearly expressed in full text, animal studies) criteria. The reference lists of the included studies also were scanned. Both adult and pediatric populations were included, in addition to both neurogenic and non-neurogenic OAB. A narrative review was then carried out. Peroneal neuromodulation, TENS, magnetic nerve stimulation, and para-sacral transcutaneous neuromodulation were the most studied investigative techniques, and were shown to yield promising results in treating OAB symptoms. Most studies showed promising results even in the complex scenarios of patients with OAB refractory to standard treatments. Comparing investigational techniques with standard of care (SOC) and their respective clinical outcomes and safety profiles, and confronting their pros and cons, these researchers believed that once such treatment modalities are further developed, they could play a role in the OAB treatment algorithm. The authors concluded that although the described neuromodulation techniques are being intensely studied, the available results are not yet sufficient for any guidelines to recommend their use.

Subfascial Tibial Nerve Stimulation (e.g., BlueWind Revi) for the Treatment of Overactive Bladder Syndrome and Urgency Urinary Incontinence

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Heesakkers et al (2018) stated that OAB is a chronic condition affecting lower urinary tract function that has a significant negative impact on QOL. In a prospective, 6-month, multi-center study, these investigators examined the performance and safety of the BlueWind implantable tibial nerve system in the treatment of refractory OAB. Objective assessment was carried out by voiding diary parameters including voids/day, volume voided/day, urgency assessment, leaking episodes/day, pads used/day, leak severity, and clinical success defined as a 50 % or greater reduction in the number of leaks/day or number of voids/day, or number of episodes with degree of urgency more than 2 or a return to less than 8 voids/day on a 3-day diary. Subjective assessment was based on OAB-q including HRQL and symptom severity score. Safety was evaluated by AE analysis. A total of 34 of the 36 implanted subjects completed the study; 1 subject withdrew voluntarily and 1 developed inflammation necessitating removal of the system. In the remaining subjects, 71 % experienced clinical success at 6 months. Leaks/day, leak severity, and pad changes/day decreased significantly over time with 27.6 % of urge incontinence subjects that became "dry". Voids/day, degree of urgency, volume/void, pads changed improved significantly. All QOL aspects (concern, coping, sleep, and social) improved as well as symptom severity scores measured by the OAB-q; AEs included: implant site pain (13.9 %), suspected infection (22.2 %), and procedural wound complications (8.3 %). The authors concluded that the BlueWind implantable tibial nerve stimulator was a safe, minimally invasive system that afforded OAB patients significant improvements. These researchers stated that these preliminary findings showed that the system exhibited a low-risk safety profile and may be considered an effective therapeutic option for OAB management.

Dorsthorst et al (2020) examined the 3-year safety and effectiveness of the BlueWind Medical RENOVA iStim system for the treatment of OAB syndrome. All patients who previously underwent implantation with the RENOVA system were offered continued participation. The primary long-term study end-point was to examine the safety profile based on incidence of serious AEs (system- and/or procedure-related), which was measured by the impact and frequency of serious AEs. The secondary end-points included clinical improvement compared to baseline and QOL improvement compared to baseline at 36 months, which was measured by 3-day voiding diary and QOL questionnaires at certain time-points. Of the 34 patients with OAB syndrome who previously underwent implantation with the RENOVA system, 20 consented to continuation in this 3-year follow-up study. Mean patient age was 56.1 years, and 16 (80 %) of the study cohort were women. The overall treatment success rate was 75 % at 36 months in the per protocol (16) and the ITT (20) analyses. A total of 73 % of the patients reported improvement in health related QOL (HR-QOL) scores above the minimal important difference of 10 points. The authors concluded that this 3-year follow-up study using the BlueWind RENOVA iStim system for the treatment of OAB syndrome symptoms confirmed the long-term good safety profile with no technical failures reported. Lasting treatment effectiveness is mirrored by a sustained positive impact on patient QOL.

Dorsthorst et al (2022) reported on features that may be predictive of treatment response for patient-tailored OAB intervention with an implantable tibial neurostimulator using patient and technical prediction factors. This study was designed as a follow-up study based on parameter settings and patients' preferences during the pilot and extended study of the implantable tibial nerve stimulator (RENOVA iStim system). For this study, these investigators compared all treatment parameters (stimulation amplitude, frequency, and pulse width) and usage data (duration of treatment) during the different follow-up visits. They obtained usage data from a total of 32 patients who were implanted with the system between February and September 2015. Age, sex, BMI, and previous experience with PTNS treatment were considered as possible prediction factors for treatment success; however, only BMI was considered a statistically significant prediction factor ($p = 0.042$). A statistically significant increase in mean treatment level was observed in the responder group during the 3 month follow-up visit (mean: 6.7 mA, SD 0.416) as compared with the initial system activation visit (mean: 5.8 mA, SD 0.400) ($p = 0.049$). No other visits showed statistically significant changes in both groups (responders and non-responders) during the defined time-points. The authors concluded that these findings underscored the need to use patient-tailored OAB treatment; BMI was found to be a negative predictive factor for treatment success. However, it was not possible to develop a specific responder model. These researchers stated that the need for patient-tailored healthcare is important, and might improve the long-term treatment outcome and compliance for each patient. They stated that multi-center studies with a larger number of patients will shed light on a better predictive model in the treatment of OAB using an implantable tibial neuro-stimulation device.

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The authors stated that a drawback of this trial was the small sample size (n = 32). A larger sample size perhaps would have allowed delineation of predictive factors correlated with treatment success. Having predictive factors could be of great importance in counseling the patient for personalized OAB treatment. These predictive factors in combination with the further ongoing research in the onset of OAB could be the cornerstone of personalized OAB treatment instead of the "one-treatment-fits-all" theory, perhaps allowing earlier utilization of this therapy in the treatment paradigm.

Heesakkers et al (2024) noted that OAB affects 1 in 6 adults in Europe and the U.S., and impairs the QOL of millions of individuals globally. When conservative management fails, 3rd-line treatments including tibial neuromodulation (TNM) is often pursued. TNM has traditionally been accomplished percutaneously in clinic. A minimally invasive implantable device activated by a battery-operated external wearable unit has been developed for the treatment of UUI, mitigating the burden of frequent clinic visits, and more invasive therapies that are currently commercially available. In a prospective, single-arm, open-label, multi-center study, these researchers examined the safety and effectiveness of the BlueWind Implantable Tibial Neuromodulation [iTNM] system in adult females with UUI (i.e., wet OAB). Results with the device were previously published under the name RENOVA iStim, which has been since renamed as the Revi System. Approximately 1-month post-implantation of the device, subjects delivered therapy at their convenience and completed a 7-day voiding diary before visits 6- and 12-months post-treatment initiation. The primary safety and effectiveness endpoints were the proportion of responders to therapy (50 % or greater improvement on average number of urgency-related incontinence episodes) and incidence of AEs from implantation to 12-month post-activation. A total of 151 subjects, mean age of 58.8 years (SD: 12.5), were implanted; 144 and 140 completed the 6- and 12-month visits, respectively. Subjects demonstrated mean baseline of 4.8 UUI/day (SD 2.9) and 10 voids/day (SD 3.3). Six and 12-months post-activation, 76.4 % and 78.4 % of participants, respectively, were responders to therapy in an ITT analysis. Of the 139 participants with completed 12-month diaries, 82 % were responders, 50 % were classified as "dry" (on at least 3 consecutive diary days), and 93.5 % of participants reported that their symptoms improved. No implanted participant experienced a serious AE related to the procedure or device. The authors concluded that iTNM, delivered and powered by a patient-controlled external wearable communicating with an implant, showed clinically meaningful and statistically significant improvement in UUI symptoms and a high safety profile. This therapy highlighted the value of patient-centric therapy for the treatment of UUI. It should be noted that this study was funded by BlueWind Medical Inc.

Kapur et al (2024) noted that 3rd-line therapies for OAB that are currently recommended include intra-vesical onabotulinumtoxin-A injections (BTX-A), PTNS, and SNM. The implantable tibial nerve stimulator (ITNS) is a novel therapy that is now available to patients with OAB. These investigators analyzed shifts in patient preference of 3rd-line therapies for OAB after introducing ITNS as an option among the previously established therapies for non-neurogenic OAB. A survey was designed and distributed via SurveyMonkey to the platform's audience of U.S. adults of age 18 years and older. Screening questions were asked to include only subjects who reported symptoms of OAB. Descriptions of current AUA/SUFU guideline-approved 3rd-line therapies (BTX-A, PTNS, and SNM) were provided, and participants were asked to rank these therapies in order of preference (stage A). Subsequently, ITNS was introduced with a description, and participants were then asked to rank their preferences among current guideline-approved therapies and ITNS (stage B). Absolute and relative changes in therapy preferences between stage A and stage B were calculated. Associations between ultimate therapy choice in stage B and participant characteristics were analyzed. A total of 485 participants completed the survey (62.5 % female); the mean age was 49.1 ± 36.5 years (SD). The most common OAB symptoms reported were UUI (73.0 %) and urinary urgency (68.0 %); 29.2 % of patients had tried medication for OAB in the past, and 8.0 % to 10.3 % of patients were previously treated with a 3rd-line therapy for OAB. In stage A, participants ranked their 1st choice of 3rd-line therapy as follows: 28 % BTX-A, 27 % PTNS, and 13.8 % SNM; and 26.6% of participants chose no therapy, and 4.5 % chose all 3 equally. In stage B, participants ranked their 1st choice as follows: 27.6 % BTX-A, 19.2 % PTNS, 7.8 % SNM, and 19.2 % ITNS; 21.9 % of participants chose no therapy, and 4.3 % chose all 4 equally as their 1st choice. There were both absolute and relative declines in proportions of patients interested in BTX-A, SNM, and PTNS as their 1st choice of 3rd-line therapy with the introduction of ITNS. Patients originally interested in PTNS in stage A had the greatest absolute change following the introduction of ITNS with 7.8 % of participants opting for ITNS in stage B. Those interested in SNM in stage A had the largest relative change in interest, with 43.5 % of those originally interested in SNM opting for ITNS in stage B.

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Lastly, with the introduction of ITNS, the number of participants initially not interested in any 3rd-line therapy declined by an absolute change of 4.7 % and relative change of 17.6 %. Participants experiencing concurrent SUI symptoms were more likely to choose a current guideline-approved 3rd-line therapy than ITNS or no therapy at all ($p = 0.047$). Those who had previous experience with 3rd-line therapies were more likely to choose a 3rd-line therapy other than ITNS as their ultimate choice of therapy in stage B. Of those who had chosen a guideline-approved 3rd-line therapy in stage B (not ITNS), 13.6 % had previous experience with BTX-A, 14.7 % with PTNS, and 32 (11.2 %) with SNM ($p < 0.001$, $p < 0.001$, $p = 0.009$, respectively). The authors concluded that from this study, it appeared that ITNS may attract a subset of patients who would not have otherwise pursued current guideline-approved 3rd-line therapies for OAB. When patients were provided with descriptions of 3rd-line OAB therapies including ITNS as an option, ITNS appeared to compete with SNM and PTNS. It is possible that ITNS will provide patients with a different phenotype of neuromodulation therapy that can appeal to a niche OAB population. Given that ITNS devices have been introduced relatively recently to the market, their application will largely depend on cost and payer coverage, provider bias, and patient co-morbidities. These researchers stated that further investigation is needed to understand how these factors interact with and influence patient preference of advanced OAB therapy to understand which patients will most benefit from this treatment modality. Furthermore ,these investigators stated that long-term studies on ITNS devices are also needed to ascertain the effectiveness of ITNS over time in treating OAB and UUI compared to established and guideline-approved 3rd-line therapies.

The authors stated that this study had several drawbacks. First, this trial did not include all types of ITNS devices that are currently available. Although names of devices were not mentioned in the survey, the type of ITNS described in stage B was based on the technology that was FDA-approved at the time of survey distribution, namely eCoin, which is leadless and has an intrinsic battery within the implant. Since then, another ITNS device called Revi (BlueWind Medical) has recently gained FDA approval for the treatment of UUI. This technology has an external pulse generator that is worn around the ankle during treatment. While not included in this study, this form of ITNS may lead to further declines in patient preference of the other guideline-approved 3rd-line therapies. Second, survey distribution was stopped after reaching a critical response rate by participants; however, study generalization may have been more feasible with a larger sample size. Additionally, this trial under-represented some racial/ethnic minorities, such as African Americans and Hispanics. This distribution may reflect the convenience sampling utilized by SurveyMonkey, which may result in demographic distributions that do not reflect the overall U.S. population. There may also be a component of selection bias since the survey participants were limited to those who are members of SurveyMonkey Audience. Future studies should expand sampling to include more participants or account for key factors such as race/ethnicity. Third, with SurveyMonkey being a digital platform with no interactive interface, any questions that participants may have had that would affect their choice of therapy could not be answered, unlike in the authors' previous study that provided time for any questions regarding therapy after patients were shown the counseling video at an office visit appointment. Physician counseling on OAB therapies heavily influences patient preference and choice of treatment. This component was not considered in this survey, which may affect these findings. Fourth, this study was designed and distributed as a market research study to gauge interest in a novel treatment modality for idiopathic OAB; thus, the influence of provider bias, medical co-morbidities, and cost/coverage of the procedure on patient preference could not be assessed. These findings were based on patient perception of each therapy via the information these researchers provided within the survey, independent of the influence of physician counseling and provider bias, cost and payer coverage, and individual patient co-morbidities. However, these confounding factors play a huge role in influencing patient perception of 3rd-line therapies for OAB and can impact follow-through with therapy. Therefore, this should be considered when interpreting these findings, and in designing future relevant studies.

Flyte System (Mechanotherapy)

The Flyte System, a FDA-cleared device, is a mechanotherapy that guides users through Kegel exercises (pelvic floor muscle contractions) for the treatment of SUI. The Flyte device is inserted into the vagina and delivers vibrations while the user contracts and relaxes their pelvic floor muscles. The device's controller has a light that indicates when to contract and relax; the standard treatment time is 5 mins per day for 12 weeks.

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Nakib e al (2024) noted that SUI presents as unintentional urine leakage associated with activities. It significantly affects QOL and is the most common type of incontinence in women. Current therapeutic options, especially non-surgical therapies, are lacking. In a randomized, controlled, double-blinded study, these researchers examined the effectiveness of mechanotherapy provided by the Flyte intra-vaginal device during PFMT. Flyte is a repeat use device for conditioning and strengthening the PFMs. It provides 2-part mechanotherapy. Part 1 is the stretching and pre-loading of the PFM from the internal wand. Part 2 integrates mechanical pulses that elicit muscle cellular and tissue level responses that trigger cellular regeneration, improve neuromuscular facilitation and motor learning. Subjects used the device for 5 mins/day for 12 weeks. Subjects (n = 144) were randomized and evaluated at 6 and 12 weeks. Arm A (72) received both Part 1 and Part 2 mechanotherapy for 12 weeks, whereas Arm B (72) received Part 1 therapy for 6 weeks, then crossed-over to full therapy. Mean age was 50, 49 years, respectively, prior pelvic/abdominal surgery 26 %, 46 %, and previous incontinence treatments 13 %, and 22 %. The primary endpoint was 24-hour pad weight (24-HR PW) at 6 weeks. Secondary endpoints were 24-HR PW at 12 weeks and QOL [ICIQ, Urinary Incontinence Quality of Life (UI-QOL)]. Part 1 therapy had a greater than anticipated therapeutic effect; therefore, the study was under-powered to identify differences between study arms. Therefore, data were pooled to examine the effects of mechanotherapy. Twenty four-HR PW was significantly reduced at 6 weeks ($p = < 0.0001$), with further reduction from 6 to 12 weeks ($p = < 0.0001$). Data were stratified based on 24-HR PW severity. Significant reductions were noted in all severity groups (mild $p = < 0.0001$, moderate $p = < 0.0001$, severe $p = < 0.01$). QOL was similarly improved at 6 weeks (ICIQ $p = < 0.0001$, UI-QOL $p = < 0.0001$), and 12 weeks (ICIQ $p = < 0.0001$, UI-QOL $p = < 0.0001$). Compliance was greater than 80 % at 6 weeks and 70 % at 12 weeks. The authors concluded that 2-part mechanotherapy significantly improved 24-HR PW and QOL across all severities of SUI. Improvements were noted in as little as 2 weeks and appeared to be sustained through 2-year follow-up. Moreover, these researchers stated that further investigations are needed to determine optimal therapy protocols, and to evaluate the effectiveness of this promising therapy in other pelvic floor disorders, or to skeletal muscle abnormalities such as post-partum SUI, mixed incontinence, urge incontinence, and fecal incontinence.

The authors stated that this study had re several drawbacks. First, it was found that the sample size was under-powered to identify superiority of mechanical pulses (Part 2 mechanotherapy) added to the muscle stretch and pre-loading provided by the wand itself (Part 1 mechanotherapy). Therefore, the study design was not adequate due to the larger than expected positive effect of the wand alone. Post-hoc calculation indicated that 50 additional subjects (25 in each arm) were needed to show superiority of the added mechanical vibratory effect. The practical consequence of this distinction between study arms was uncertain, as the commercial device provides both components of the mechanotherapy. When the study data were pooled for both groups, the benefit of this therapy over baseline was highly significant for both objective and subjective endpoints. Second, some participants experienced discomfort using the wand. While this was anticipated, as with muscle soreness noted after any muscle exercise, a smaller wand configuration is now the standard product available. Third, the study did not examine variations in therapy delivery, (e.g., duration of muscle contraction against the device, length of each therapy session, and length of therapy beyond 12 weeks).

Furthermore, an UpToDate review on “Female urinary incontinence: Treatment” (Lukacz, 2024) does not mention mechanotherapy as a management / therapeutic tool.

Electro-Acupuncture for the Treatment of Neurogenic Bladder

Zeng and Li (2024) noted that neurogenic bladder dysfunction is a common consequence of stroke, and it substantially impacts the QOL and functional independence of affected individuals. Traditional treatment modalities have limitations in achieving optimal outcomes. In a retrospective, single-center study, these researchers examined the effectiveness of electro-acupuncture on bladder function and neurogenic bladder urodynamic characteristics in stroke patients. This trial was carried out on 100 stroke patients with neurogenic bladder admitted to the authors’ hospital from January 2020 to October 2023. It compared traditional treatment (n = 51) with electro-acupuncture treatment (n = 49). Baseline characteristics, urodynamic parameters, bladder function parameters, bladder symptoms, QOL assessments, AEs as well as patient satisfaction were collected from medical records and compared between the 2 groups. No significant

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difference was observed in age, sex distribution, BMI, duration of stroke, and alcohol intake between the 2 groups ($p > 0.05$). Compared with the traditional treatment group, the electro-acupuncture group showed significant improvements in urodynamic parameters such as maximum cytometric capacity (MCC), detrusor pressure at maximum capacity, post-void residual (PVR) volume, bladder compliance, Qmax, and average flow rate, and the differences were significant between groups ($p < 0.05$). The electro-acupuncture treatment group showed a significant reduction in bladder symptoms such as frequency of micturition and incontinence episodes ($p < 0.001$) and a significant improvement in cognitive function and social return function ($p < 0.05$). The incidence rates of UTI, hematuria, skin allergy and treatment discontinuation in the electro-acupuncture treatment group (6.12 %, 2.04 %, 4.08 %, and 12.24 %, respectively) were significantly lower than those in the traditional treatment group (23.53 %, 27.45 %, 29.41 %, and 35.29 %, respectively) ($p < 0.05$). The patient satisfaction score in the electro-acupuncture treatment group (97.96 %) was significantly higher than that in the traditional treatment group (70.58 %) ($p < 0.001$). The authors concluded that electro-acupuncture exhibited certain clinical value and holds promise as an adjunctive treatment for the treatment of neurogenic bladder dysfunction in stroke patients. Well-designed studies are needed to validate these preliminary findings.

High-Intensity Focused Electromagnetic Therapy in the Treatment of Stress Urinary Incontinence

In a single-center study, Long et al (2024) examined the effectiveness of high-intensity focused electromagnetic (HIFEM) technology in the treatment of female SUI. A total of 20 women with SUI received a treatment course with HIFEM technology. Patients attended 6 therapies scheduled twice-weekly. Validated questionnaires were assessed, including the OABSS, UDI-6, incontinence impact questionnaire-7 (IIQ-7), ICIQ, and valued living questionnaire (VLQ). Some urodynamic parameters, such as Qmax, residual urine (RU), and bladder volume at 1st sensation to void (Vfst). Bladder neck mobility in ultrasound (US) topography was also collected pre- and post-treatment at 1- and 6-month follow-up visits. HIFEM treatment significantly improved SUI symptoms on pad tests from 4.2 ± 5.5 to 0.6 ± 1.3 and patients' self-assessment in the 6-month follow-up. Furthermore, the data from urinary-related questionnaires, including OABSS (5.3 ± 3.9 to 3.9 ± 3.6), UDI-6 (35.7 ± 22.3 to 15.2 ± 10.6), IIQ-7 (33.1 ± 28.7 to 14.3 ± 17.2), and ICIQ (9.4 ± 5.0 to 5.4 ± 3.6), all showed a significant reduction. Then, the analysis of the urodynamic study showed that only maximum urethral closure pressure (MUCP) (46.4 ± 25.2 to 58.1 ± 21.2) and urethral closure angle (UCA) (705.3 ± 302.3 to 990.0 ± 439.6) significantly increased after the 6 sessions of HIFEM treatment. The urethral and vaginal topography were carried out and found that HIFEM mainly worked on PFM and enhanced their function and integrity. The authors concluded that the findings of this study suggested that HIFEM technology was an effective therapy for the treatment of SUI. Moreover, these investigators noted that future studies should examine several areas to strengthen the clinical implications of HIFEM technology in the treatment of SUI. The longer follow-up periods should be included, and the observation period should be extended to 1 or more years, which could provide more robust data regarding the long-term effectiveness of HIFEM treatment effects. Furthermore, a comparative study could also be conducted with other non-invasive or minimally invasive treatments, such as fractional CO2 laser therapy or PRP, which could contextualize its relative benefits as well as limitations. Lastly, the investigation of the accurate molecular and cellular mechanisms by HIFEM on PFM might be beneficial to its application on other pelvic floor disorders.

The authors stated that this trial had several drawbacks. First, the small sample size ($n = 20$) might reduce the statistical power and might not represent the whole population of SUI patients. Second, the lack of a control group might limit the attributed observed improvements solely to the HIFEM treatment. Third, the short observation period (up to 6 months). Fourth, the study was carried out in a single center, which may have introduced selection bias and limited the generalizability of findings to other settings.

Leonardo et al (2025) noted that many treatments have been employed to treat UI; however, all these therapeutic approaches have limitations. Recently, a new non-invasive treatment for PFM using a HIFEM field was unveiled. In a systematic review and meta-analysis, these investigators examined the available evidence on the safety and effectiveness of HIFEM. The databases used were PubMed, Cochrane, Embase, and SCOPUS. The literature search was performed using strategic keywords (women) AND ((High Intensity

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Electromagnetic Field) OR (Electromagnetic Stimulation)) AND ((urinary incontinence) OR (overactive bladder) OR (pelvic floor dysfunction)). Studies that meet the inclusion and exclusion criteria were then analyzed. A total of 7 studies were included in this review, most of the studies concluded that the usage of HIFEM could reduce the symptoms related to UI significantly and improve QOL. There was higher decrease of UI episodes (MD: -4.10, 95 % CI: -7.34 to -0.85, p = 0.01) and improvement of ICIQ-UI SF score (MD: -3.03, 95 % CI: -3.27 to -2.79, p = < 0.00001) in HIFEM group compared to control. Sub-group analyses showed better QOL parameter (MD -3.40; p = 0.01, MD -0.70; p = 0.04) compared to control, albeit statistically comparable overall (p = 0.09). However, both pooled analyses for contraction and resting tone changes showed that there were no significant differences between the 2 groups (SMD: 0.98, 95 % CI: -0.70 to 2.660, p = 0.25; and SMD: 0.20, 95 % CI: -0.18 to 0.58, p = 0.30, respectively). Lastly, there was no safety issue highlighted in most of the studies included. The authors concluded that available evidence suggested that HIFEM may be a safe and effective non-invasive treatment for female UI by promoting QOL. Moreover, these researchers stated that as a consequence of high heterogeneity and possible bias, future well-designed studies with proper blinding and standardized outcomes are needed to ascertain the applicability of HIFEM for the treatment of UI.

Implantable Sacral Nerve Stimulators for the Treatment of Neurogenic Bladder

In a meta-analysis, Sun and Song (2025) examined the safety and effectiveness of SNM in the treatment of neurogenic bladder (NB) and neurogenic bowel dysfunction (NBD). These investigators carried out a systematic literature search using PubMed and Web of Science up to August 2024, focusing on studies related to SNM treatment for NB or NBD. After assessing the quality of the studies, data were extracted and analyzed using Review Manager 5.3. A total of 15 studies involving 573 patients were included. After SNM treatment, the patients showed significant improvements in key outcome measures, including voiding frequency per 24 hours (V24) (weighted mean difference [WMD] -4.08; 95 % CI: -6.80 to -1.35; p = 0.003), mean urine volume per micturition (MUV) (WMD 123.60; 95 % CI: 93.17 to 154.03; p < 0.001), number of leakage episodes per 24 hours (L24) (WMD -4.27; 95 % CI: -5.79 to -2.74; p < 0.001), number of nocturia (WMD -2.48; 95 % CI: -2.62 to -2.35; p < 0.001), clean intermittent self-catheterization per 24 hours (WMD -2.35; 95 % CI: -2.98 to -1.71; p < 0.001), bladder compliance (WMD 9.09; 95 % CI: 2.31 to 15.87; p = 0.009), maximum detrusor pressure (MDP) during storage phase (WMD -14.76; 95 % CI: -18.63 to -10.88; p < 0.001), maximum urine flow rate (Qmax) (WMD 6.50; 95 % CI: 4.21 to 8.80; p < 0.001), MCC (WMD 66.28; 95 % CI: 2.83 to 129.73; p = 0.04), Wexner score (WMD -9.98; 95 % CI: -13.65 to -6.31; p < 0.001), and NBD score (WMD -6.31; 95 % CI: -6.89 to -5.73; p < 0.001). The authors concluded that these findings indicated that SNM was safe and effective in treating NB or NBD.

In a meta-analysis, Yu et al (2025) examined available evidence of non-invasive or minimally invasive neuromodulation therapies in improving urodynamic outcomes, voiding diaries, and quality of life (QOL) in patients with neurogenic lower urinary tract dysfunction (NLUTD) after spinal cord injury (SCI). These researchers carried out a comprehensive search of 10 data-bases from inception until August 30, 2023. Randomized controlled trials (RCTs) examining the effects of conventional treatment (CT) and CT combined with sham stimulation (SS), transcranial magnetic stimulation (TMS), sacral nerve magnetic stimulation (SNMS), TMS+SNMS, sacral pulsed electromagnetic field therapy (SPEMFT), sacral transcutaneous electrical nerve stimulation (STENS), sacral dermatomal transcutaneous electrical nerve stimulation (SDTENS), bladder & sacral transcutaneous electrical nerve stimulation (B&STENS), transcutaneous tibial nerve stimulation (TTNS), transcutaneous electrical acupoint stimulation (TEAS), pelvic floor electrical stimulation (PFES), or pelvic floor biofeedback therapy (PFBFBT) on PVR volume, MCC, V24, MUV, Qmax, MDP, maximum voiding volume (MVV), L24, lower urinary tract symptoms (LUTS) score, and spinal cord injury-QOL (SCI-QOL)score in patients with NLUTD after SCI were included. Two researchers independently extracted data on study characteristics and outcomes following the PRISMA statement. The Cochrane risk of bias tool (2.0) was used to assess the quality of RCTs. A total of 52 RCTs with 2,884 participants were included. CT+TMS was able to remarkably decrease PVR (mean difference [MD], -132.14; 95 % CI: -230.97 to -33.31) and increase MUV (MD, 147.79; 95 % CI: 64.51 to 231.06). CT+SNMS ranked high in improving V24 (MD, 2.76; 95 % CI: 1.26 to 4.25) and reducing L24 (MD, -2.73; 95 % CI: -4.46 to -1.01); CT+TMS+SNMS maximized the reduction of SCI-QOL scores (MD, -1.52; 95 % CI: -2.97 to -0.25) and ranked 2nd in both reducing PVR and improving MCC; CT+SPEMFT had a

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significant advantage in improving MCC (MD, 83.31; 95 % CI: 39.73 to 126.88) and increasing Qmax (MD, 5.91; 95 % CI: 2.99 to 8.84); improvement in MDP was highly ranked by CT+TTNS (MD, 9.46; 95 % CI: 2.15 to 16.76). The authors concluded that CT combined with magnetic stimulation therapy provided more benefits than its combination with electrical stimulation. These researchers stated that TMS+SNMS appeared to be a promising non-invasive neuromodulation technique in managing NLUTD after SCI. Moreover, these researchers stated that high-quality RCTs are needed to validate these findings.

Buford et al (2024) stated that NLUTD is highly prevalent among patients with neurologic disorders. Some studies have reported that implantable neuromodulation could improve symptoms of NLUTD. In a retrospective, single-center study, these investigators described their experience with sacral and pudendal neuromodulation in patients with NLUTD. They carried out a retrospective chart review of patients with "neurogenic bladder" ICD-9/10 (International Classification of Diseases, 9th Revision/10th Revision) code was carried out at a single institution. This included patients from 2008 to 2020 who underwent stage 1 neuromodulation trial. Demographic and clinical information was collected, including neurologic diagnosis, the character of patients' voiding symptoms, the presence or absence of fecal incontinence, the need for intermittent catheterization, and whether patients had sufficient (greater than 50 %) improvement in their symptoms to undergo stage 2 IPG placement. These researchers identified 82 patients with neurologic diagnoses who underwent stage 1 neuromodulation. The most common diagnoses were diabetic cystopathy (17.07 %), spinal surgery (17.07 %), and SCI (12.20 %). The most commonly reported symptoms were urinary urgency, and urge urinary incontinence. Overall, 59 patients (71.95 %) advanced to stage 2 IPG placement including 72 % of patients with sacral leads and 76 % with pudendal leads. The authors concluded that neuromodulation was feasible and effective in the treatment of NLUTD. Moreover, these researchers stated that further investigations into its utilization are needed.

The authors stated that drawbacks of this trial included its retrospective design; thus, they could not examine potential causality, delve deeply into patient history, and standardize data collection. The small sample size (n = 82 subjects; 10 patients with SCI) limited the generalizability of the findings. However, the sample size was on par with similar studies that examined neuromodulation for the management of NLUTD. Furthermore, the study was carried out at a single-single trial, which may have restricted the diversity of the patient population. These investigators also stated that it was important to acknowledge that due to the focus of this study on the short trial phase, long-term effectiveness and AEs were not specifically recorded or analyzed. In a subsequent meta-analysis that included SNM for non-neurogenic voiding symptoms, the incidence of AEs was comparable to those reported in the van Ophoven meta-analysis; thus, it was reasonable to assume potential long-term AEs would be consistent with other studies. This highlighted the need for future research with prospective designs, larger sample sizes, multi-center collaborations, validated questionnaires, and longer follow-up periods to provide more robust and comprehensive evidence.

Furthermore, an UpToDate review on “Chronic complications of spinal cord injury and disease” (Abrams and Wakasa, 2024) states that “Studies of implanted sacral nerve modulators show promise as a treatment for urinary incontinence following SCI. For patients with unsatisfactory response to medical and catheter management, other treatments, bladder augmentation, urinary diversion, sphincterotomy, urethral stent, and electrical implantation devices can be considered in selected cases, although there is limited evidence of their comparative efficacy”.

Micro-Ablative Radiofrequency for the Treatment of Over-Active Bladder

Slongo et al (2025) stated that energy therapies have been suggested as potential treatments for OAB; however, there are few studies examining their effectiveness. In a pilot study, these researchers compared the effects of fractional micro-ablative RF to sham treatment. This trial was carried out with 77 women diagnosed with OAB, randomized into 2 groups: one receiving 3 monthly sessions of fractional micro-ablative RF and the other receiving sham treatment, both combined with behavioral therapy. Assessments were carried out at baseline and 30 days after therapy using validated questionnaires for urinary symptoms (ICIQ-Short Form, ICIQ-OAB, and ICIQ-QOL), as well as vaginal and sexual function, and pelvic floor muscle functionality. Both treatment groups showed significant improvements in all validated questionnaires evaluating urinary and vaginal symptoms ($p < 0.001$), with no significant differences between them. The ICIQ-OAB scores improved significantly in both the RF group (-4.5 points; $p < 0.001$) and the sham group

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(-4.5 points; $p < 0.001$), with no significant differences between the groups ($p = 0.812$). Furthermore, there were no improvements noted in sexual function or vaginal trophism in the RF group. However, assessments of endurance, resistance, and fast contractions of the pelvic floor muscles revealed improvement only in the RF group, with no changes observed in power or perineometer measurements. The authors concluded that micro-ablative RF treatment combined with behavioral therapy did not show benefits over sham treatment with behavioral therapy for the treatment of OAB symptoms.

Percutaneous Tibial Nerve Stimulation (PTNS) for the Treatment of Idiopathic Overactive Bladder

Jericevic et al (2024) stated that OAB patients who do not achieve satisfactory results with 2nd-line OAB medications should be offered 3rd-line therapies (e.g., BTX-A, PTNS, and SNM). These researchers determined which clinical factors affect progression from 2nd- to 3rd-line OAB therapy. Between 2014 and 2020, the American Urological Association (AUA) Quality Registry was queried for adult patients with idiopathic OAB. For the primary outcome, patient and provider factors associated with increased odds of progression from 2nd- to 3rd-line therapy were assessed. Secondary outcomes included median time for progression to 3rd-line therapy and 3d-line therapy utilization across subgroups. A total of 641,122 patients met inclusion criteria and were included in analysis. Of these, only 7,487 (1.2 %) received 3rd-line therapy after receiving 2nd-line therapy. On multi-variate analysis, patients aged 65 to 79, women, White race, history of dual anti-cholinergic and β3 agonist therapy, metropolitan area, government insurance, as well as single specialty practice had the greatest odds of progressing to 3rd-line therapy. Black and Asian race, male gender, and rural setting had lower odds of progressing to 3rd-line therapy. BTX-A was the most common therapy overall (40 % BTX-A, 32 % SNM, 28 % PTNS). The median time of progression from 2nd- to 3rd-line therapy was 15.4 months (IQR 5.9 to 32.4). Patients less than 50 years old and women progressed fastest to 3rd-line therapy. The authors concluded that very few patients received 3rd-line therapies, and the time to progression from 2nd- to 3rd-line therapies was greater than 1 year. These researchers stated that the findings of this study high-lighted a potential need to improve 3rd-line therapy implementation.

Furthermore, the American Urological Association/Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (AUA/SUFU)'s guideline on "Diagnosis and treatment of idiopathic overactive bladder" (Cameron et al, 2024) provided the following recommendation:

> In patients with OAB who have an inadequate response to, or have experienced intolerable side effects from, pharmacotherapy or behavioral therapy, clinicians should offer SNM PTNS, and/or intra-detrusor BTX injection. (Moderate Recommendation; Evidence Level: Grade A)

Jefferson et al (2025) stated that procedural therapies for OAB most often include intra-detrusor BTX injections, SNM, and PTNS. Despite their proven effectiveness, real-world data regarding therapy continuation and cross-over rates between these modalities remain limited. In a retrospective, multi-center study, these investigators examined longitudinal patterns of treatment continuation and therapy cross-over among women receiving procedural OAB treatments. This trial enrolled women with idiopathic OAB who underwent index BTX, SNM, or PTNS between 2014 and 2022. Subjects were identified using electronic medical records, and follow-up data were collected through December 31, 2023. The primary outcomes included duration of maintenance treatments and cross-over to another procedural therapy for OAB. Kaplan-Meier survival analysis estimated rates of continued treatment, and a log-rank test examined differences in cross-over rates among treatment groups. A total of 506 women met inclusion criteria: 266 (52.6 %) received BTX, 114 (22.5 %) underwent SNM, and 126 (24.9 %) received PTNS. Ongoing treatment rates were lowest for PTNS, with only 17.1 % maintaining therapy between 3 and 5 years and 26.1 % crossing over to another procedure within 3 years. Among BTX patients, 65.2 % underwent repeat treatment within 3 years, and 9.5 % of patients crossed-over to another procedure within 3 years. Of those with 5 or more years of follow-up, 39.1 % of patients continued with BTX procedures in the 3- to 5-year follow-up time-point. Among SNM patients, the cumulative revision rates were 7.6 % at 1 year and 10.9 % at 2 years; and 13.1 % of patients crossed-over from SNM to another procedure within 3 years (the vast majority to BTX). The authors concluded that PTNS had the lowest ongoing therapy rates and the highest cross-over rates. BTX and SNM

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showed similar cross-over rates. These researchers stated that these findings underscored the importance of individualized patient counseling and treatment selection based on expectations of therapy persistence.

Furthermore, an UpToDate review on "Urgency urinary incontinence/overactive bladder (OAB) in females: Treatment" (Lukacz, 2025) states that "Patients who have persistent urgency urinary incontinence or other irritative symptoms despite an adequate trial of initial treatments and pharmacotherapies, are unable to tolerate pharmacologic therapy, or decline the above treatments after discussion of risks and benefits, can be referred to a specialist to discuss further options for treatment. Advanced therapies range from noninvasive office-based acupuncture-like nerve stimulation (PTNS) and office-based injections of botulinum toxin A to surgically implanted nerve stimulation devices (tibial nerve stimulation and SNM)".

Protect PNS System for the Treatment of Overactive Bladder

Protect PNS (Uro Medical, Boca Raton, FL) is a minimally invasive, wirelessly powered, micro-technology neurostimulator, currently being studied for the treatment of OAB and is under regulatory review by the FDA.

Sirls et al (2023) noted that tibial nerve stimulation is an effective treatment for OAB and has been employed as an in-person recurring session therapeutic option for many years. In a pilot study, these investigators examined the safety and effectiveness of a long-term implantable device and the method of using a retrograde approach to place the device (a percutaneous implantable pulse generator [pIPG] with integrated quadripolar electrodes) at the tibial nerve (Protect PNS; Uro Medical). This novel retrograde implant technique was developed via multiple cadaveric dissections to percutaneously implant a chronic, wireless, minimally invasive pIPG device with integrated quadripolar electrodes at the tibial nerve. A proof-of-concept (POC) pIPG device approved as part of an FDA IDE was designed to gain early experience in subjects with refractory OAB. The pIPG was implanted in the office under local anesthesia using the novel retrograde approach, and stimulation was activated by means of an external wireless energy source. Initially, a pilot study was designed to compare outcomes in subjects randomized to either PTNS or Protect PNS. However, due to the small sample sizes available at this time, it was not possible to compare the 2 groups; therefore, the objective of this manuscript was to describe the outcome of subjects who underwent implantation of the Protect PNS system; 12-month safety and effectiveness were evaluated. A total of 9 subjects were enrolled in the randomized pilot study; 5 to the pIPG group and 4 to PTNS, and all completed the 13-week primary end-point. Subsequently, 2 subjects in the PTNS group chose to cross-over and had the pIPG implanted after 13 weeks. Outcomes of the 7 subjects who underwent implantation of the pIPG were described. No complications related to the office procedure were noted. Two of the older model pIPG devices became non-responsive at 1 and 4 weeks and were replaced. Six minor AEs were reported and resolved. Participants reported improvement in UUI episodes, OAB symptoms, and QOL. Subjects implanted with a pIPG reported a 50 % reduction in UUI as early as 1 week. The authors concluded that the findings of this pilot study suggested that retrograde percutaneous implantation of a pIPG was a safe, minimally invasive, 1-stage office procedure for the treatment of urge incontinence-related OAB symptoms, without significant complications after 12 months follow-up. These researchers stated that future studies are needed to compare outcomes among treatment modalities.

The PROTECT Trial is a prospective, randomized-controlled, non-inferiority, multi-center clinical trial of the Protect PNS System of the tibial nerve versus SNS in the treatment of urinary urgency and incontinence resulting from refractory OAB. Another clinical trial, entitled "Guardian" is designed to examine the safety and effectiveness of Protect PNS in the treatment of OAB, compared to traditional medical management. The Guardian Trial enrolls about 600 subjects with refractory OAB at multiple centers across the U.S. While the GUARDIAN Trial has been completed, the Protect PNS is still considered investigational and is not yet available for general use. Uro Medical is expected to use the study results to support long-term, nationwide payor reimbursement coverage for the Protect PNS System as a mainline therapeutic option for OAB.

Amundsen et al (2025) stated that implantable tibial nerve neuromodulation (iTNM) systems have recently become commercially available in the U.S., and offer a new method of neurostimulation for the treatment of OAB. In the absence of head-to-head studies, the objective of this meta-analysis was to indirectly compare

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the safety and effectiveness of SNM and iTNM for the treatment of OAB. These investigators carried out a comprehensive search using terms for OAB and neuromodulation. Primary effectiveness measures included a 50 % or greater reduction in UUI episodes, urinary frequency, and/or OAB symptoms. Primary safety measures included the rate of device-related AEs. A total of 20 studies fulfilled selection criteria, totaling 1,416 patients treated with SNM and 350 patients treated with iTNM. No comparative or placebo-controlled studies for SNM and iTNM were identified; thus, the analysis was completed using single-arm results. Weighted averages showed that the UUI responder rate was similar for both SNM and iTNM (71.8 % and 71.3 %, respectively). Similarly, weighted averages of OAB responder rates were 73.9 % for SNM and 79.4 % for iTNM. Similar rates of device-related AEs were also observed. The authors concluded that this meta-analysis found similar safety and effectiveness of SNM and iTNM for the treatment of OAB and UUI, including UUI and OAB symptom response rates, reduction in UUI episodes, significant improvements in QOL, and low rates of procedure and device-related AEs. Notably, this comparable effectiveness was observed without the use of a trial phase of neuromodulation in the iTNM studies versus SNM studies. These researchers stated that current findings suggested that iTNM may also have lower surgical re-intervention rates; however, additional follow-up time is needed to confirm if this trend will continue.

The authors stated that this meta-analysis had several drawbacks. First, while this meta-analysis featured a large number of studies and corresponding patients, 11 studies were considered to have a moderate risk of potential bias, most often related to challenges with randomization and blinding due to patient sensation of neurostimulation, along with the retrospective and single-center nature of several of these studies. Differences in study populations, geography, study methods, effectiveness definitions, and stage of device development (i.e., pilot study versus pivotal study versus study completed post-marketization) may also impact generalizability of the results. Patient level data were not available for analysis; thus, data for all end-points were not always available for each study even if they had been collected. These elements, though possible with any meta-analysis and not specific to this review, should be considered when interpreting the presented results. Second, the expected difference between the length of follow-up observed in the selected iTNM and SNM studies (weighted average of 13.0 months and 39.2 months, respectively). While SNM devices have been available for commercial use in the U.S. for almost 30 years, iTNM was first FDA-approved less than 2 years ago. Accordingly, the average length of follow-up data available for SNM is much longer than iTNM. More importantly, most device- and procedure-related AEs notably occurred within the first several months following implantation for both iTNM and SNM devices. Further comparability between later revisions and/or explant rates will certainly need to be assessed. Clearly, devices designed with implanted batteries -- both iTNM and SNM -- will require eventual surgical replacement. Both the original implant and surgical revisions result in costs to medical systems and patients; published data comparing cost between iTNM and SNM are not yet available. Third, challenges in comparing frequency of urination were also identified in this analysis, primarily because of consistent differences in urinary frequency reported in the studies completed in Asia, which ultimately may describe a more severe population of patients with OAB. While the reason for this difference is unclear, the frequency was reported with range of 10 to 14.9 in iTNM and SNM studies completed in North America, Europe, and the Middle East, compared with 21.6 to 29.2 in SNM studies in Asia, resulting in challenges in comparison of results related to urinary frequency. The impact of iTNM on urinary frequency needs to be further examined in future studies. Fourth, none of the studies identified was head-to-head trials, limiting the ability for direct comparisons. Future RCTs are needed to provide additional clarity regarding comparative differences in safety and effectiveness, and to provide further guidance regarding ideal patient selection for both methods of neuromodulation, tibial versus sacral.

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The above policy is based on the following references:

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Implantable / Subfascial Tibial Nerve Stimulation (e.g., BlueWind Revi) for the Treatment of Overactive Bladder Syndrome and Urgency Urinary Incontinence

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